



BOM Groups

BOM GROUP	BOM OPTIONS
J44_COMMON	ALTERNATE, COMMON, J44_COMMON1, J44_COMMON2, J44_COMMON3, J44_COMMON4, J44_PROGPARTS
J44_COMMON1	TBTHV:P15V, SKIP_5V3V3:AUDIBLE, SPI:DUAL_IO
J44_COMMON2	EDP, EDP_LS_CAP, CAMERA_3V3:S0, CAM_WAKE:NO, CAM_XTAL:NO, MEM_ODT:PU, VCORE_FETS
J44_COMMON3	XDP, LPCPLUS, BKL T:PROD, CPUTHRM:ALRT, LOADRC:NO, OTHERRC:NO, DDRRC:NO, TBTRC:NO, BMONRC:NO
J44_PROGPARTS	SMC_PROG:PVT, BOOTROM:PVT, TBTRM:PVT, TPAD_PSOC:PROG
ENGISNS	LOADISNS, OTHERISNS, DDRISNS, TBTISNS, BMONISNS

Programmables (All Builds)

TBT

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S3918	1	EPROM, FALCON RIDGE (V13.7) J44	U2890	CRITICAL	TBTROM:PVT

SMC

341S3922	1	IC, SMC-B1, EXT(V2.16F39), PVT, J44	U5000	CRITICAL	SMC_PROG:PVT
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Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4596	1	HSWULT, SR18A, PRQ, C0, 2.4, 28W, 2+3, 3M, BGA	U0500	CRITICAL	CPU_HSW:2.4G
337S4597	1	HSWULT, SR189, PRQ, C0, 2.6, 28W, 2+3, 3M, BGA	U0500	CRITICAL	CPU_HSW:2.6G
337S4598	1	HSWULT, SR188, PRQ, C0, 2.8, 28W, 2+3, 4M, BGA	U0500	CRITICAL	CPU_HSW:2.8G
338S1247	1	IC, TBT, FR-4C, A0, PRQ, C10, SR13C, FCBGA288	U2800	CRITICAL	
338S1186	1	IC, BCM15708A2, S2 PCIE CAMERA PROCESSOR	U3900	CRITICAL	
376S1194	2	MOSFET, N-CH, 30V, 15.3A, 12M, 8P 3.3X3.3 DFN	Q7310, Q7320	CRITICAL	VCORE_FET:VSHY
376S1193	2	MOSFET, N-CH, 30V, 22A, 6.0M, 8P 3.3X3.3 DFN	Q7311, Q7321	CRITICAL	VCORE_FET:VSHY
376S0964	2	MOSFET, N-CH, 25V, 30A, 9.6M, 8P 3.3X3.3 DFN	Q7310, Q7320	CRITICAL	VCORE_FET:REN
376S1104	2	MOSFET, N-CH, 25V, 30A, 6.1M, 8P 3.3X3.3 DFN	Q7311, Q7321	CRITICAL	VCORE_FET:REN

EFI ROM

341S3924	1	IC, EFI ROM (V0116), PVT, J44	U6100	CRITICAL	BOOTROM:PVT
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PSOC

341S3862	1	IC, TRKPD/KYBD PSOC, CU ONLY(V224) J44	U4801	CRITICAL	TPAD_PSOC:PROG
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Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	
128S0311	128S0329		ALL	
138S0739	138S0706		ALL	
197S0481	197S0480		ALL	
152S0461	152S1645		ALL	
376S1080	376S0820		ALL	
155S0667	155S0583		ALL	
138S0725	138S0724		ALL	
376S1032	376S0855		ALL	
376S1129	376S0855		ALL	
376S1089	376S1128		ALL	
353S3452	353S1286		ALL	
376S1180	376S0761		ALL	
128S0364	128S0264		ALL	
107S0254	107S0241		ALL	
138S0843	138S0674		ALL	
138S0803	138S0639		ALL	
138S0846	138S0811		ALL	
197S0542	197S0544		ALL	
197S0545	197S0544		ALL	
152S1876	152S1804		ALL	
107S0255	107S0240		ALL	
107S0250	107S0248		ALL	
127S0164	127S0162		ALL	
353S4070	353S4069		ALL	
353S4068	353S4069		ALL	
353S3814	353S3812		ALL	
311S0649	311S0541		ALL	
128S0436	128S0392		ALL	

Diodes alt to Fairchild
NEC alt to Sanyo
Samsung alt to Murata
Epson alt to NDK
Cyntec alt to Vishay
Diodes alt to On Semi
Panasonic alt to TDK
Samsung alt to Murata
Toshiba alt for Diodes Dual
NXP Alt for Diodes Dual
NXP Alt for Diodes Single
Maxim alt to Microchip
Renesas alt to Vishay
Sanyo 2nd Factory alt
Cyntec alt to TFT
Samsung alt to Murata (BKLT)
Samsung alt to Murata (BKLT)
Samsung alt to Murata (BKLT)
NDK alt to TXC
Epson alt to TXC
TDK alt to Toko
Cyntec alt to TFT
Cyntec alt to TFT
Rohm alt to Vishay
Pericom alt to TI DP Mux U9750
NXP alt to TI DP Mux U9750
TI alt to NXP
ONsemi alt to Toshiba
Kemet alt to Sanyo

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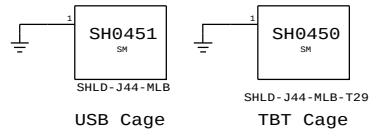
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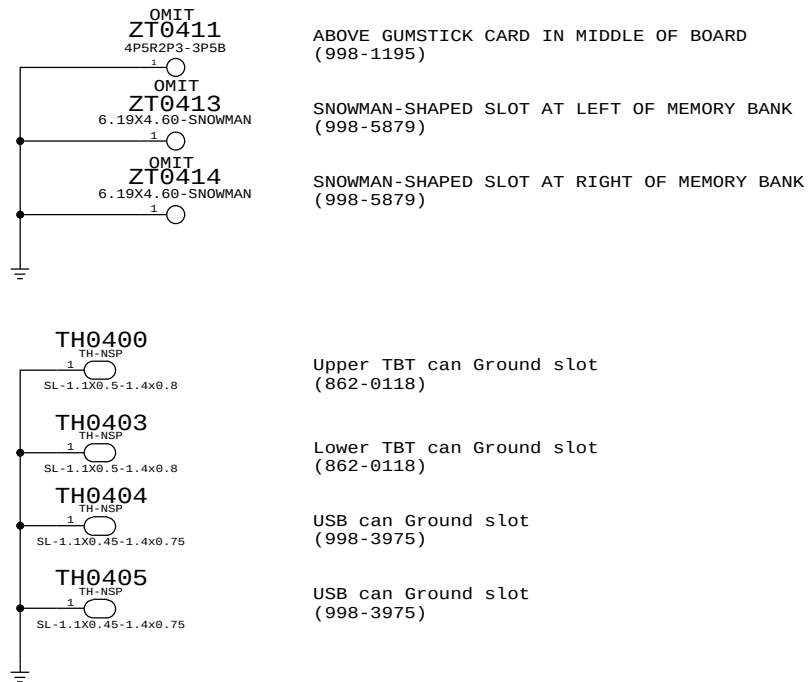
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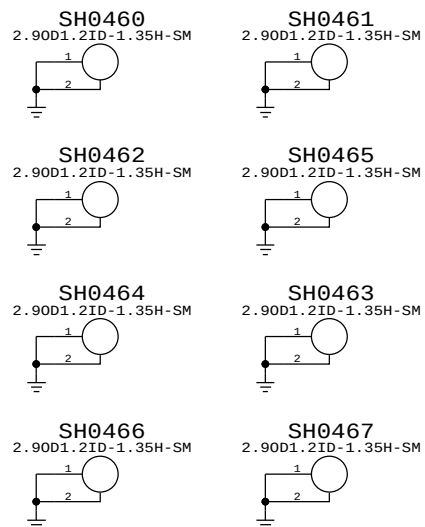
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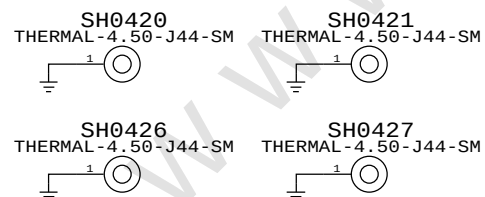
Mounting Holes & Slots



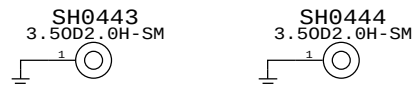
Rubber Mount Standoffs (860-1448)



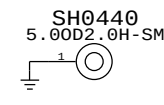
THERMAL MODULE STANDOFF (860-1645)



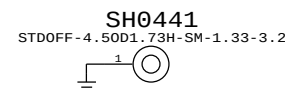
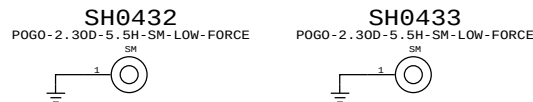
RIO FLEX BRACKET BOSSES (860-2354)



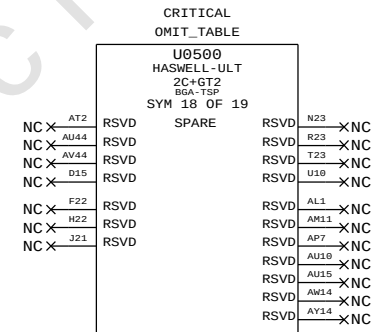
SSD STANDOFF (806-5375)



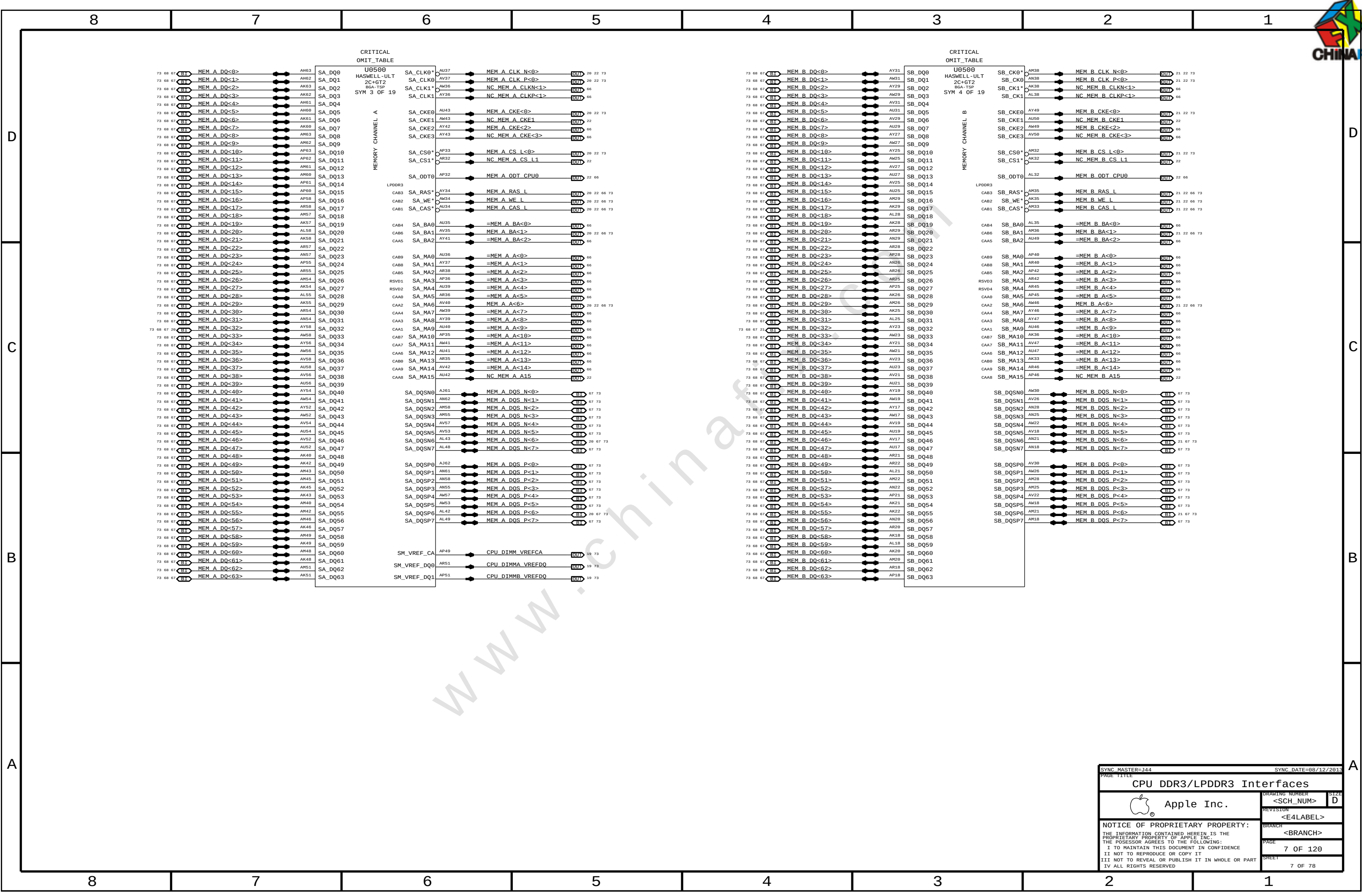
FAN STANDOFF (806-5376)

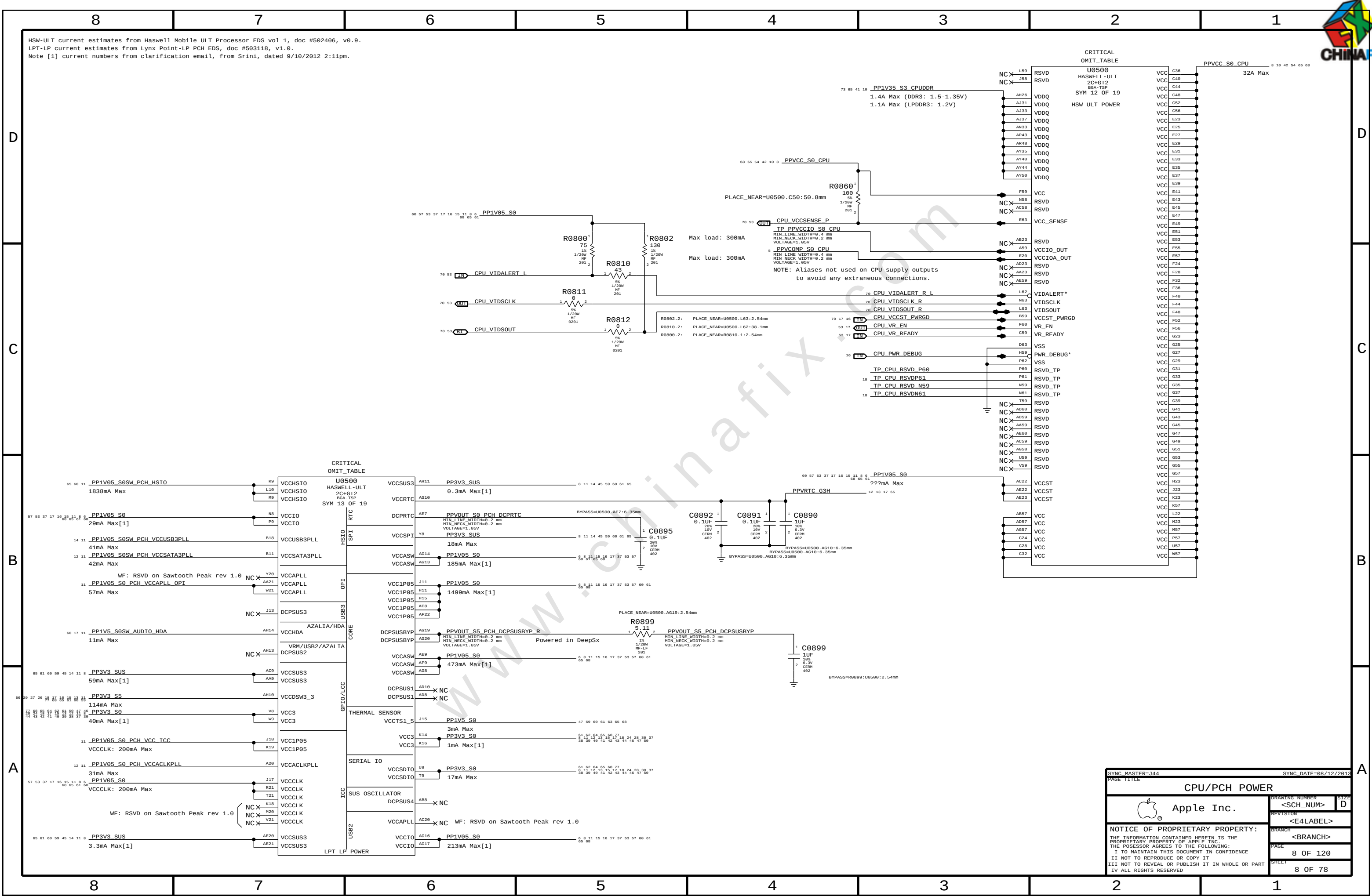
POGO PINS (870-2451)
SH0435 & SH0436 removed.

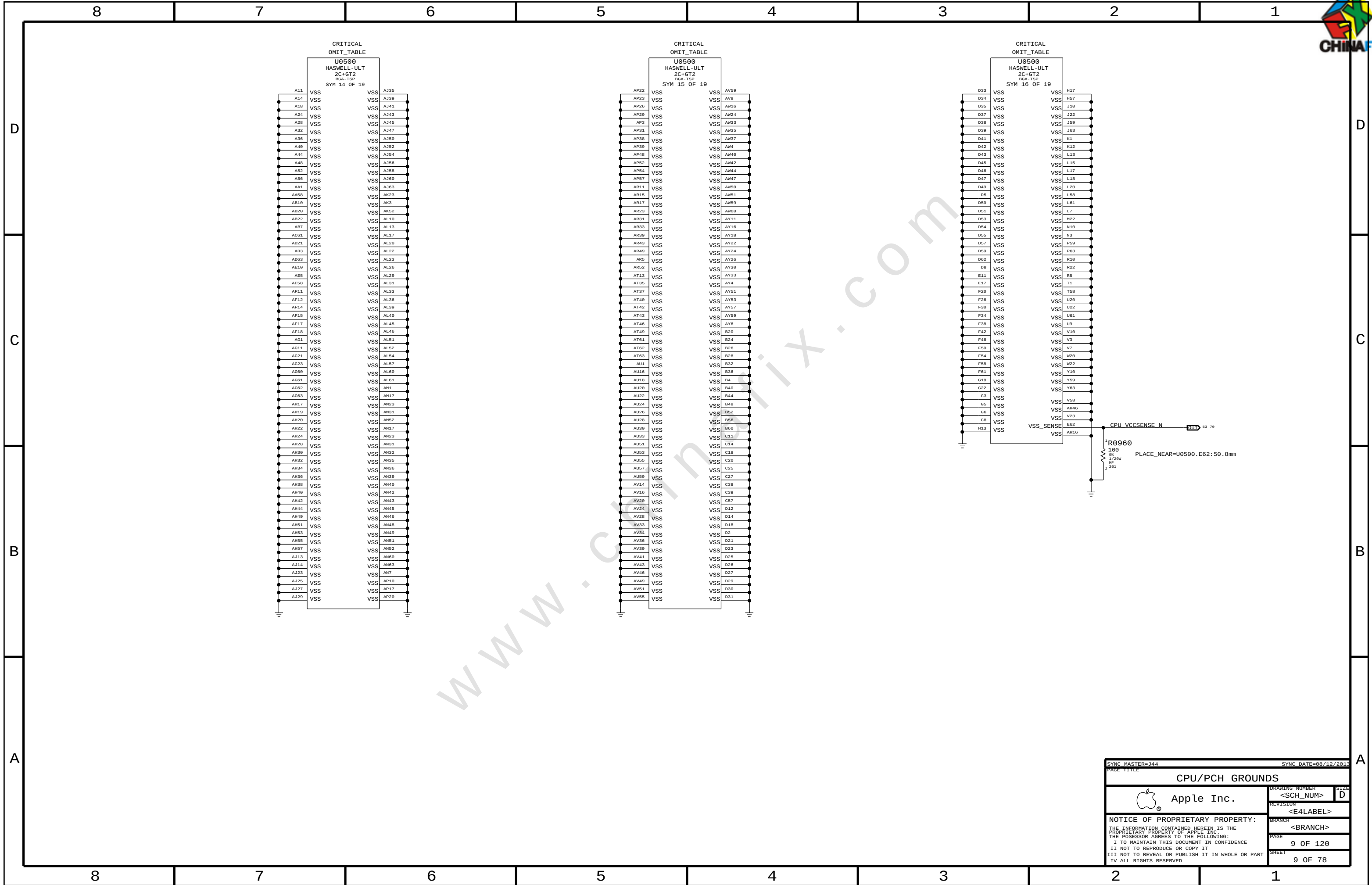
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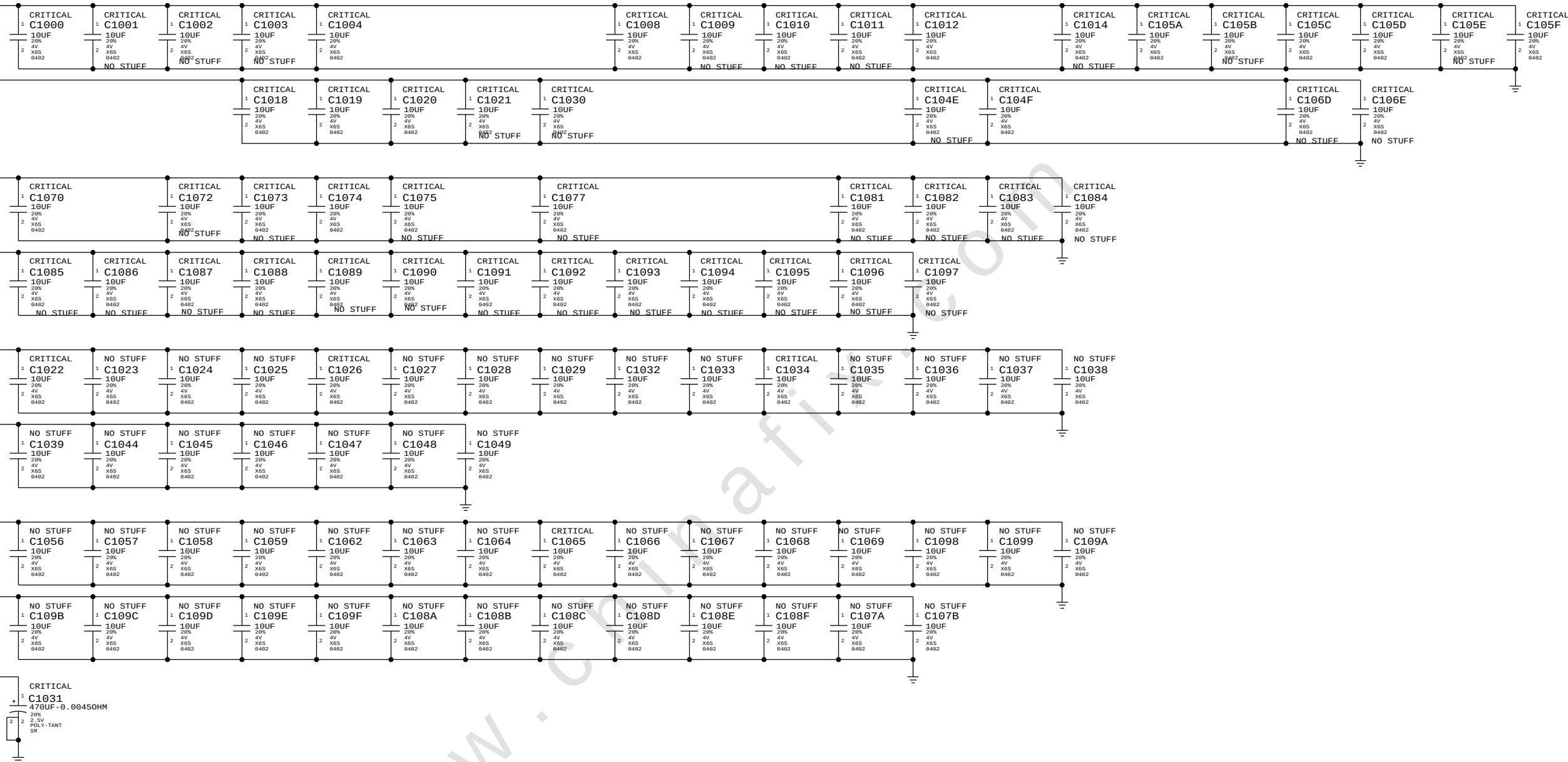


All Intel recommendations from Intel doc #603160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 0.9 unless stated otherwise

CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff
Apple implementation : 18x 22uF 0603 stuff, 80x 22uF 0603 nostuff

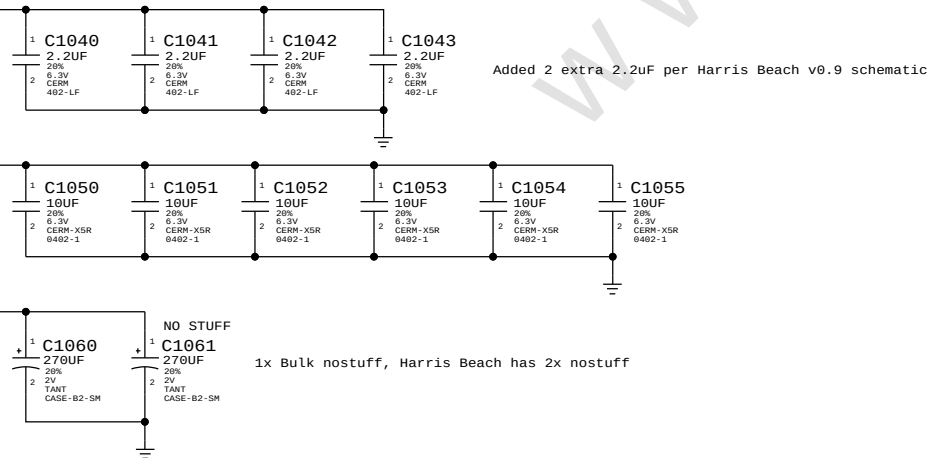
68 65 54 42 8 PPVCC_S0_CPU



CPU VDDQ DECOUPLING

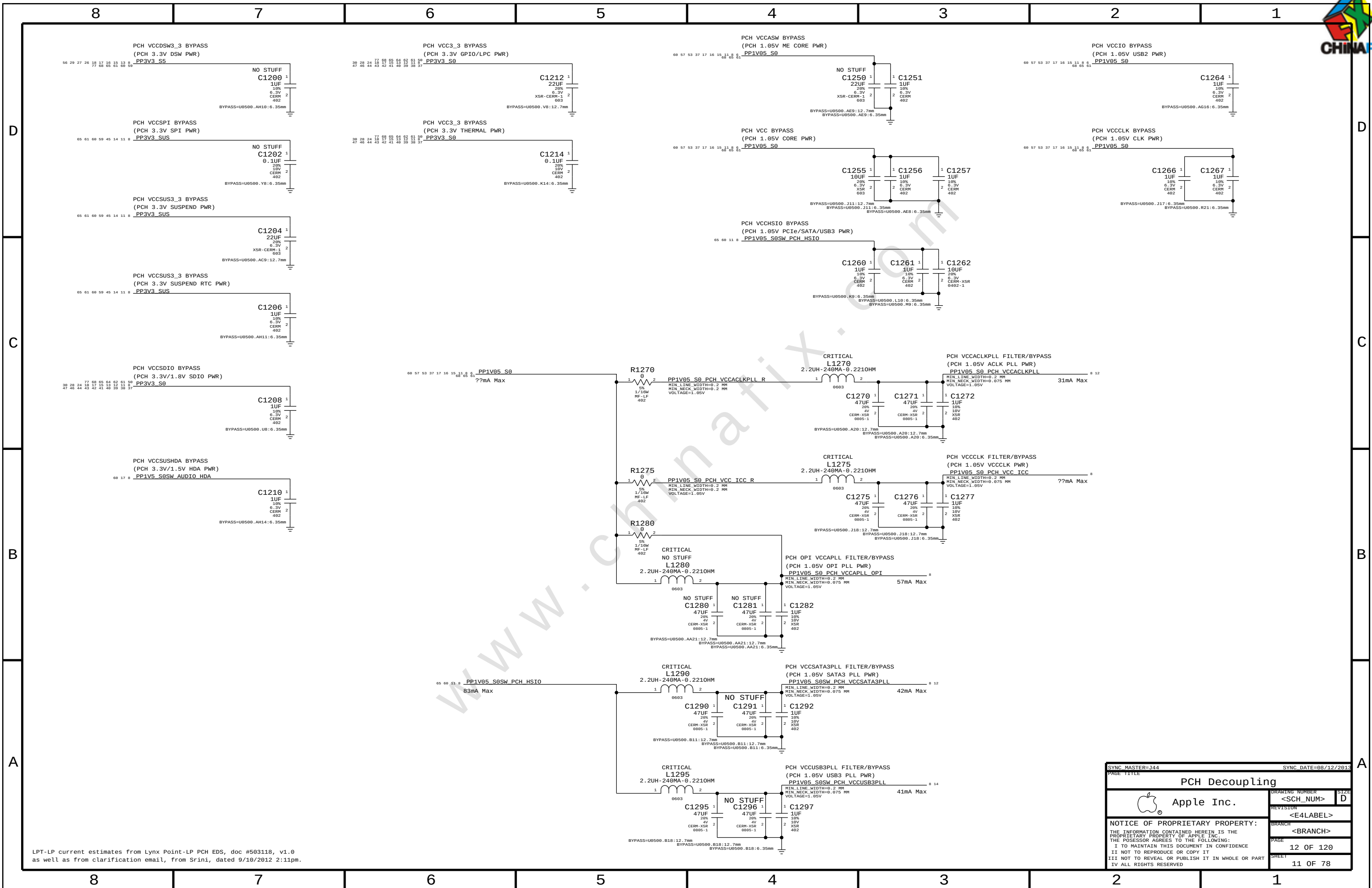
Intel recommendation (Table 5-4): 2x 2.2uF 0402, 6x 10uF 0603
Apple implementation : 2x 2.2uF 0402, 6x 10uF 0402

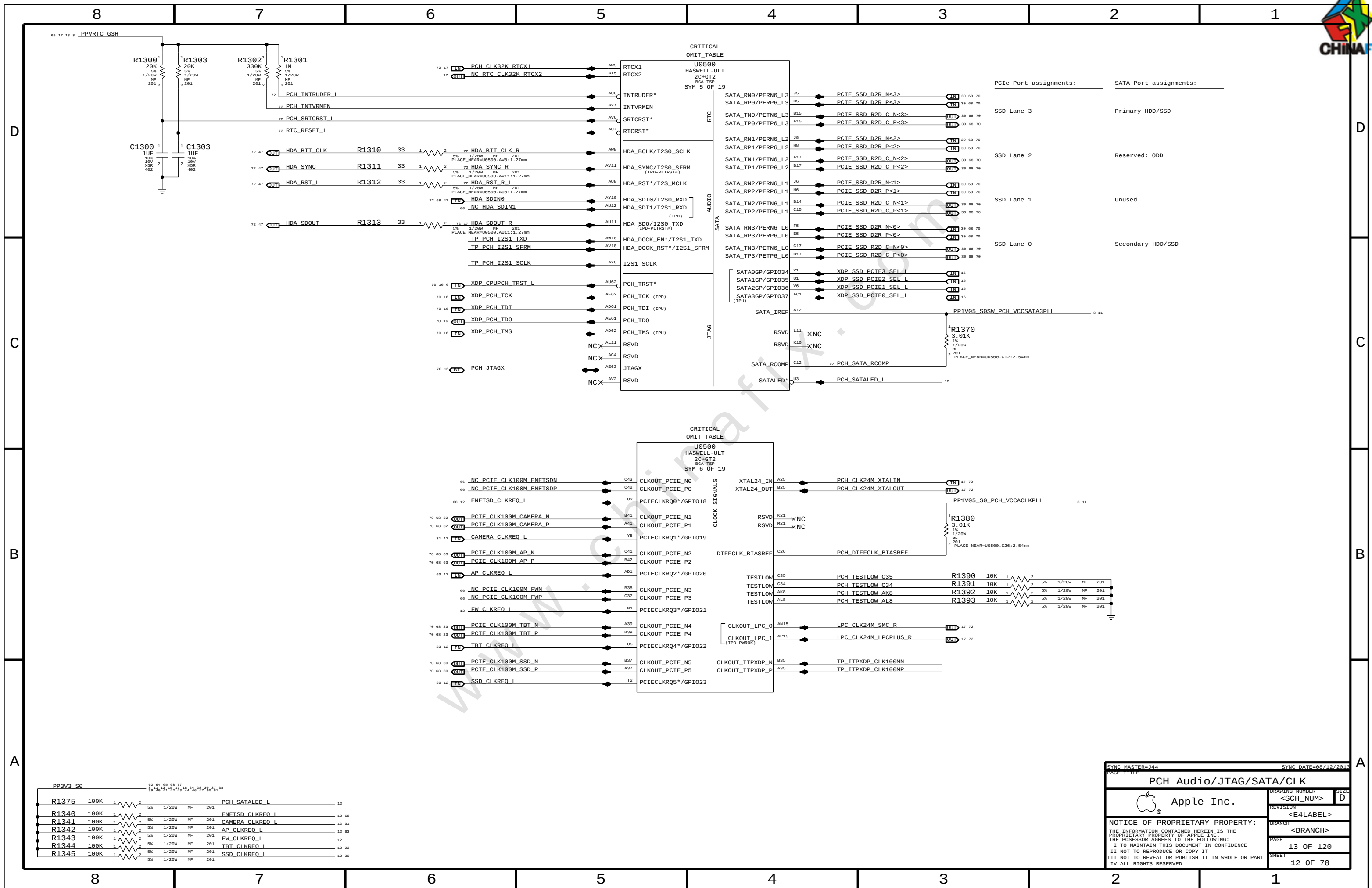
73 65 41 8 PP1V35_S3_CPUDDR

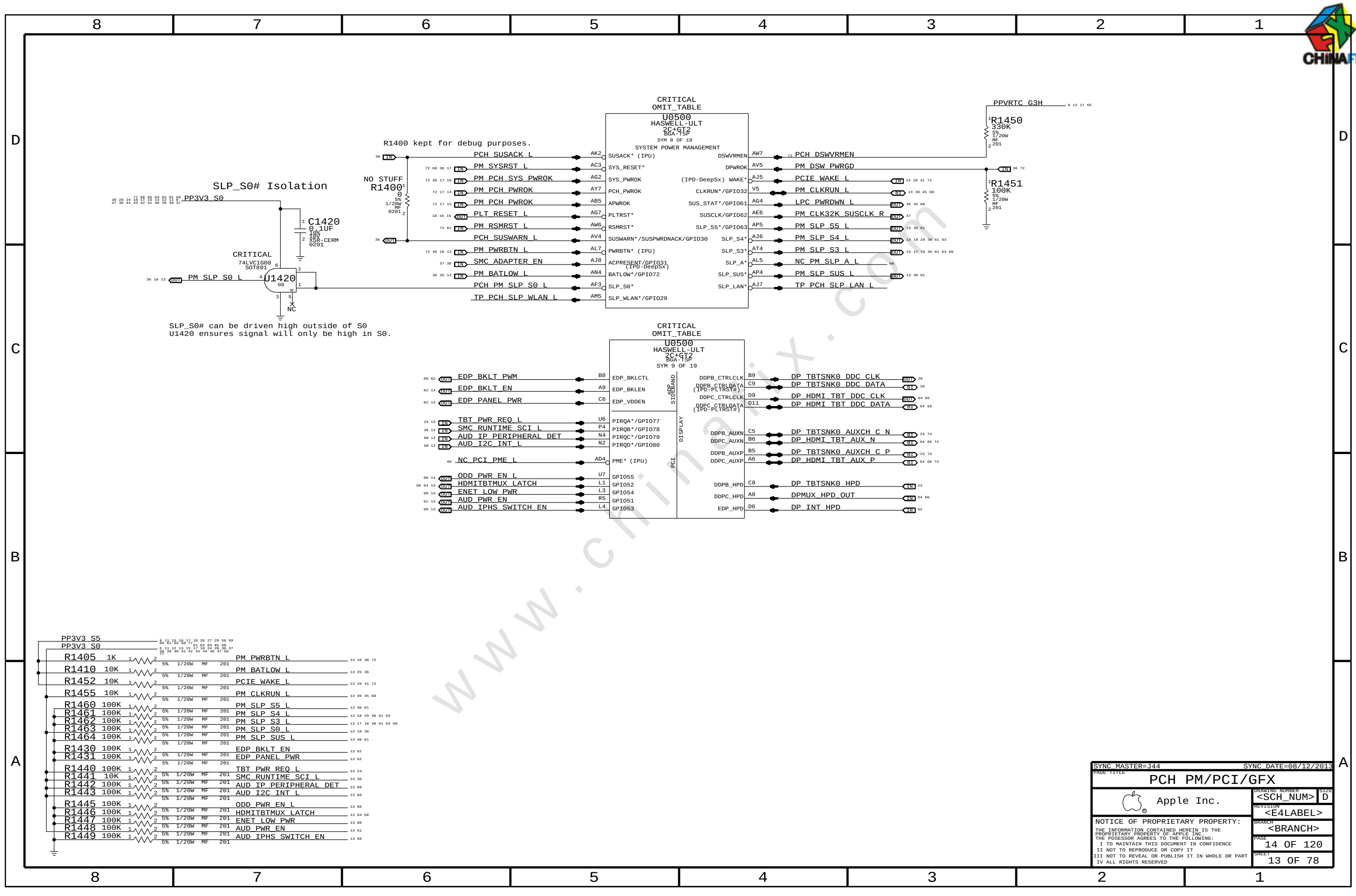


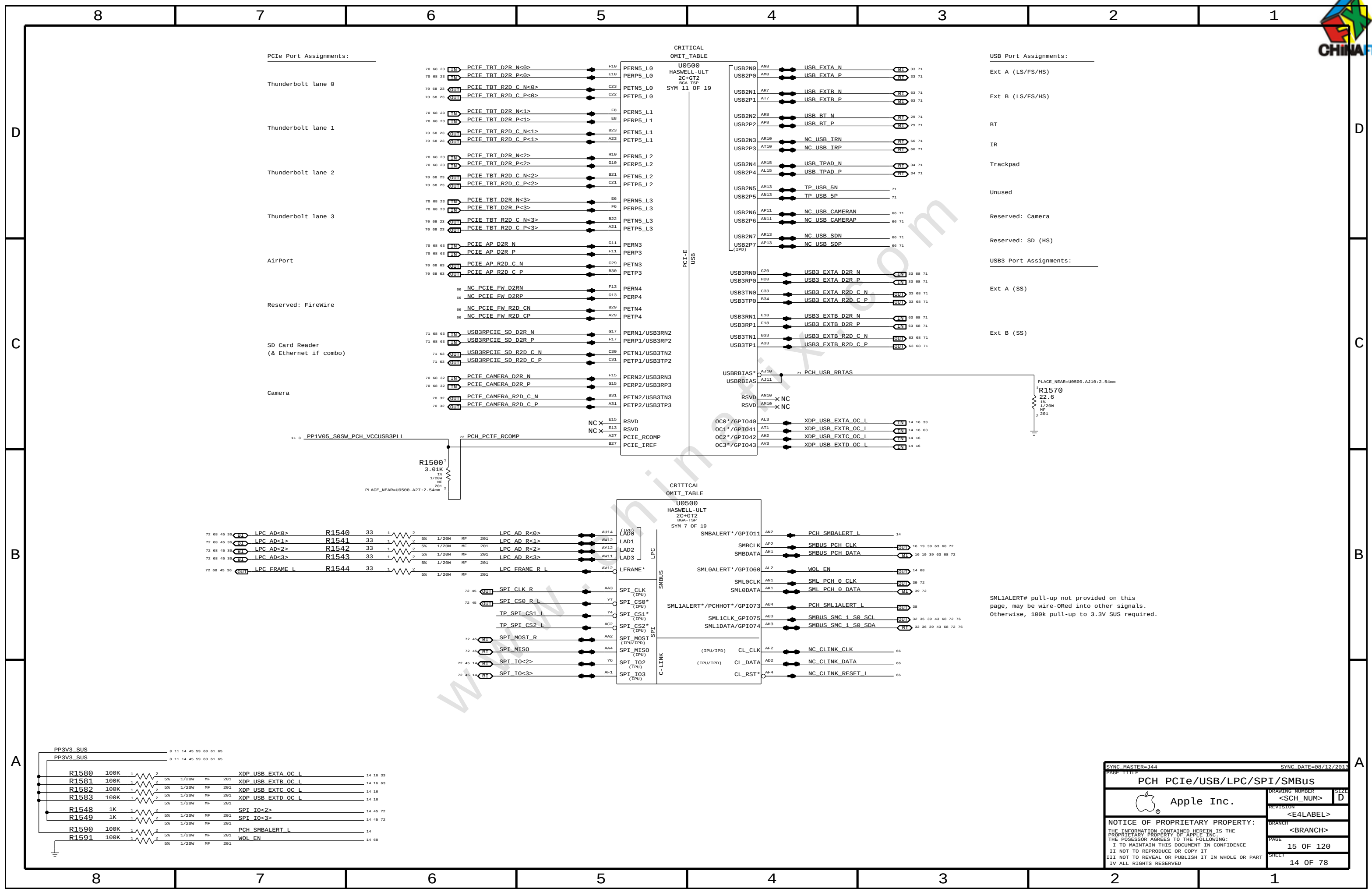
CPU VCC Decoupling

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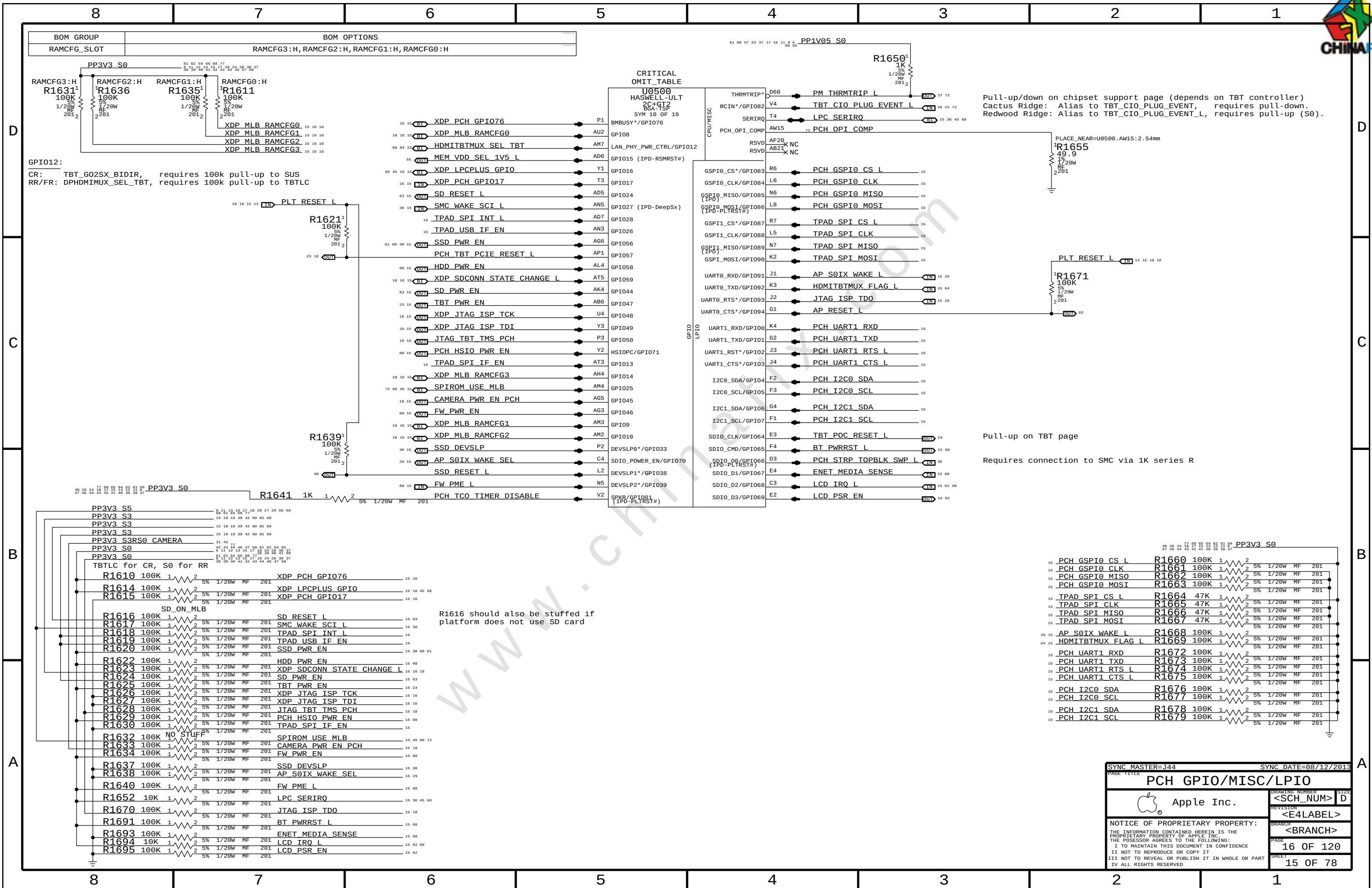


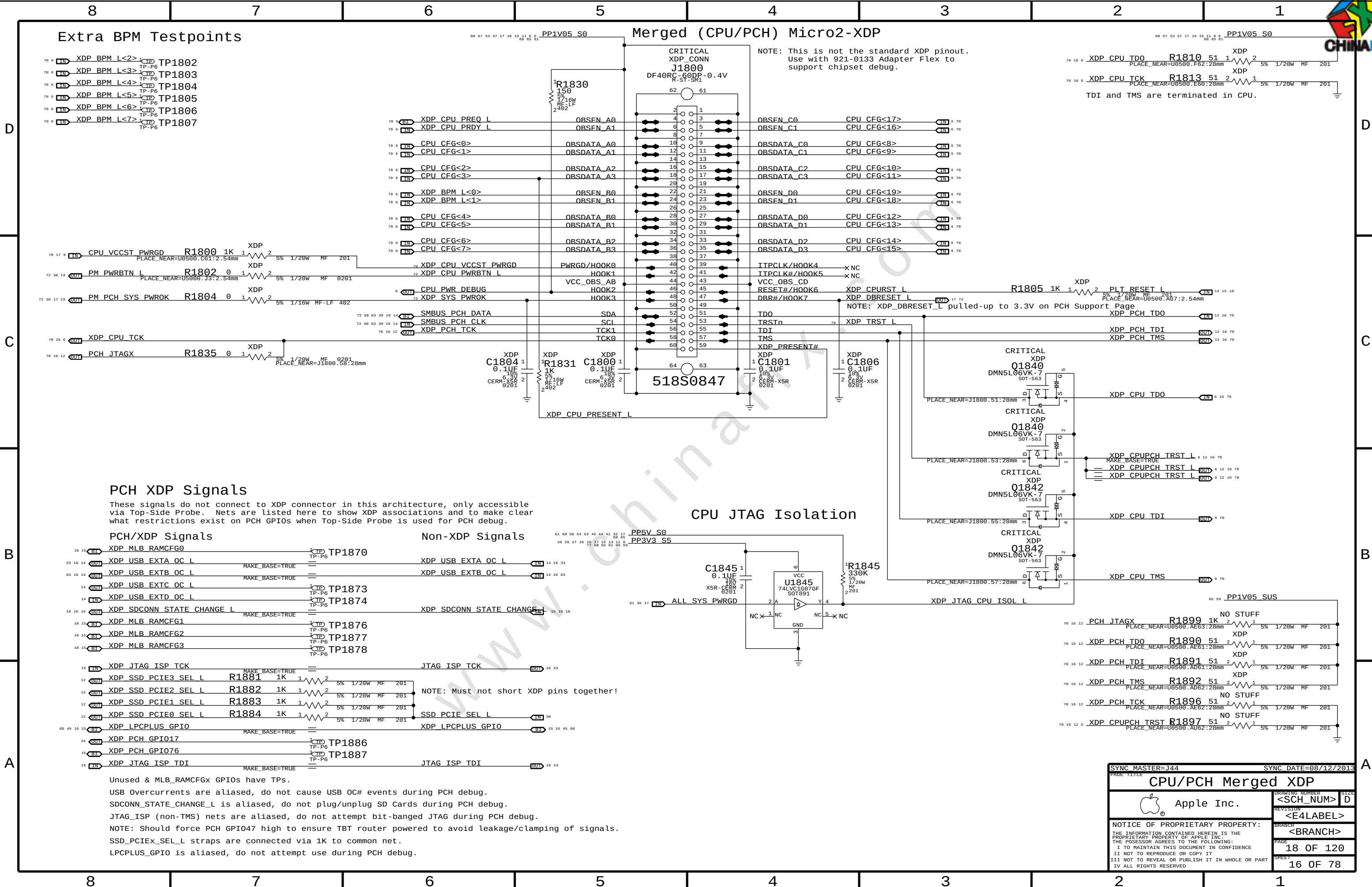






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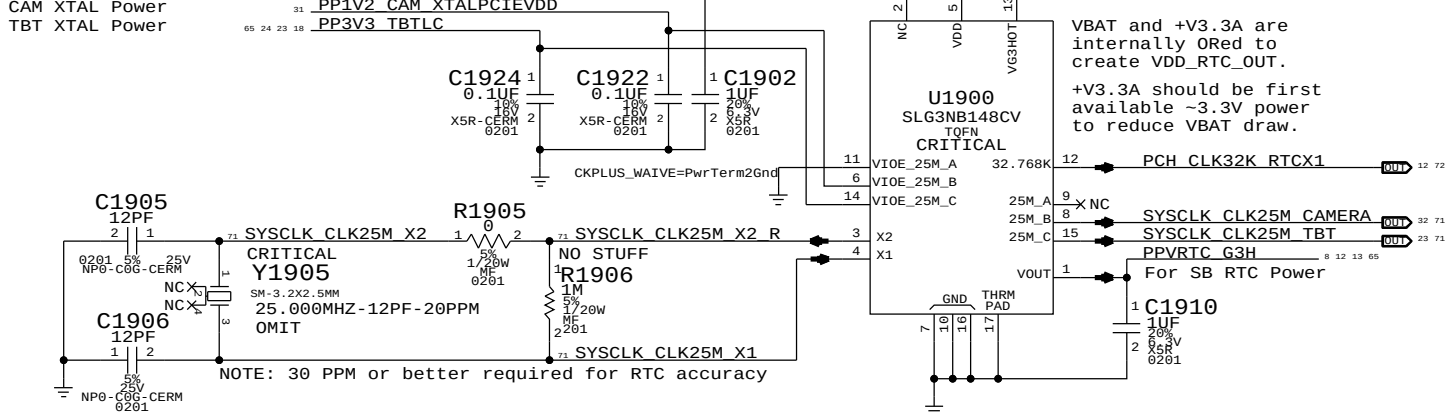
System RTC Power Source & 32kHz / 25MHz Clock Generator

Chipset uses 24MHz crystal, GreenCLK kept to save 1x 25MHz crystal & 1x 32kHz crystal

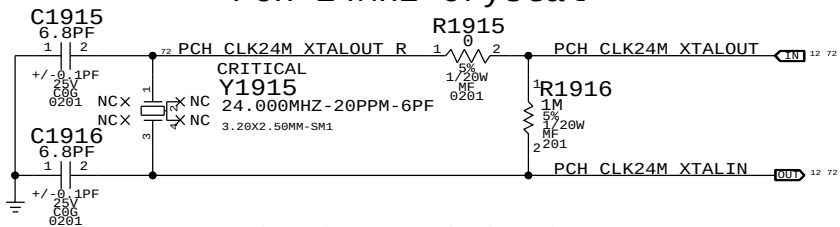
This looks a little ugly to support new and old parts. With GreenCLK Rev C pin 5 must receive S5 power (Stuff R2042)

GreenCLK 25MHz Power Must be powered if any VDDIO is powered.

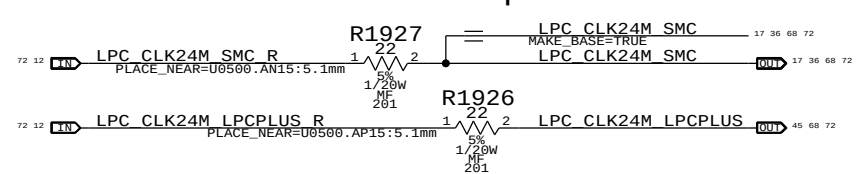
CAM XTAL Power
TBT XTAL Power



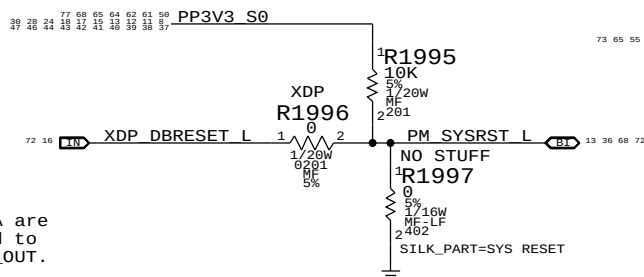
PCH 24MHz Crystal



PCH 24MHz Outputs

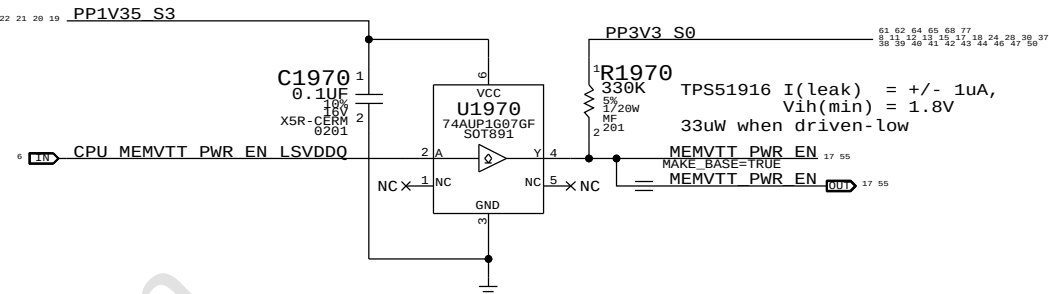


PCH Reset Button

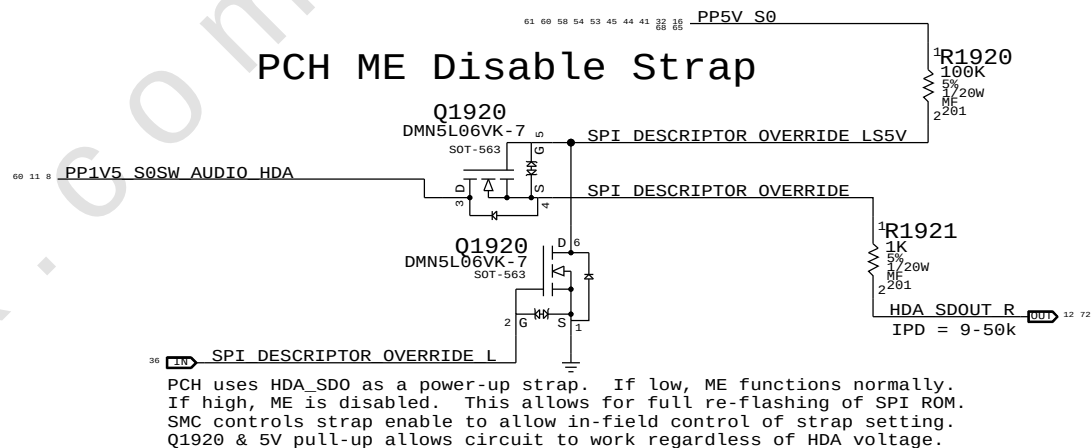


Memory VTT Enable Level-Shifter

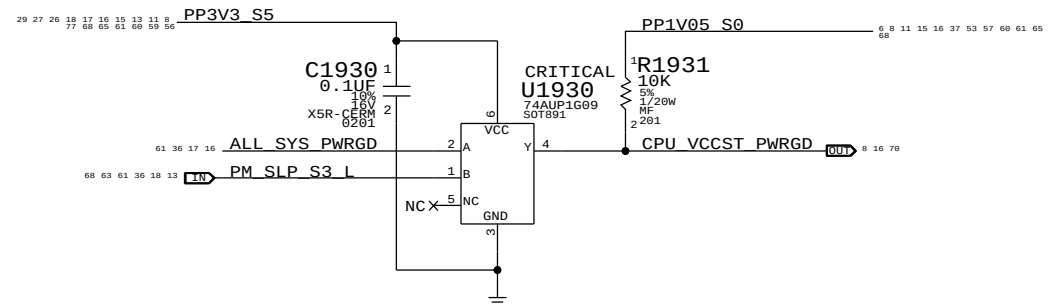
CPU output is on VDDQ rail (1.2V), TPS51916 has 1.8V Vih(min).



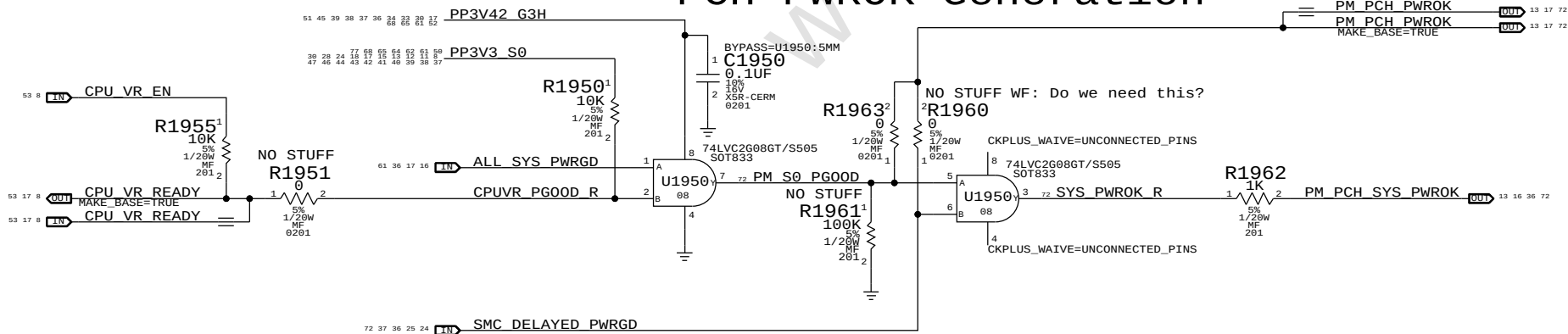
PCH ME Disable Strap



VCCST (1.05V S0) PWRGD

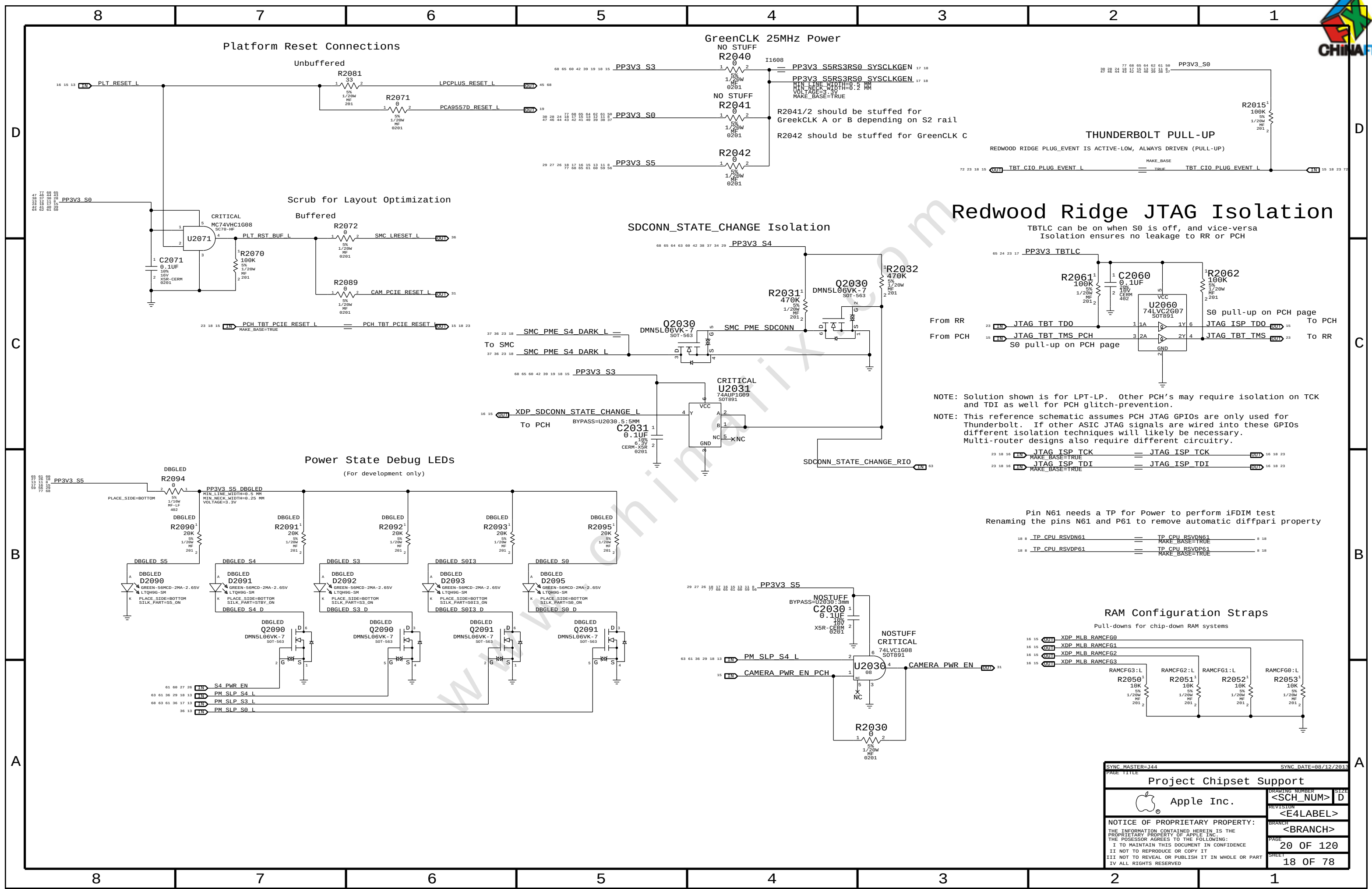


PCH PWR0K Generation



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
197S0480	1	XTAL, 25MHZ, 20PPM, 12PF, 3.2X2.5X.6MM, 85C	Y1905		

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BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

FETs for CPU isolation during DAC margining

NOTE: CPU DAC output step sizes:

DDR3	(1.5V)	7.70mV per step
DDR3L	(1.35V)	6.99mV per step
LPDDR3	(1.2V)	?..?mV per step

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure VREFMRGN_CPU_EN is low to remove short due to CPU.

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.

R2218
SHORT
1 ^ ^ ^ 2 PP3V3 S3 VREFMRGN DAC

Addr=0x98(WR)/0x99(RD)

Addr=0x30(WR)/0x31(RD)

RST* on 'platform reset' so that system watchdog will disable margining.

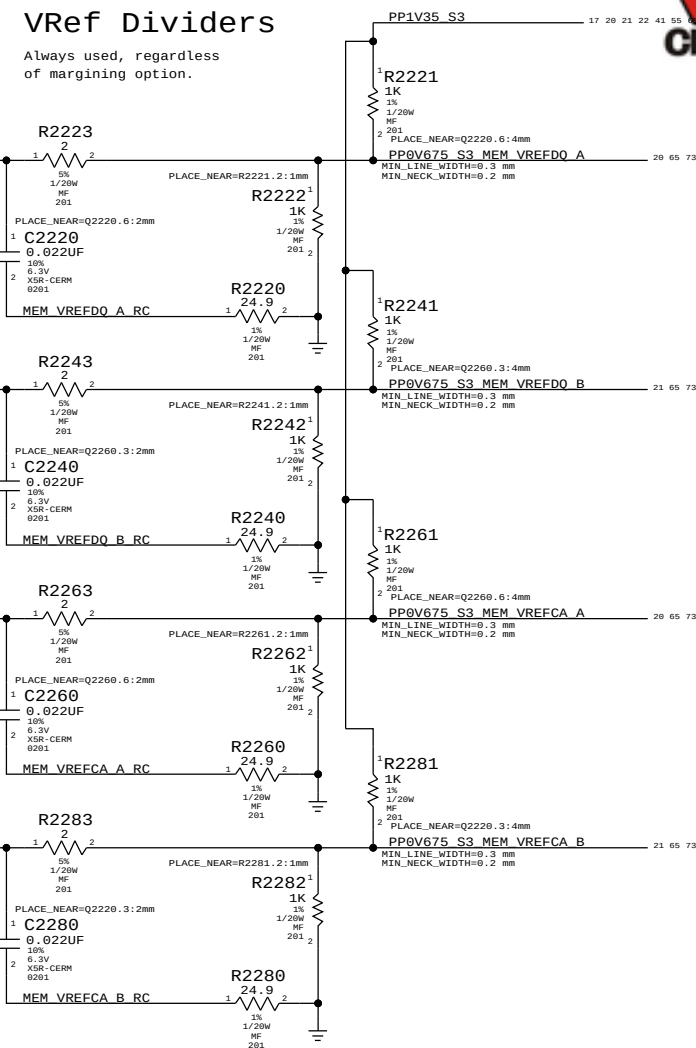
NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PCA9557D_RESET

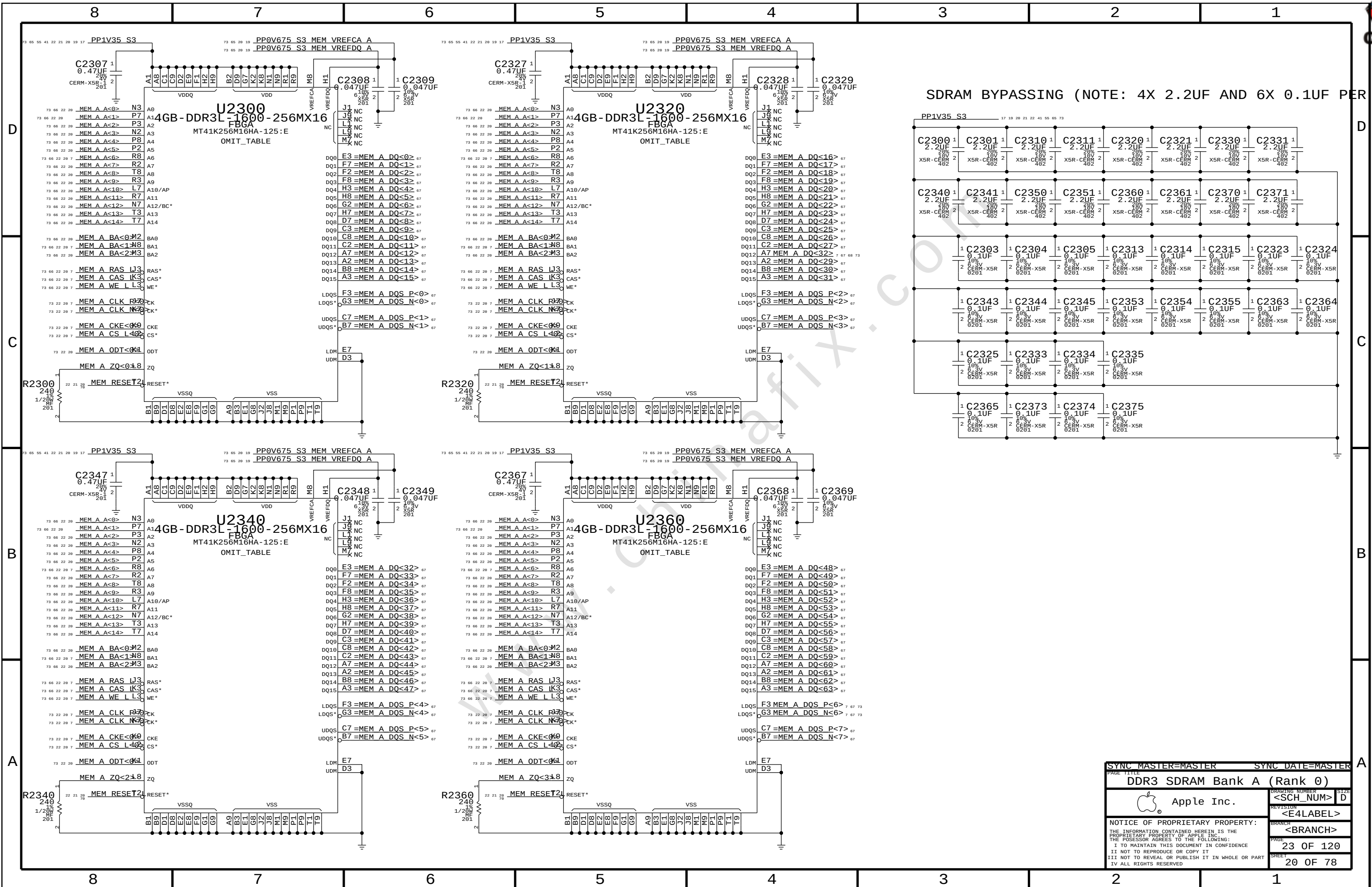
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	LPDDR3 (1.2V)		DDR3L (1.35V)		LPDDR3 (1.2V) DDR3L (1.35V)
Nominal value	0.600V (DAC: 0x2E.5)		0.675V (DAC: 0x34)		1.200V (DAC: 0x5D) 1.343V (DAC: 0x68)
Margined target:	0.300V - 0.900V (+/- 300mV)		0.337V - 1.013V (+/- 337.5mV)		0.800V - 1.600V (+/- 400mV) 0.972V - 1.714V (+/- 371mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 2.397V (0x00 - 0xBA) 0.000V - 2.694V (0x00 - 0xD1)
VREF current:	+73uA - -73uA (- = sourced)		+82uA - -82uA (- = sourced)		+21uA - -21uA (- = sourced) +25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output		6.36mV / step @ output		4.28mV / step @ output 3.53mV / step @ output

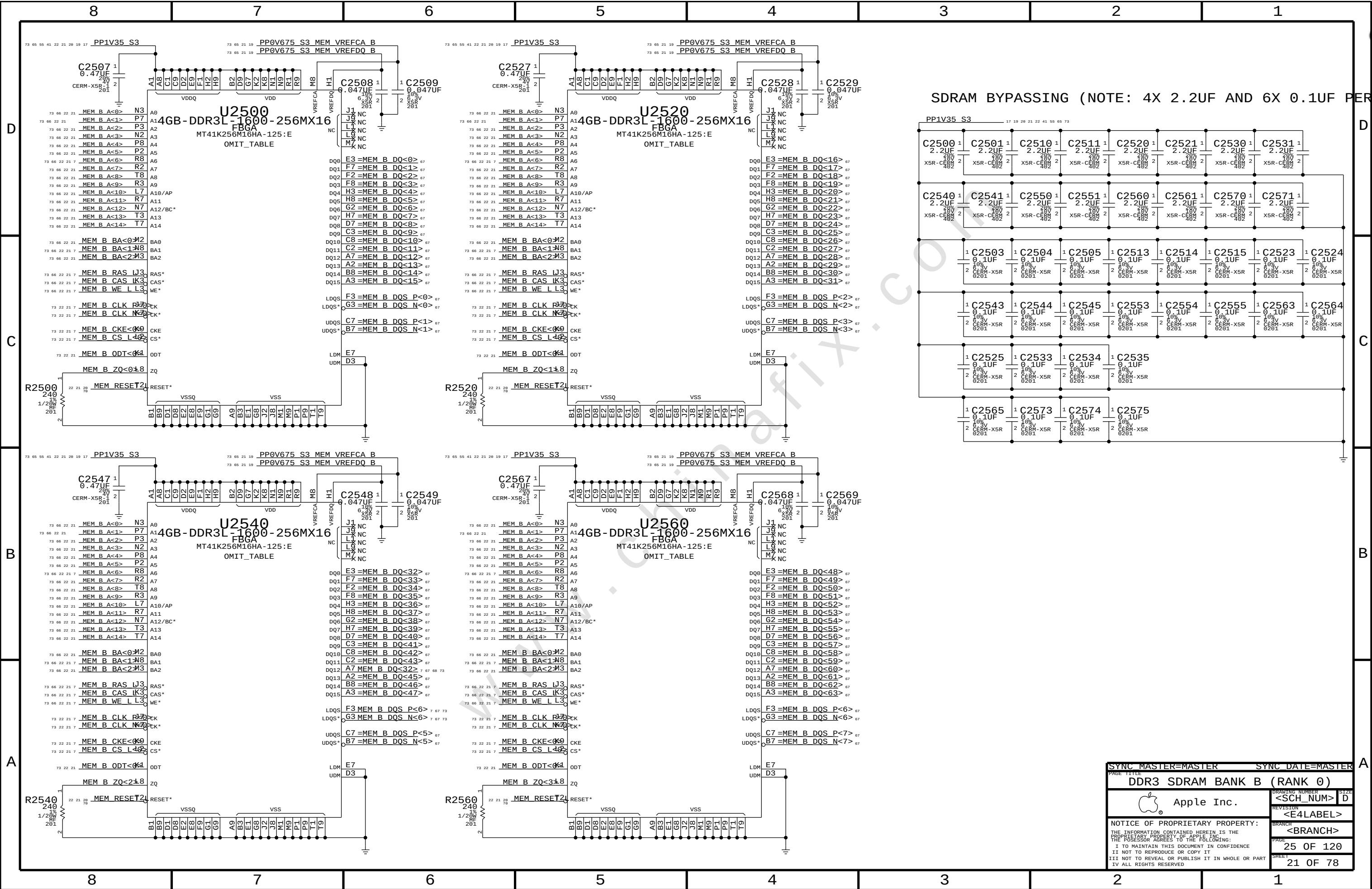
NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

Always used, regardless of margining option.



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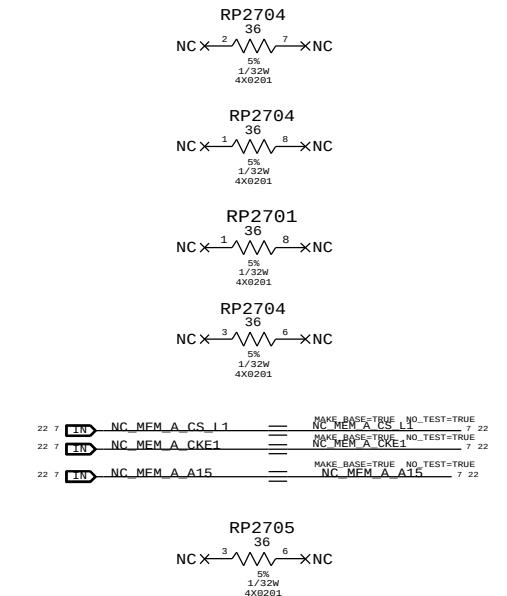
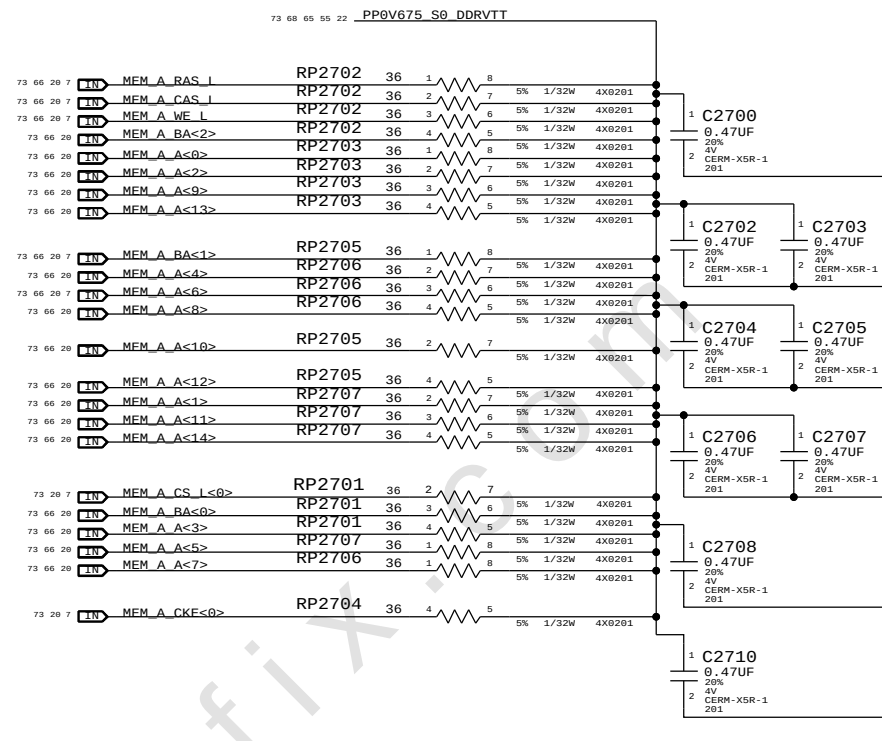
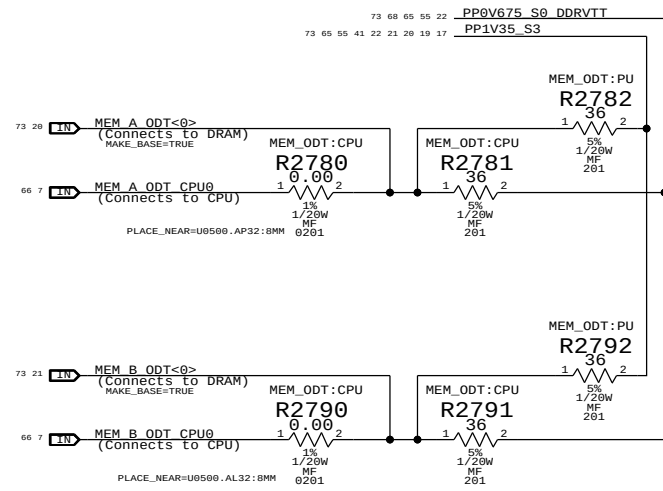
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Memory CMD/CTL Termination - Channel A

MEMORY RPACK SPARES

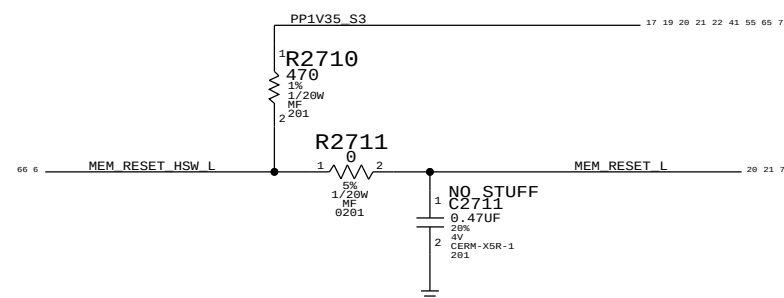
Memory ODT Option

MEM_ODT:CPU drives ODT from CPU, terminated to 0.675V VTT.
MEM_ODT:PU disconnect ODT from CPU, ODT pins on DRAM pulled up to 1.35V VDDQ.

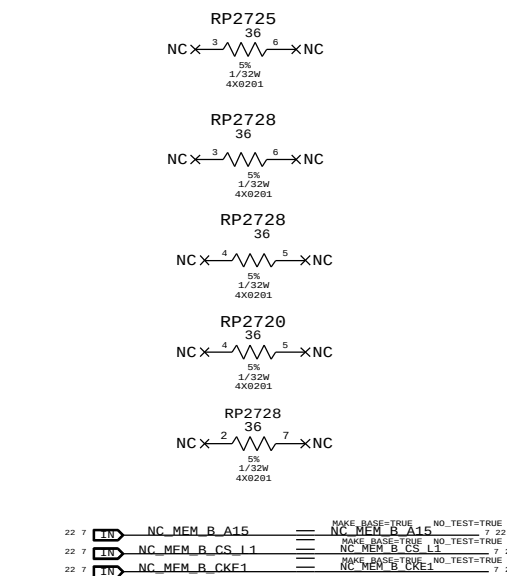
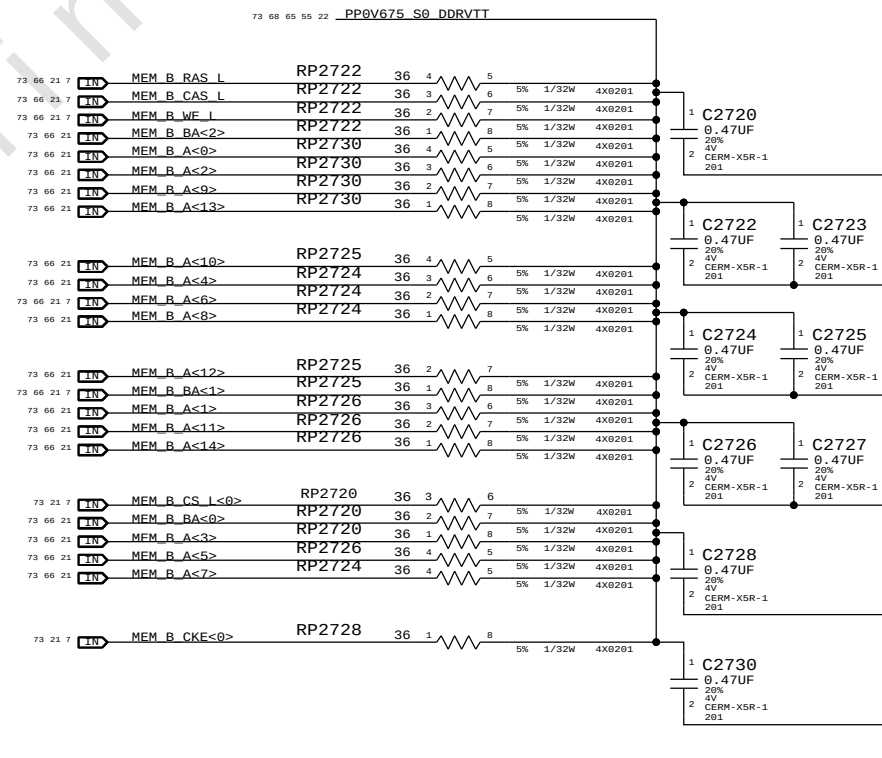


Memory Reset Pull Up

Reset is an open drain in Haswell ULT and needs pull up

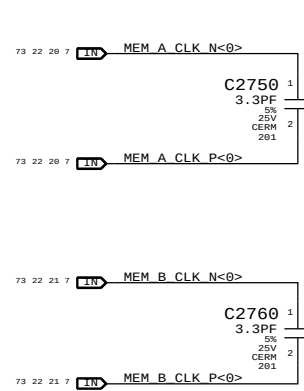


Memory CMD/CTL Termination - Channel B



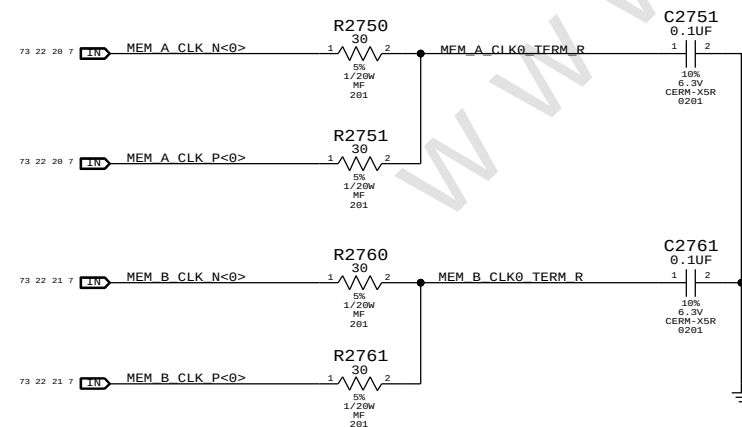
Memory Clock Near-End Termination

Place Source C termination before first DRAM

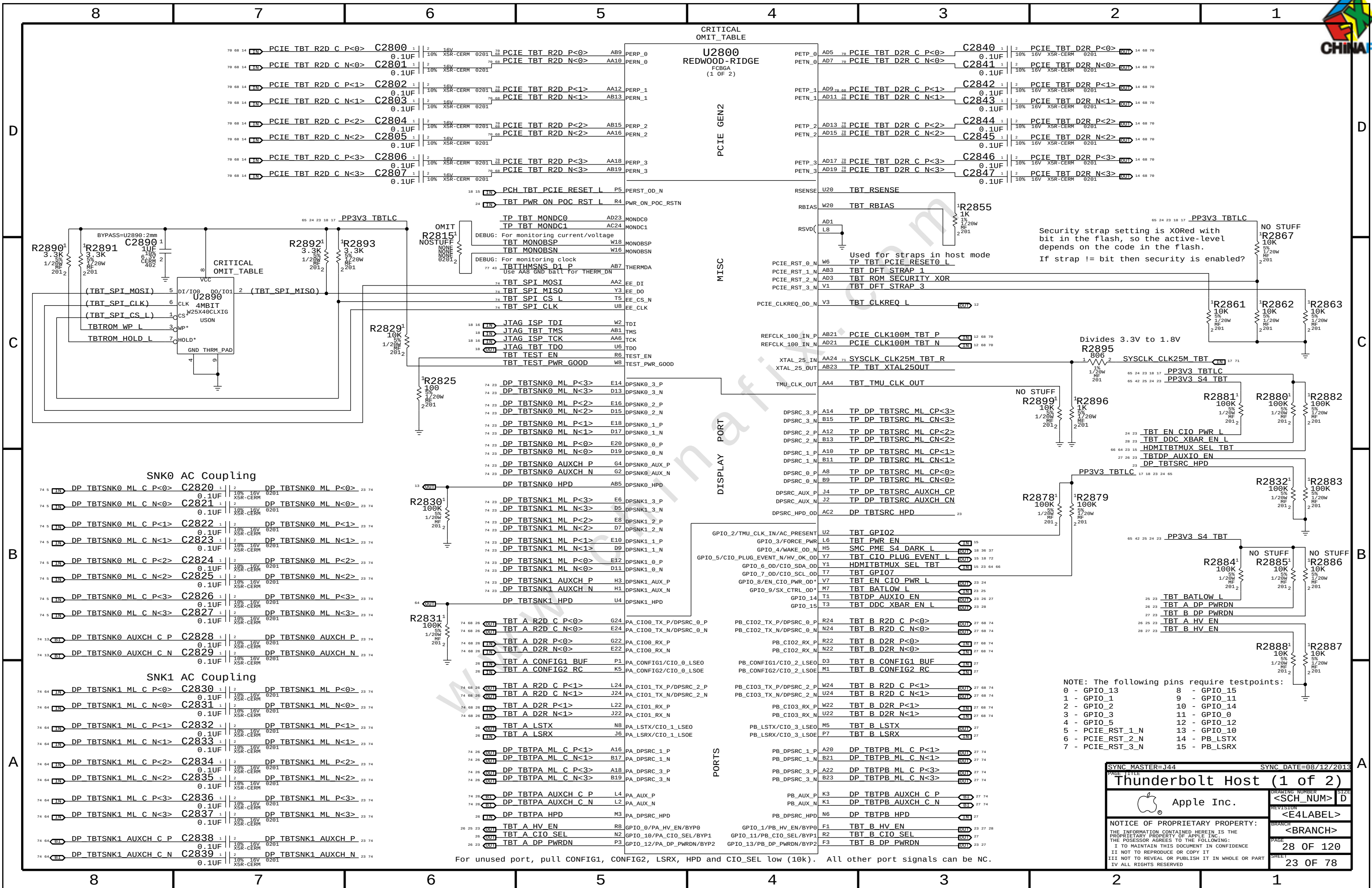


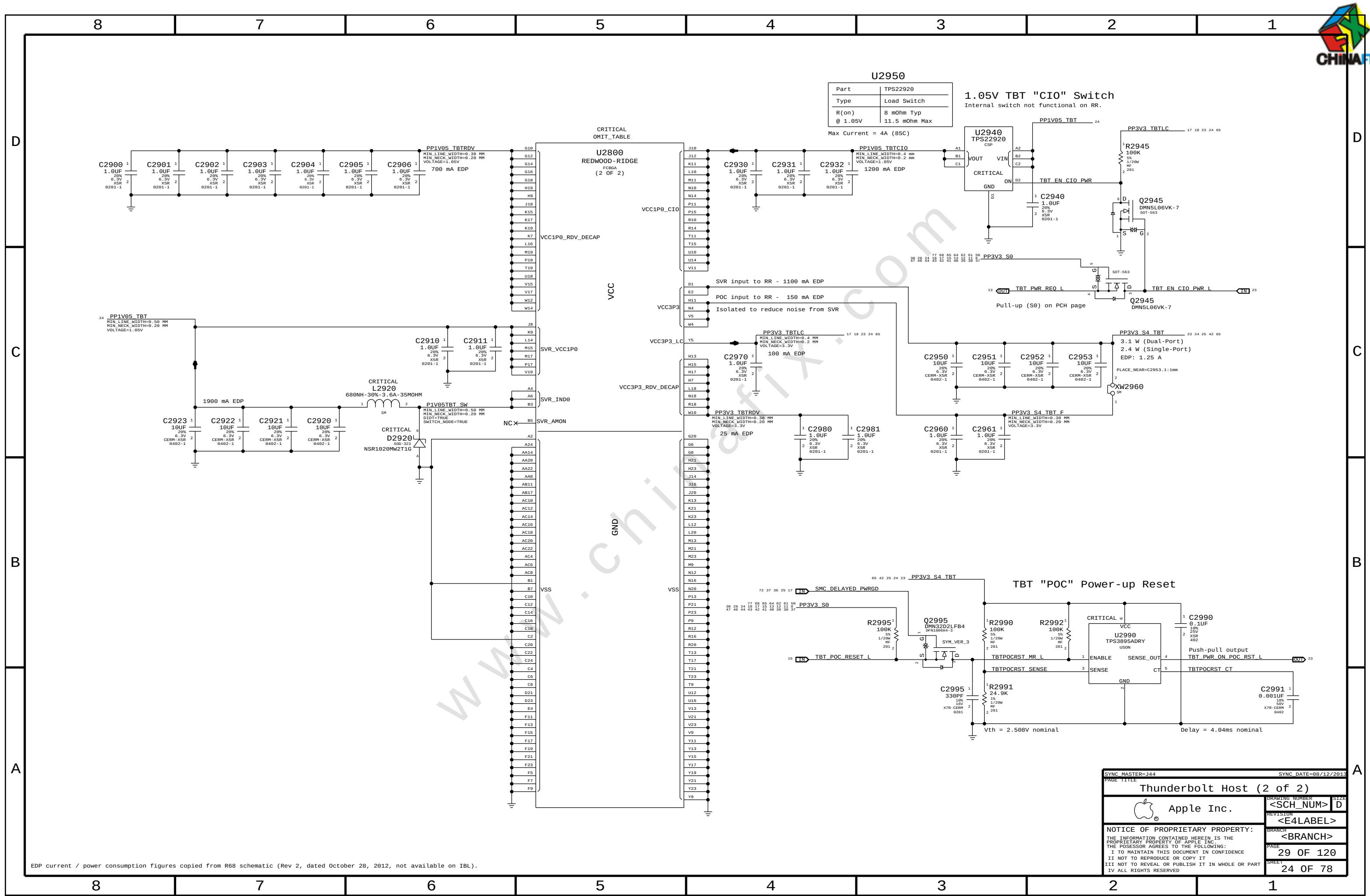
Memory Clock Far-End Termination

Place RC end termination after last DRAM



SYNC MASTER=J44 YONAS-4GB		SYNC DATE=04/02/2013	
PAGE TITLE			
DDR3 Termination			
Apple Inc.		DRAWING NUMBER	
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EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE			
Thunderbolt Host (2 of 2)			
 Apple Inc.	DRAWING NUMBER	SIZE	
	<SCH_NUM>	D	
	REVISION	<E4LABEL>	
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		SHEET	24 OF 78

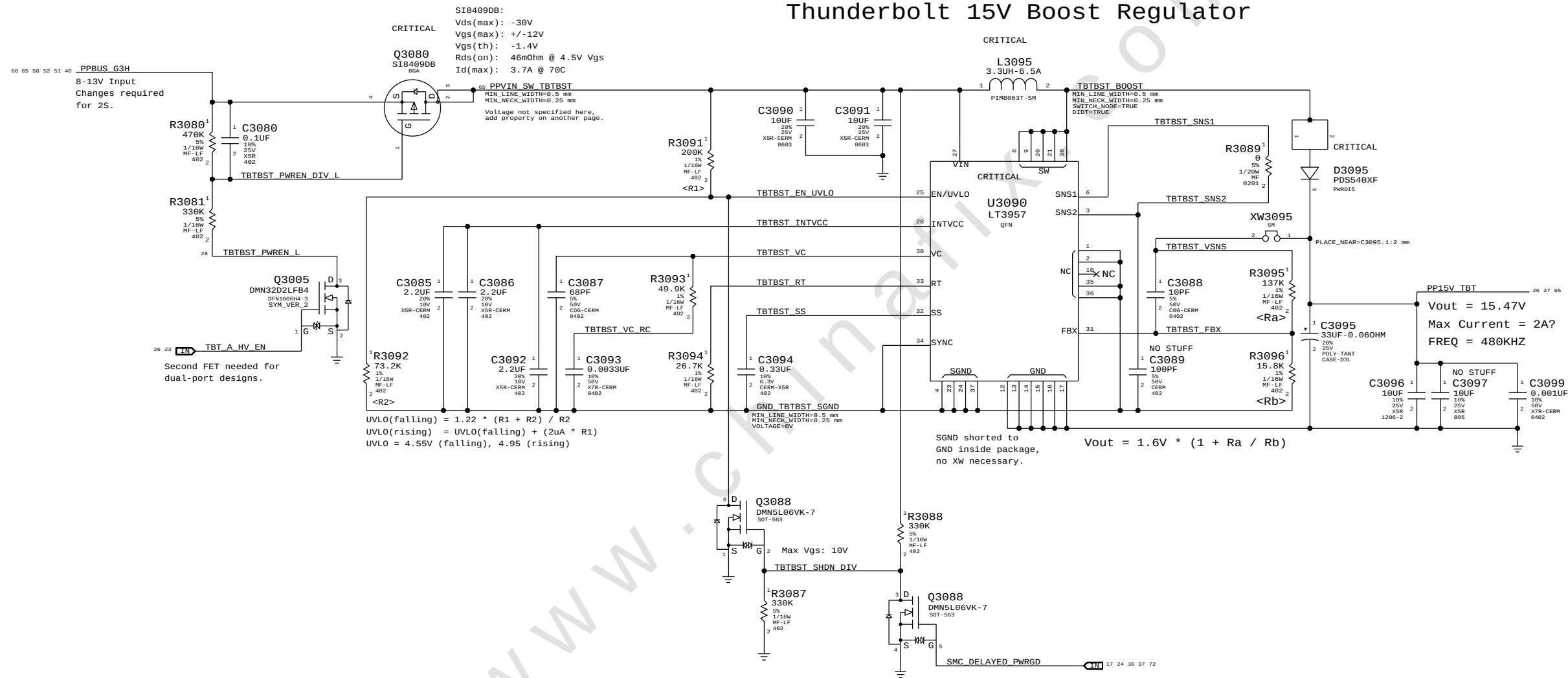
Page Notes

Power aliases required by this page:
 - =PPVIN_SW_TBTBST (8-13V Boost Input)
 - =PP15V_TBT_REG (15V Boost Output)

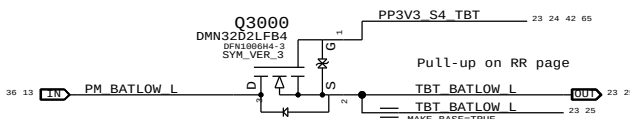
Signal aliases required by this page:
 (NONE)

BOM options provided by this page:
 (NONE)

Thunderbolt 15V Boost Regulator



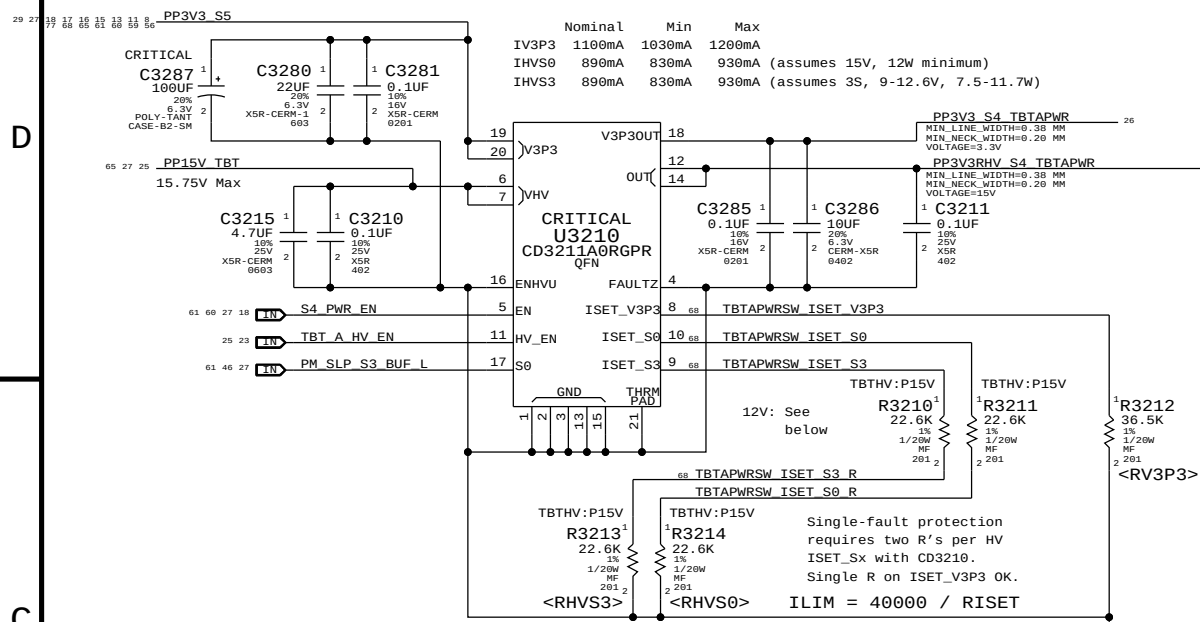
BATLOW# Isolation



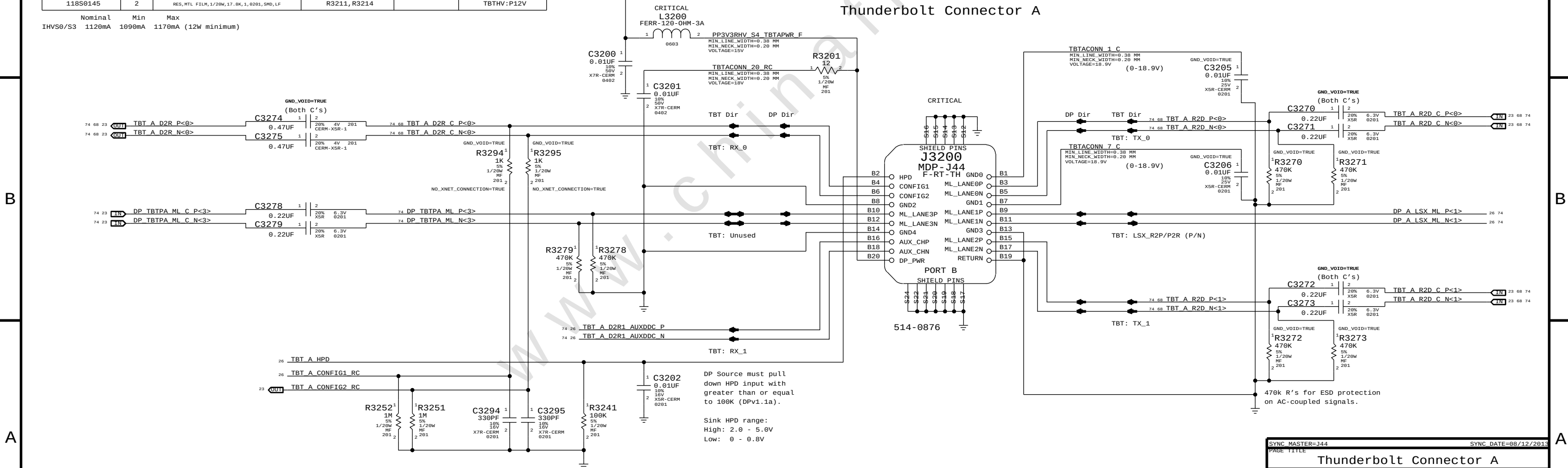
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Apple Inc.		DRAWING NUMBER	SIZE
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3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.



Thunderbolt Connector A

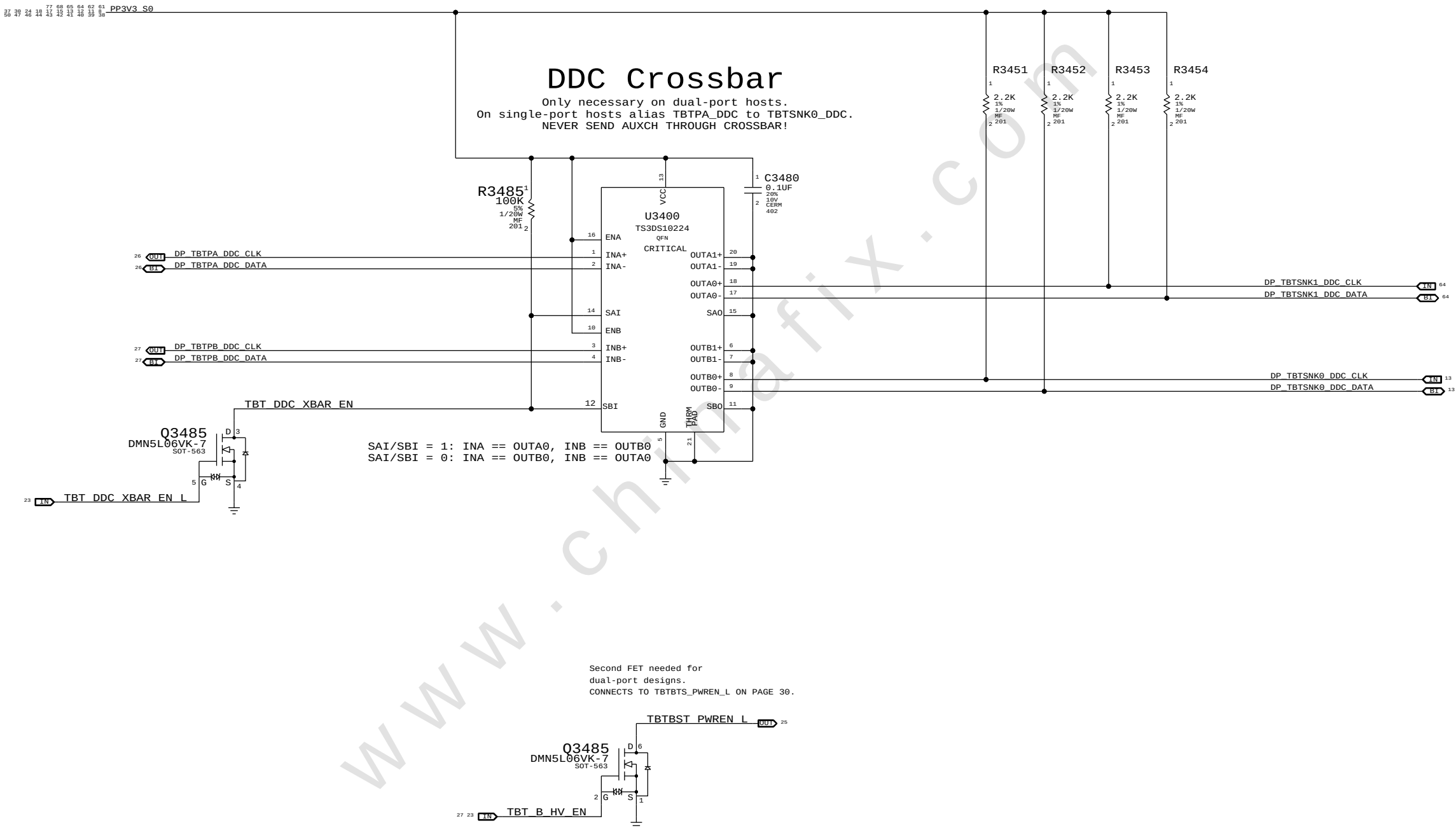




DDC Pull-Ups

2.2k pull-ups are required by PCH
to indicate active display interface.
DP++ spec violation, should remove!

NOTE: Only DDC_DATA is sensed, so DDC_CLK
pull-ups are unstuffed.



SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE			
DDC Crossbar			
 Apple Inc.		DRAWING NUMBER	SIZE
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		PAGE	34 OF 120
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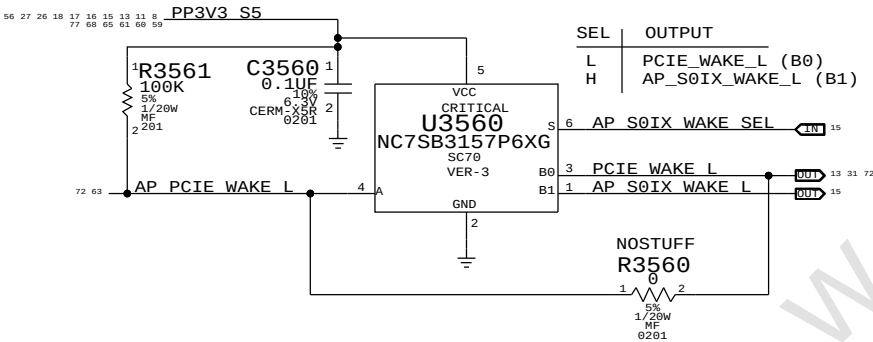
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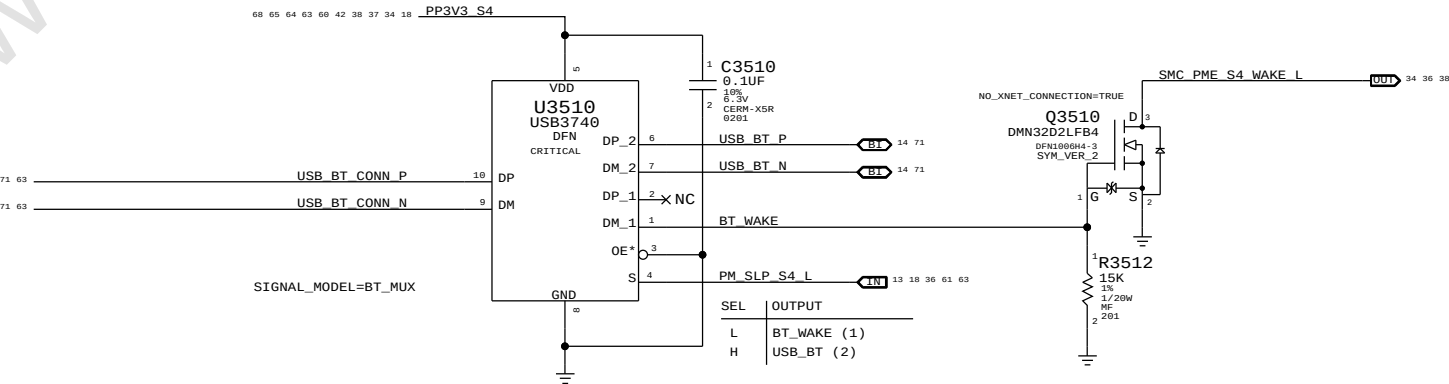
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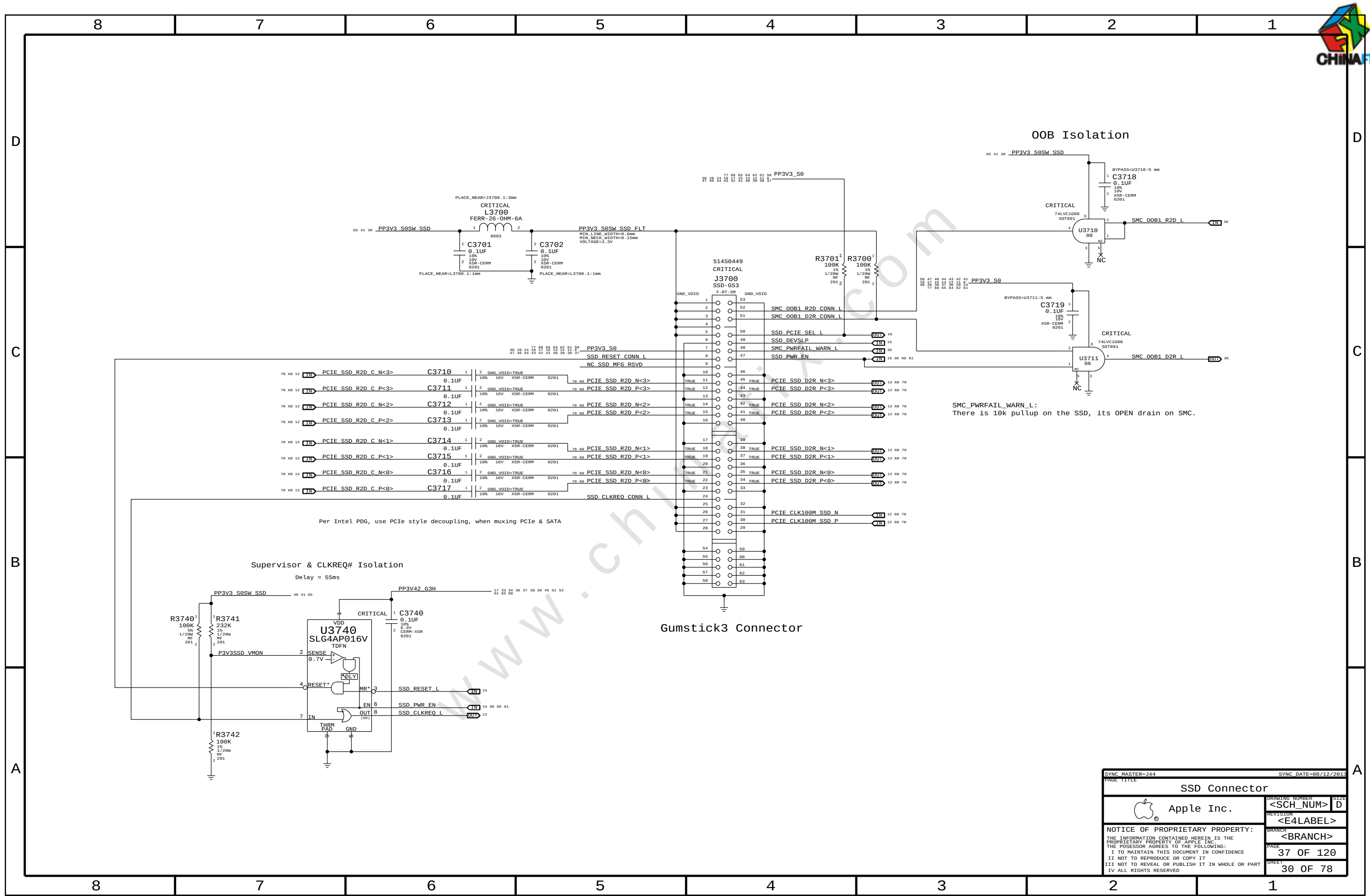
PCIe Wake Muxing



BLUETOOTH

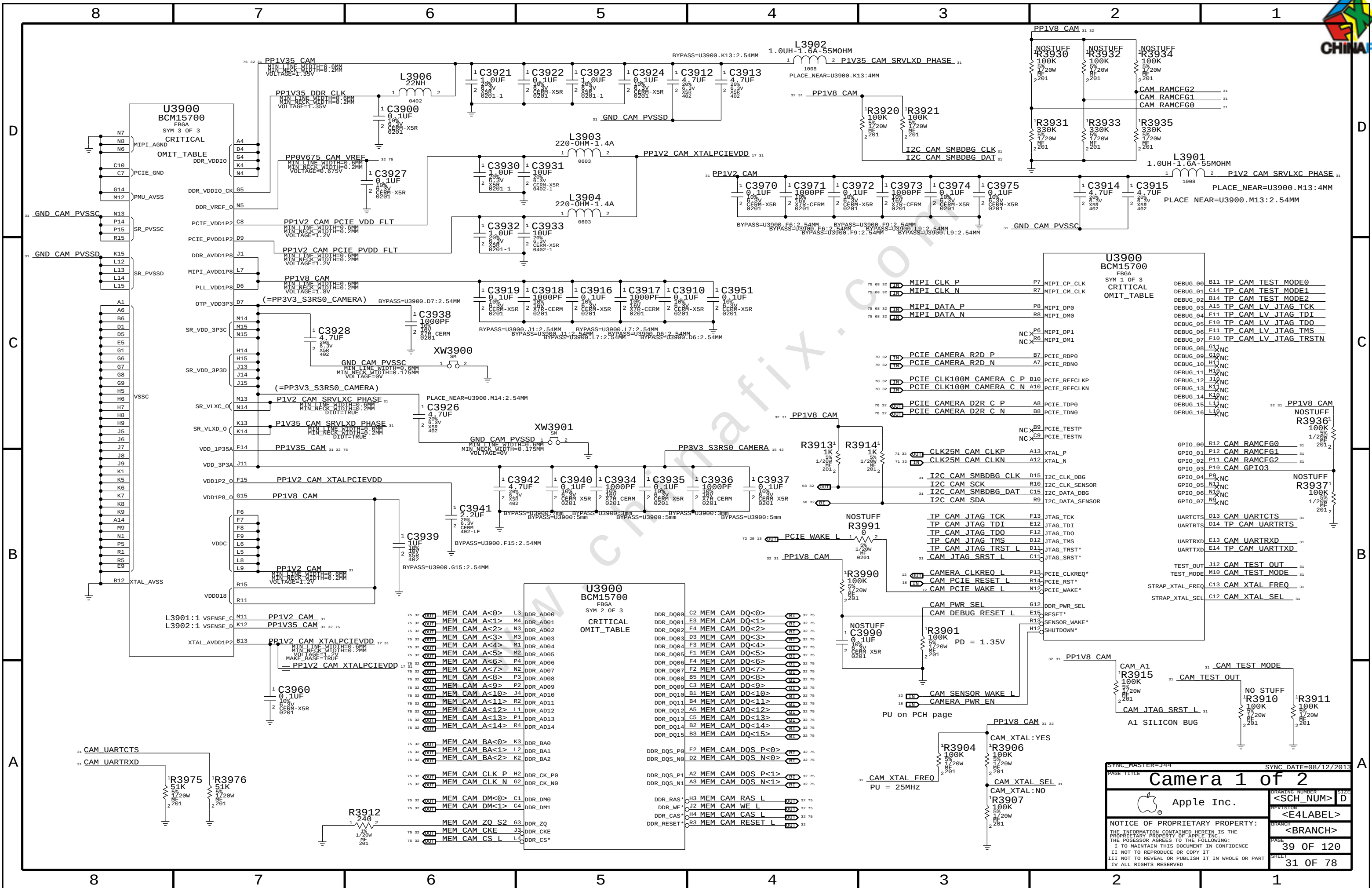


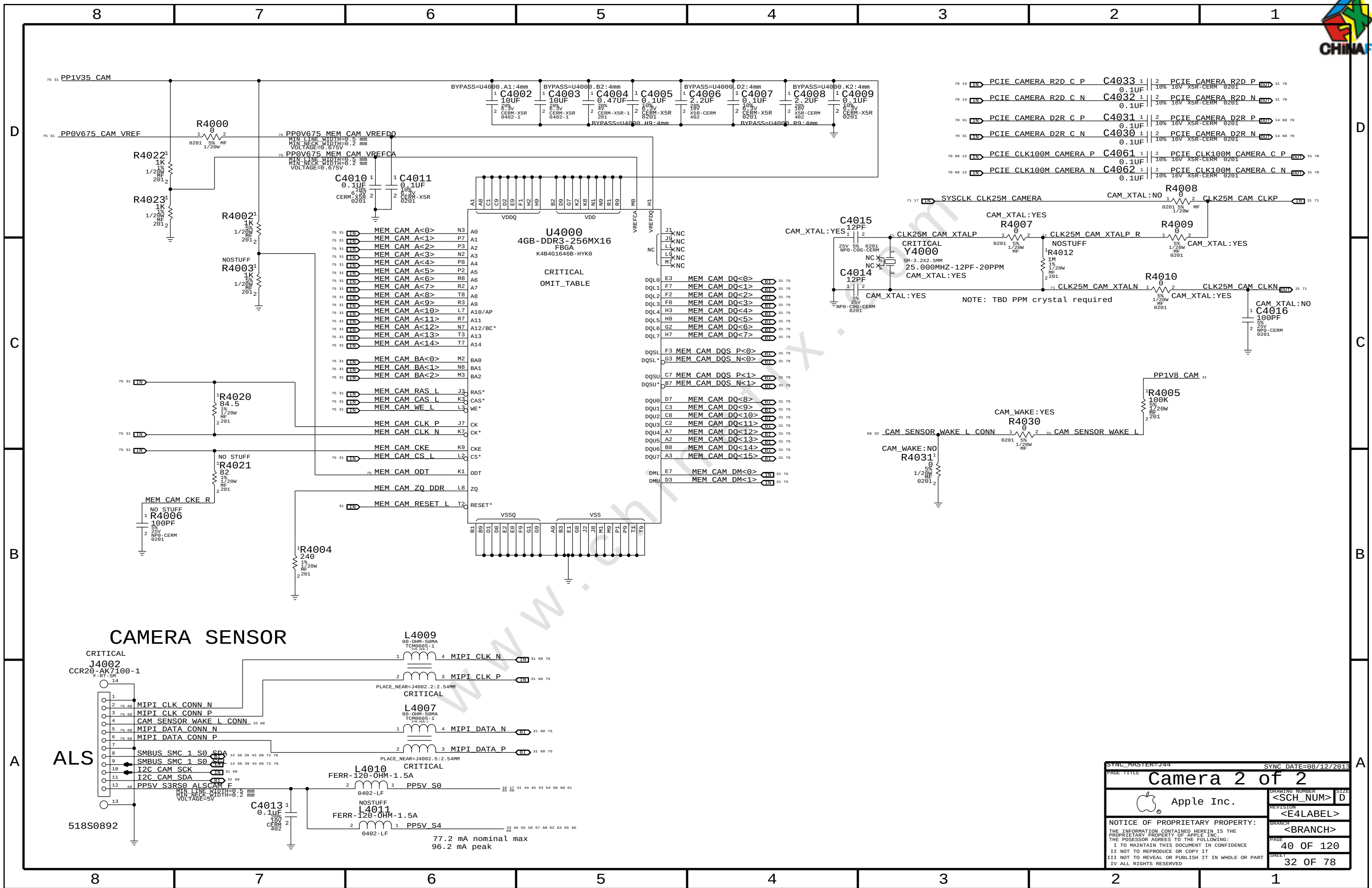
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SMC_PWRFAIL_WARN_L:
There is 10k pullup on the SSD, its OPEN drain on SMC.

SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE		SSD Connector	
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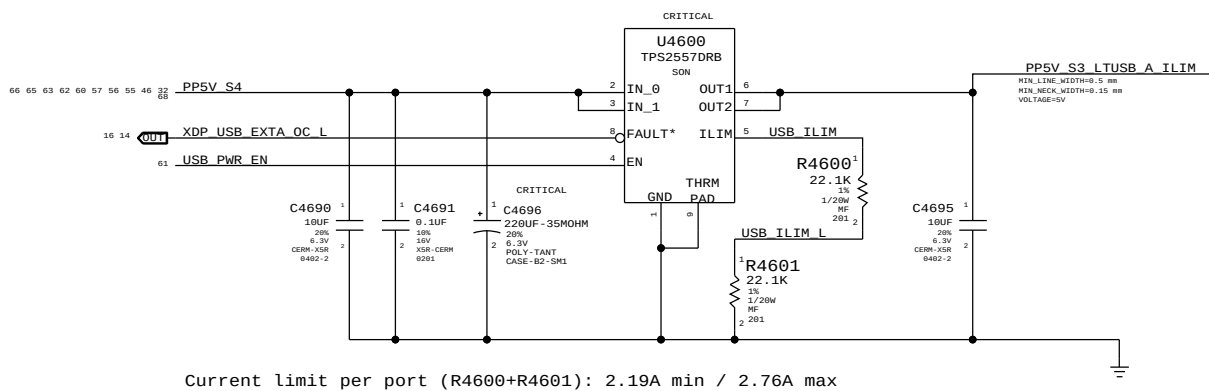






RIGHT USB PORT A

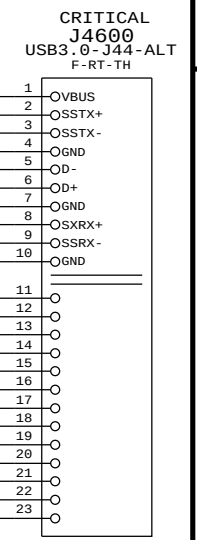
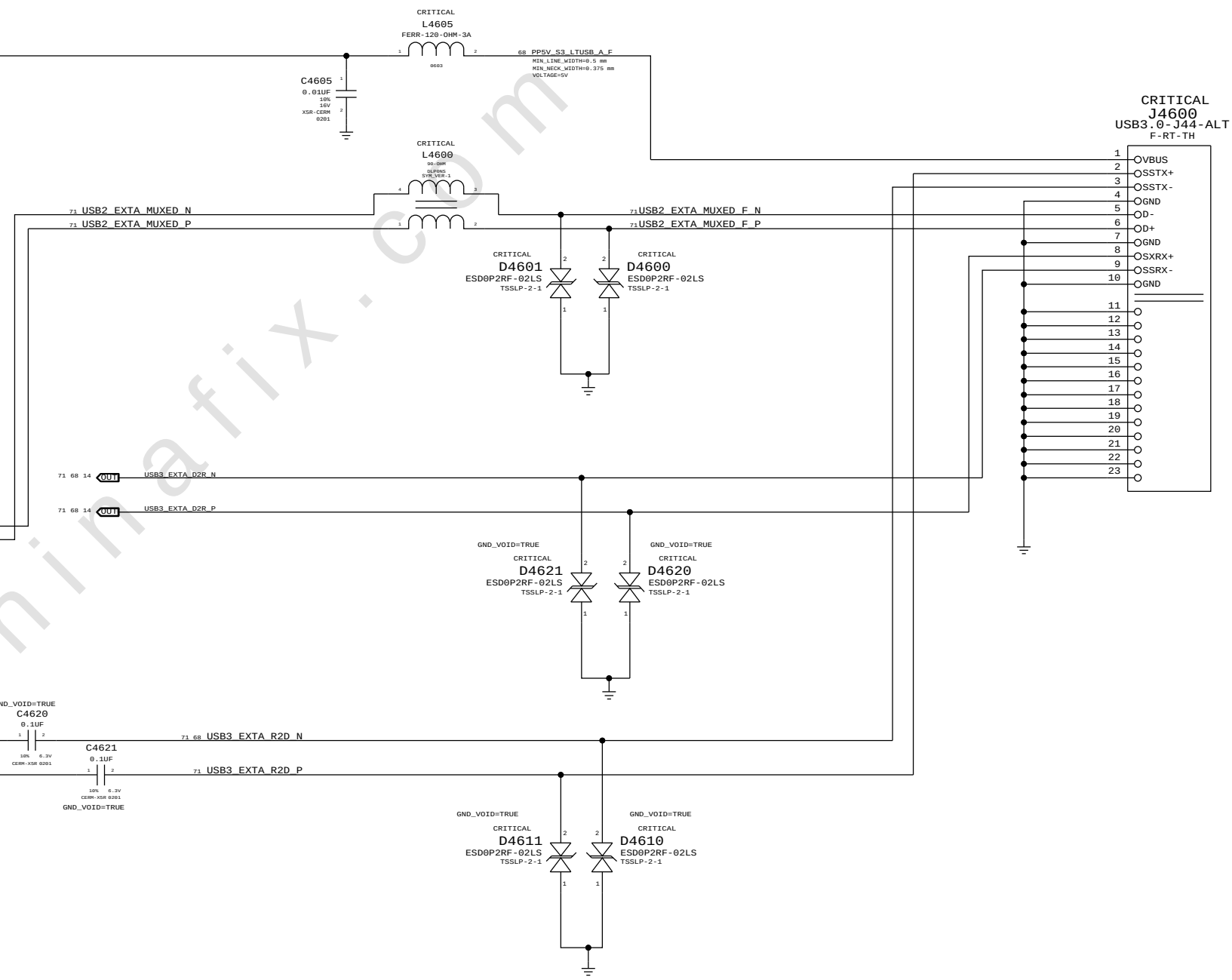
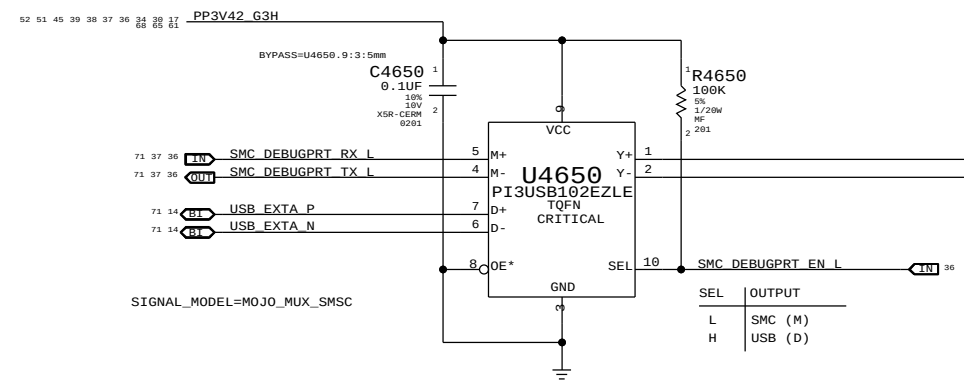
USB Port Power Switch



Current limit per port (R4600+R4601): 2.19A min / 2.76A max

Mojo SMC Debug Mux

THE PI3USB102E CAN CLAMP VOLTAGE IN THE INTERNAL USB PINS



SYNC MASTER=J44

SYNC DATE=08/12/2013

PAGE TITLE

External A USB3 Connector



Apple Inc.

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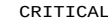
CRITICAL



No IPD on OE input pin PP3V3_S4 (symbol error).
PP3V42 G3H



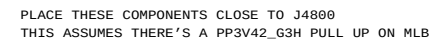
C



518S0752
CRITICAL
J4813



Spare MOSFET symbol

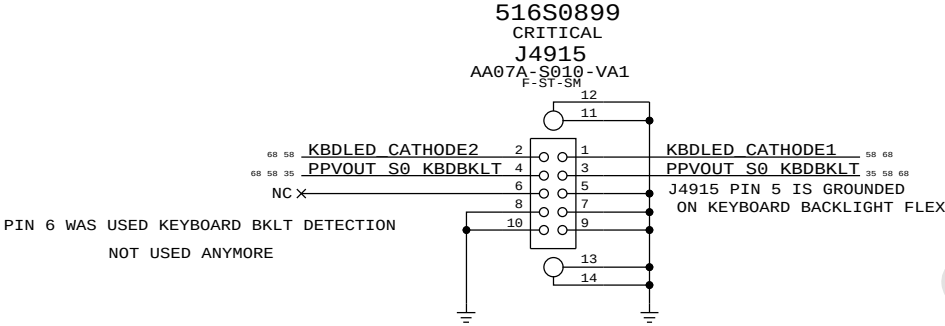


C	PIN	NAME	CURRENT	R_SNS	V_SNS	POWER
C	V+	100A		2.55 KOHM	0.0255 V	0.255E-6
		800A			0.204 V	16.32E-6
	VDD	60MA (MAX)	10 OHM		0.6 V	36E-3
	VOUT	60MA (MAX)	0.2 OHM		0.012 V	0.72E-3
	VDD	10MA (TYP)	1.5 OHM		0.012 V	90E-6
		14MA (MAX)			0.021 V	294E-6
FER	VTN	40MA (MAX)	4.7 OHM		0.018V V	75.2E-6

SYNC MASTER=J44		SYNC DATE=08/12/2013	
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KEYBOARD/TRACKPAD (1 OF 2)			
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Keyboard Backlight Connector



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KEYBOARD/TRACKPAD (2 OF 2)			
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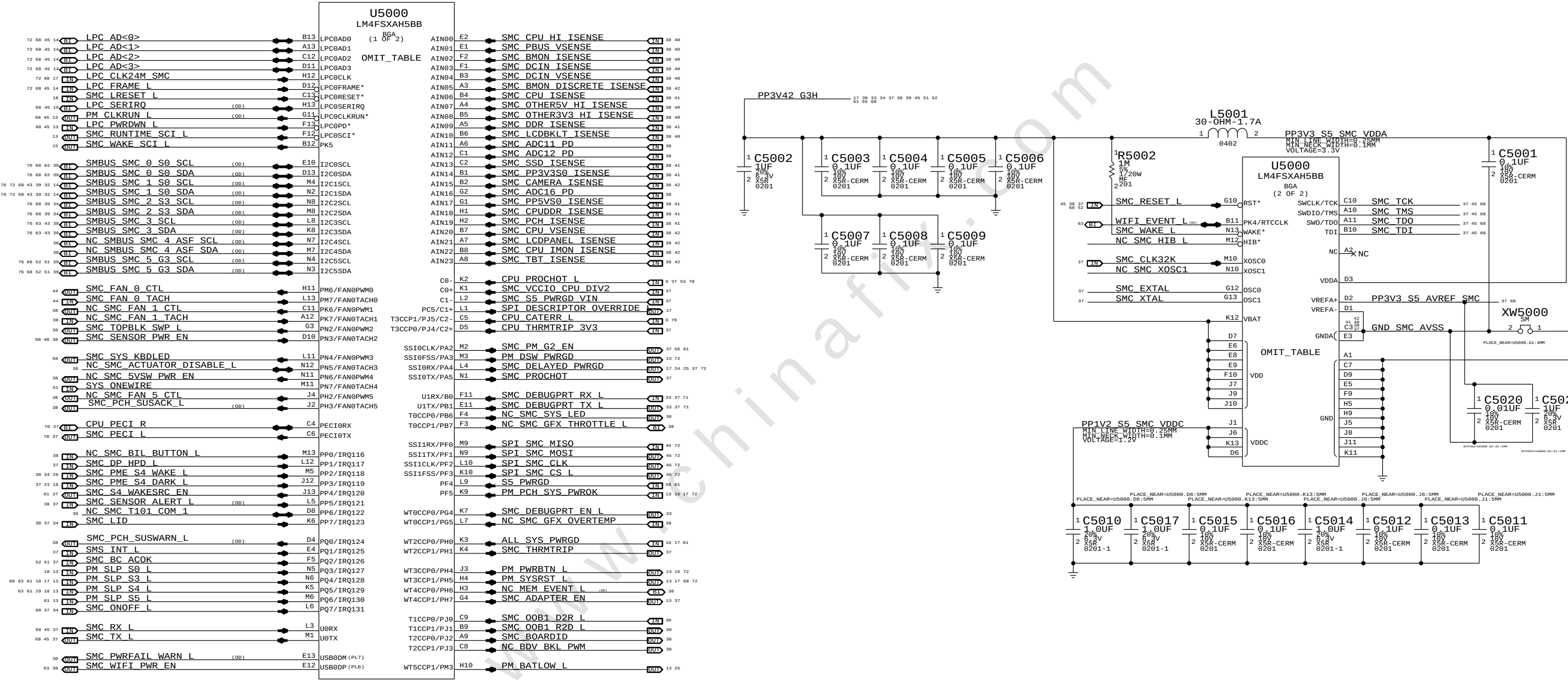
C

B

B

A

A



NOTE:
SMS INTERRUPT IS NOT USED, PULL UP TO SMC RAIL.

NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE			
SMC		DRAWING NUMBER	
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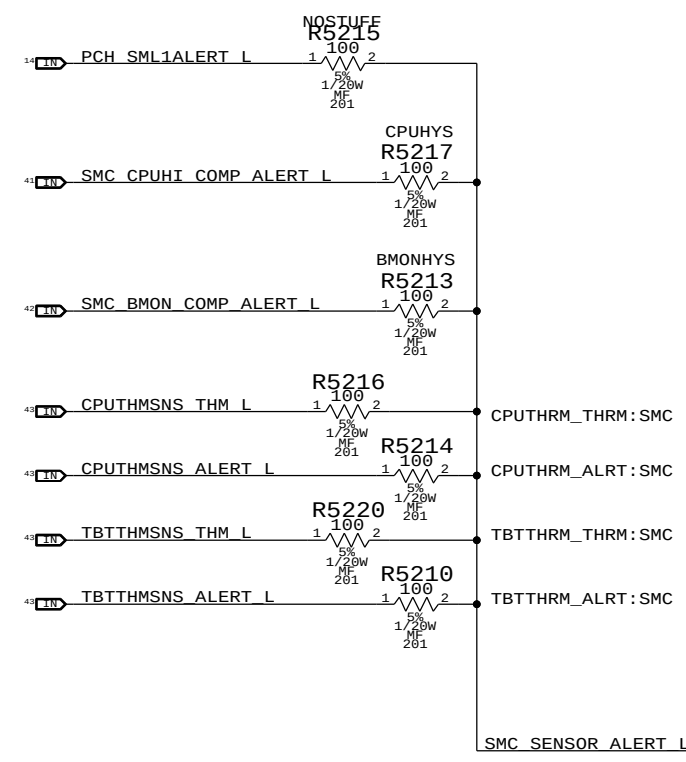
SMC12 ADC Assignments

36	OUT	SMC CPU HI ISENSE	=	SMC CPU HI ISENSE	IN	36	38	40
36	OUT	SMC PBUS VSENSE	=	SMC PBUS VSENSE	IN	36	38	40
36	OUT	SMC BMON ISENSE	=	SMC BMON ISENSE	IN	36	38	40
36	OUT	SMC DCIN ISENSE	=	SMC DCIN ISENSE	IN	36	38	40
36	OUT	SMC DCIN VSENSE	=	SMC DCIN VSENSE	IN	36	38	40
36	OUT	SMC BMON DISCRETE ISENSE	=	SMC BMON DISCRETE ISENSE	IN	36	38	42
36	OUT	SMC CPU ISENSE	=	SMC CPU ISENSE	IN	36	38	41
36	OUT	SMC OTHER5V HI ISENSE	=	SMC OTHER5V HI ISENSE	IN	36	38	40
36	OUT	SMC OTHER3V3 HI ISENSE	=	SMC OTHER3V3 HI ISENSE	IN	36	38	40
36	OUT	SMC DDR ISENSE	=	SMC DDR ISENSE	IN	36	38	41
36	OUT	SMC LCDBKLT ISENSE	=	SMC LCDBKLT ISENSE	IN	36	38	40
36	OUT	SMC ADC11 PD	=	SMC ADC11 PD	IN	36	38	38
36	OUT	SMC ADC12 PD	=	SMC ADC12 PD	IN	36	38	38
36	OUT	SMC SSD ISENSE	=	SMC SSD ISENSE	IN	36	38	41
36	OUT	SMC PP3V3S0 ISENSE	=	SMC PP3V3S0 ISENSE	IN	36	38	41
36	OUT	SMC CAMERA ISENSE	=	SMC CAMERA ISENSE	IN	36	38	42
36	OUT	SMC ADC16 PD	=	SMC ADC16 PD	IN	36	38	38
36	OUT	SMC PP5VS0 ISENSE	=	SMC PP5VS0 ISENSE	IN	36	38	41
36	OUT	SMC CPUDDR ISENSE	=	SMC CPUDDR ISENSE	IN	36	38	41
36	OUT	SMC PCH ISENSE	=	SMC PCH ISENSE	IN	36	38	41
36	OUT	SMC CPU VSENSE	=	SMC CPU VSENSE	IN	36	38	42
36	OUT	SMC LCDPANEL ISENSE	=	SMC LCDPANEL ISENSE	IN	36	38	42
36	OUT	SMC CPU IMON ISENSE	=	SMC CPU IMON ISENSE	IN	36	38	42
36	OUT	SMC TBT ISENSE	=	SMC TBT ISENSE	IN	36	38	42

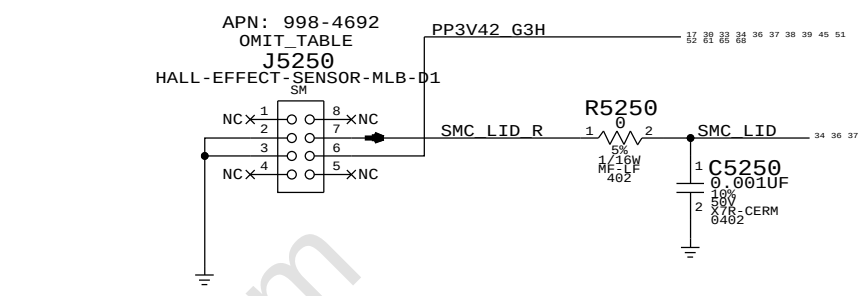
SMC12 Pin Assignments

36	NC	SMC 4 ASF SCL	=	NC SMC 4 ASF SCL	36	38
36	NC	SMC 4 ASF SDA	=	NC SMC 4 ASF SDA	36	38
36	NC	BDV BKL PWM	=	NC BDV BKL PWM	36	38
36	NC	SMC SYS LED	=	NC SMC SYS LED	36	38
36	NC	SMC GFX THROTTLE L	=	NC SMC GFX THROTTLE L	36	38
36	NC	SMC GFX OVERTEMP	=	NC SMC GFX OVERTEMP	36	38
36	NC	SMC FAN 1 CTL	=	NC SMC FAN 1 CTL	36	38
36	NC	SMC FAN 1 TACH	=	NC SMC FAN 1 TACH	36	38
36	NC	SMC 5VSW PWR EN	=	NC SMC 5VSW PWR EN	36	38
36	NC	SMC FAN 5 CTL	=	NC SMC FAN 5 CTL	36	38
36	NC	SMC BIL BUTTON L	=	NC SMC BIL BUTTON L	36	38
36	NC	MEM EVENT L	=	NC MEM EVENT L	36	38
36	NC	SMC T101 COM 1	=	NC SMC T101 COM 1	36	38
36	NC	SMC ACTUATOR DISABLE	=	NC SMC ACTUATOR DISABLE	36	38

Thermal Alerts



Hall Effect Pads



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
677-0912	1	SUBASSY,PCBA HALL EFFECT, J44	J5250	CRITICAL	

639-4502 (J44 HALL EFFECT BOARD) REPORTS TO 677-0912

Specify one of these BOM GROUPS.

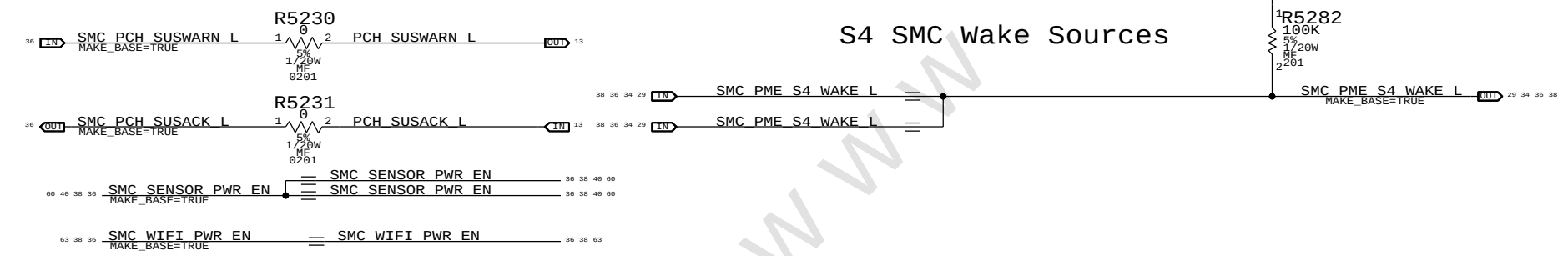
BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC, CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC, CPUTHRM_ALRT:PU
CPUTHRM:ALRT	CPUTHRM_THRM:PU, CPUTHRM_ALRT:SMC
CPUTHRM:NONE	CPUTHRM_THRM:PU, CPUTHRM_ALRT:PU

Specify one of these BOM GROUPS.

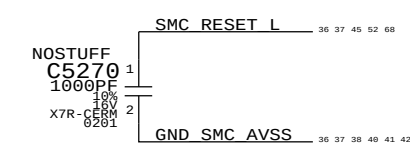
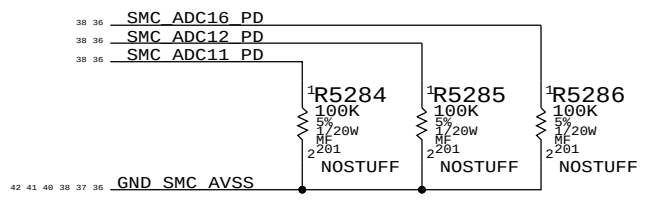
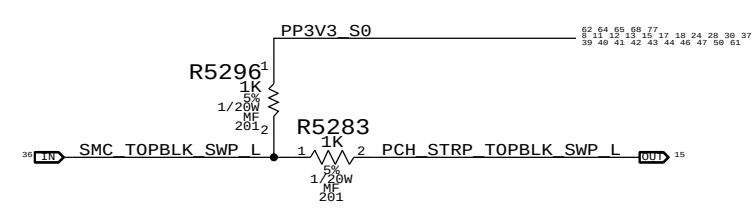
BOM GROUP	BOM OPTIONS
TBTTHRM:BOTH	TBTTHRM_THRM:SMC, TBTTHRM_ALRT:SMC
TBTTHRM:THRM	TBTTHRM_THRM:SMC, TBTTHRM_ALRT:PU
TBTTHRM:ALRT	TBTTHRM_THRM:PU, TBTTHRM_ALRT:SMC
TBTTHRM:NONE	TBTTHRM_THRM:PU, TBTTHRM_ALRT:PU

Requires EMC1412-1 or EMC1412-2 instead of EMC1412-A, new APN needs to be created.

S4 SMC Wake Sources



Top Block Swap



SMC Project Support

Apple Inc.

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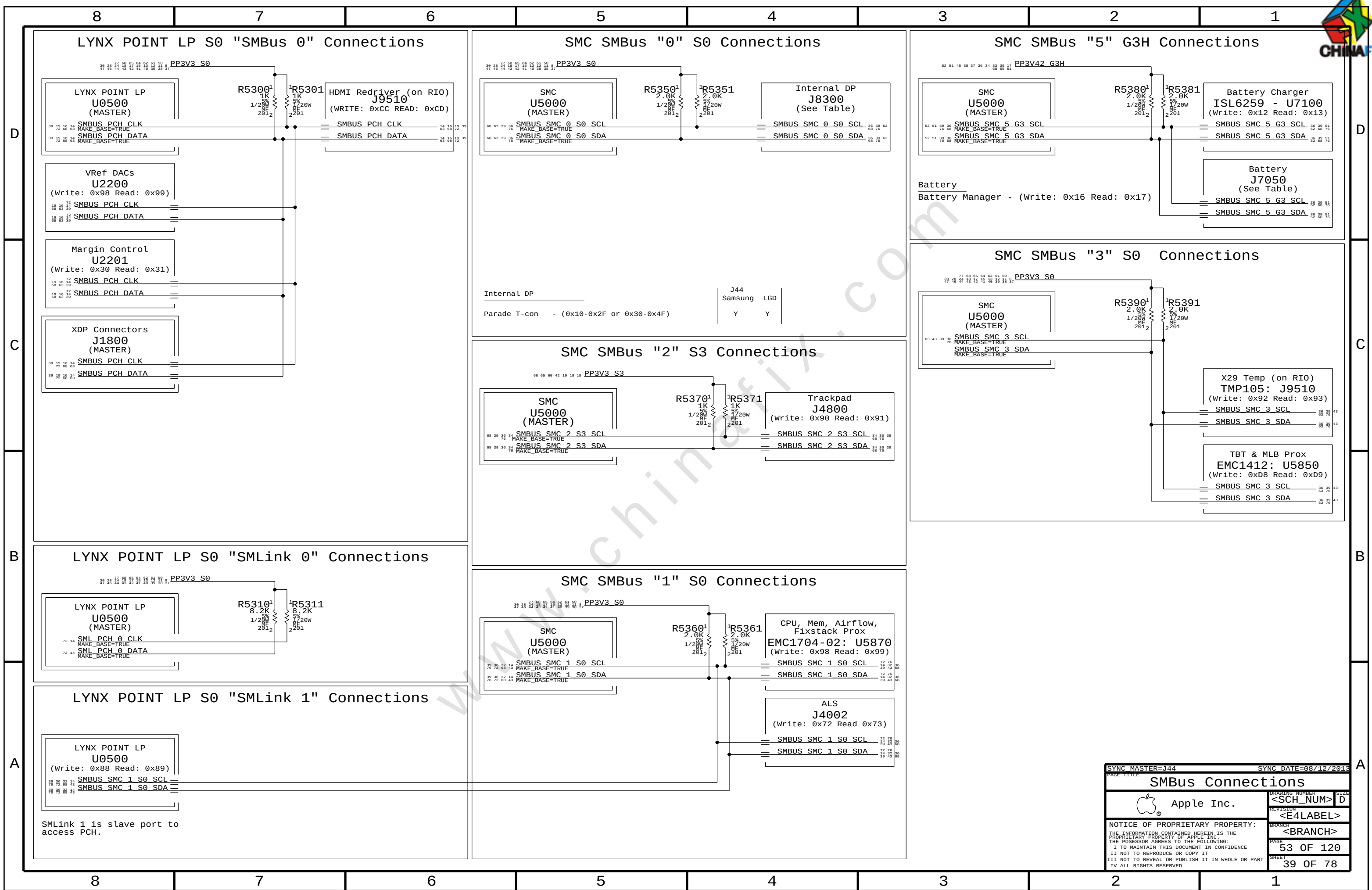
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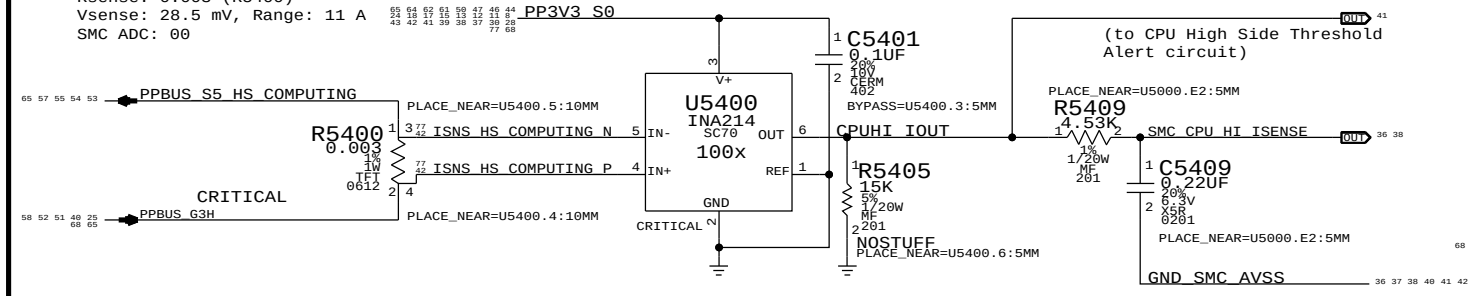


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SMBus Connections			
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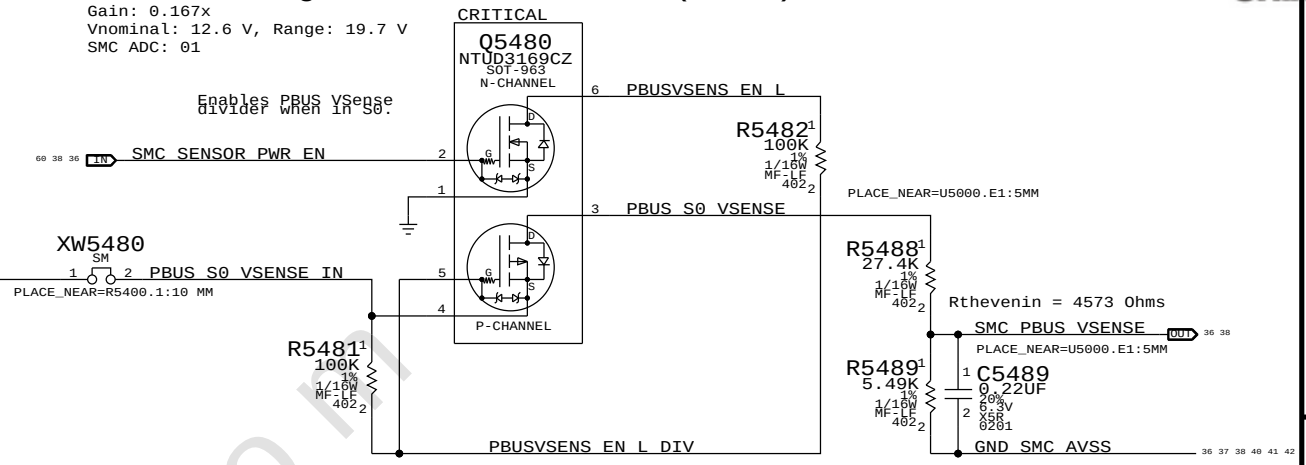
CPU High Side Current Sense (IC0R)

Gain: 100x, EDP: 9.5 A
Rsense: 0.003 (R5400)
Vsense: 28.5 mV, Range: 11 A
SMC ADC: 00



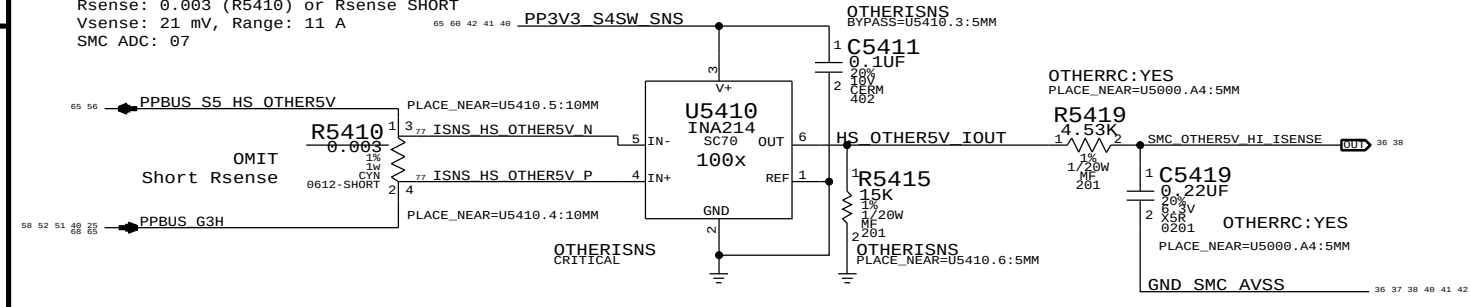
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
Vnominal: 12.6 V, Range: 19.7 V
SMC ADC: 01



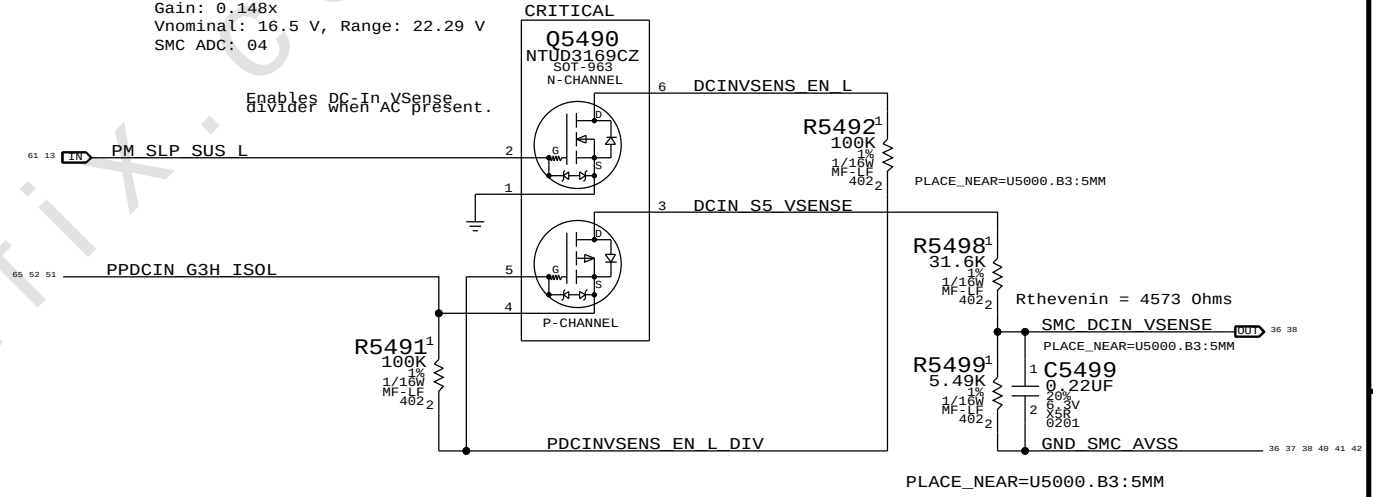
OTHER 5V High Side Current Sense (IO5R)

Gain: 100x, EDP: 7 A
Rsense: 0.003 (R5410) or Rsense SHORT
Vsense: 21 mV, Range: 11 A
SMC ADC: 07



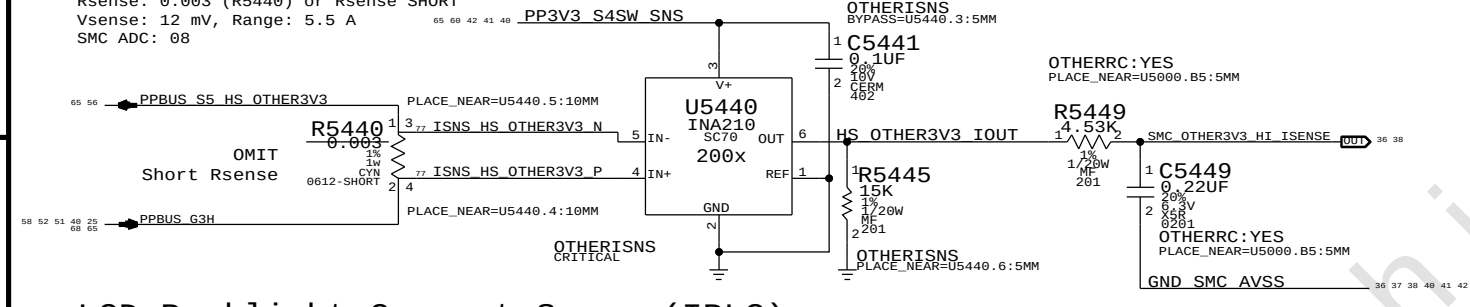
DC In Voltage Sense & Enable (VD0R)

Gain: 0.148x
Vnominal: 16.5 V, Range: 22.29 V
SMC ADC: 04



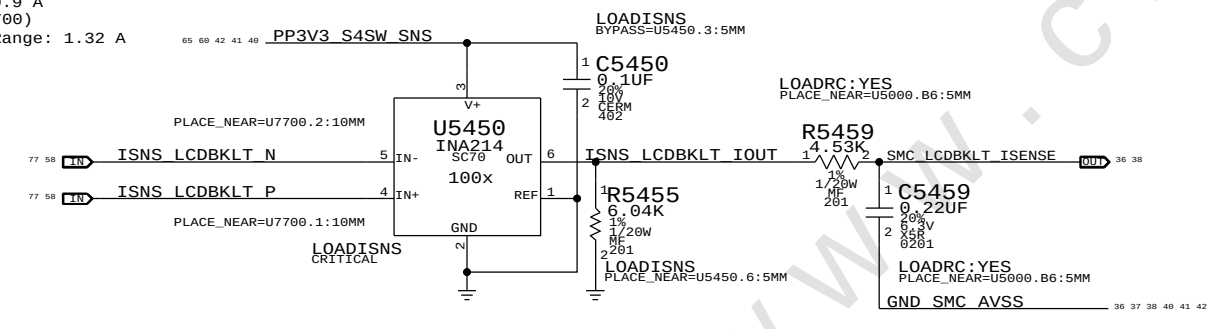
OTHER 3.3V High Side Current Sense (IO3R)

Gain: 200x, EDP: 5 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 12 mV, Range: 5.5 A
SMC ADC: 08



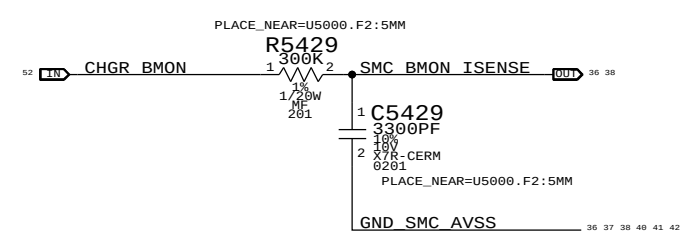
LCD Backlight Current Sense (IBLC)

Gain: 100x, EDP: 0.9 A
Rsense: 0.025 (R7700)
Vsense: 22.5 mV, Range: 1.32 A
SMC AD: 10



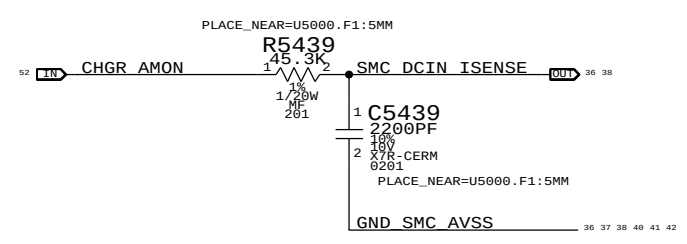
Charger (BMON) Current Sense (IPBR)

Charger Gain: 36x, EDP: 8 A
Rsense: 0.005 (R7150)
SMC ADC: 02



DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
Rsense: 0.020 (R7120)
SMC ADC: 03



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5419, C5449		OTHERRC: NO
117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5459		LOADRC: NO

SYNC_MASTER=J44

SYNC_DATE=08/12/2013

Power Sensors: High Side

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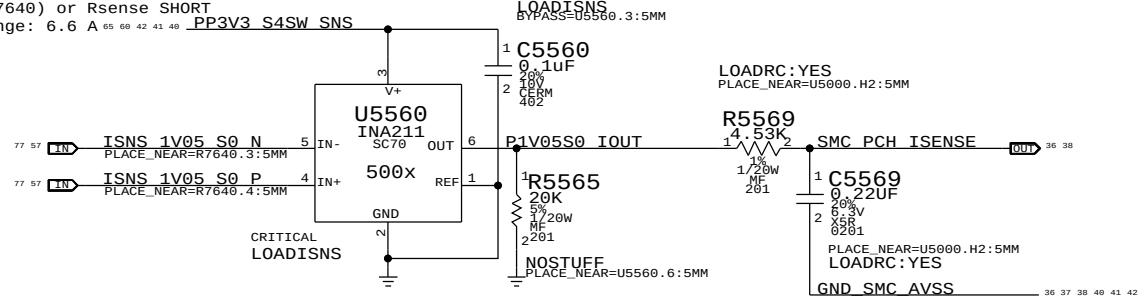
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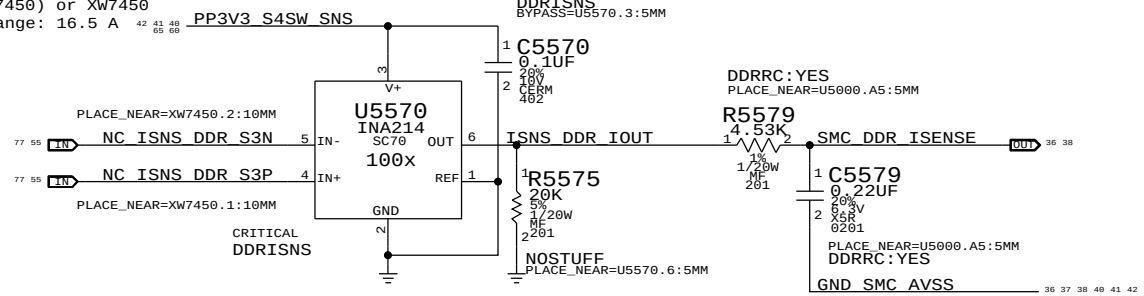
PCH 1.05V Current Sense (IC1C)

Gain: 500x, EDP: 5 A
Rsense: 0.001 (R7640) or Rsense SHORT
Vsense: 5 mV, Range: 6.6 A
SMC ADC: 19



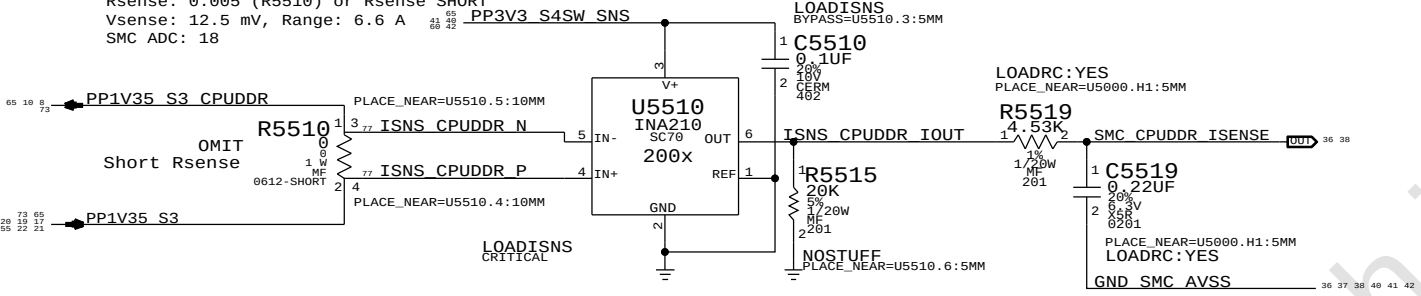
DDR 1.35V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 100x, EDP: 9 A
Rsense: 0.002 (R7450) or XW7450
Vsense: 21 mV, Range: 16.5 A
SMC ADC: 09



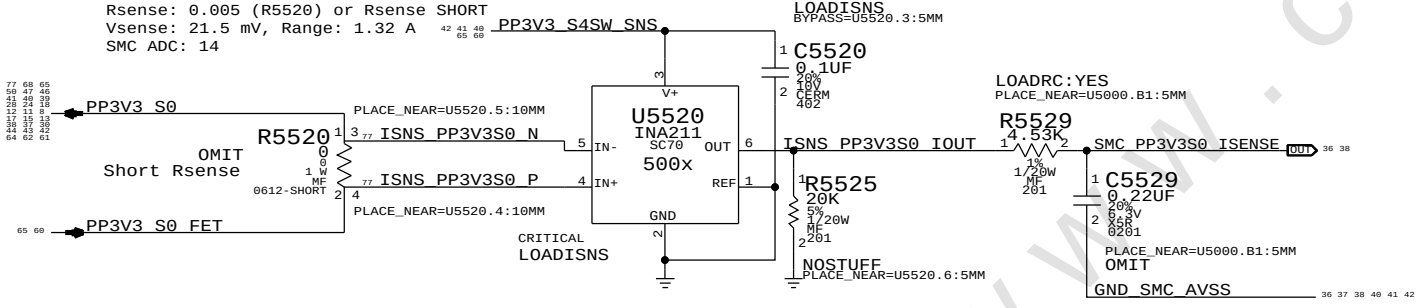
CPU DDR 1.35V S3 (CPU Only) Current Sense (IM1C)

Gain: 200x, EDP: 2.5 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 12.5 mV, Range: 6.6 A
SMC ADC: 18



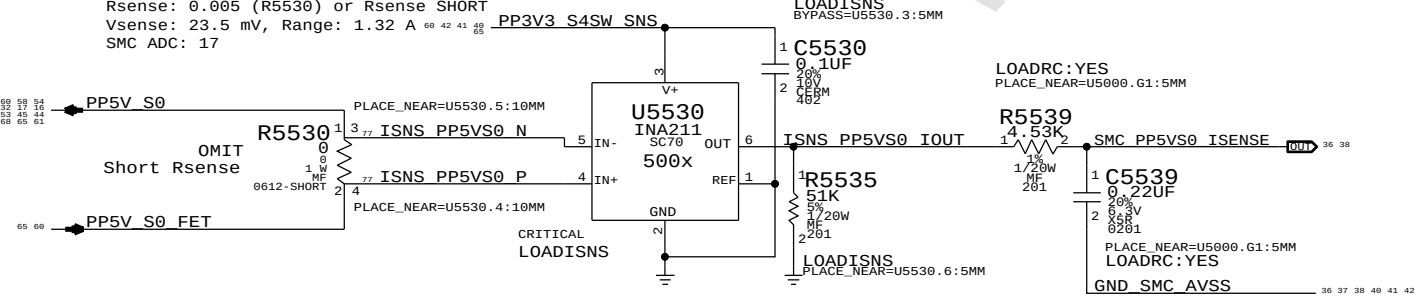
3.3V S0 Rail Current Sense (IR3C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5520) or Rsense SHORT
Vsense: 21.5 mV, Range: 1.32 A
SMC ADC: 14



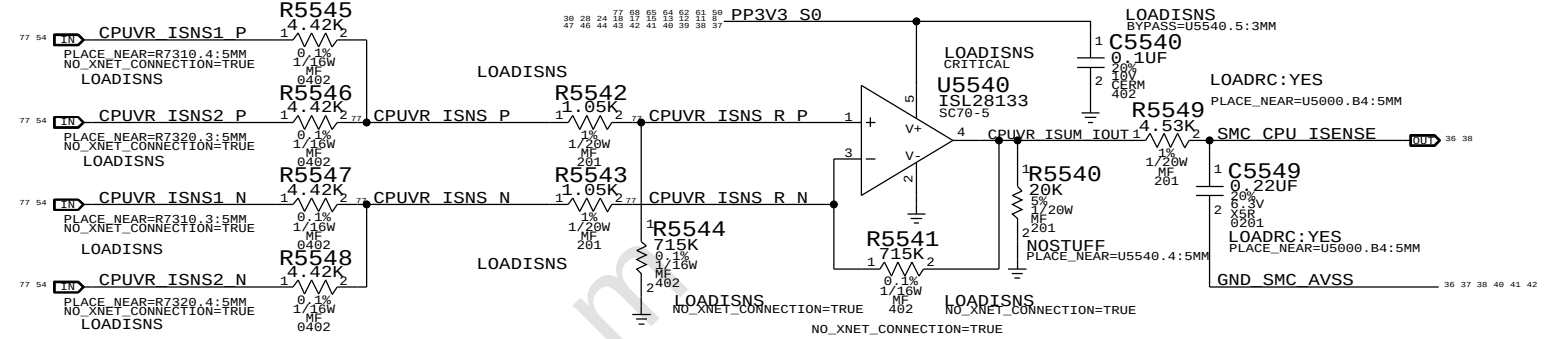
5V S0 Rail Current Sense (IR5C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5530) or Rsense SHORT
Vsense: 23.5 mV, Range: 1.32 A
SMC ADC: 17



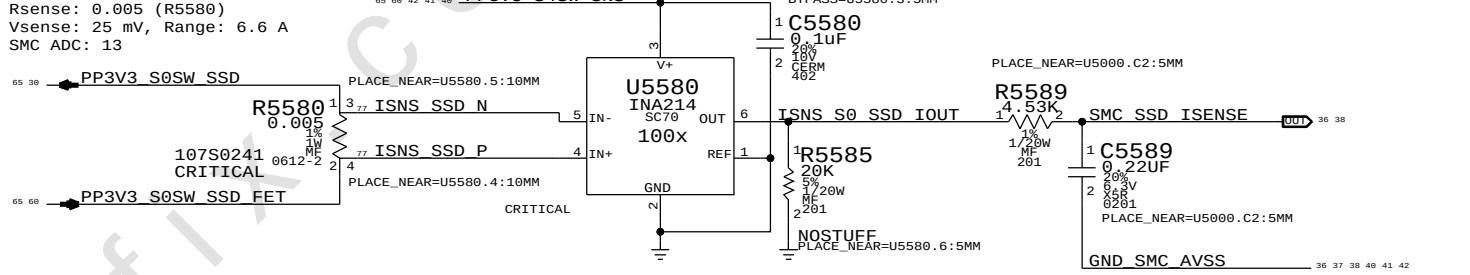
CPU Fixed Current Sense (IC0C)

Gain: 219.33x, EDP: 40 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 15 mV, Range: 40.12 A
SMC ADC: 06



SSD Current Sense (ISDC)

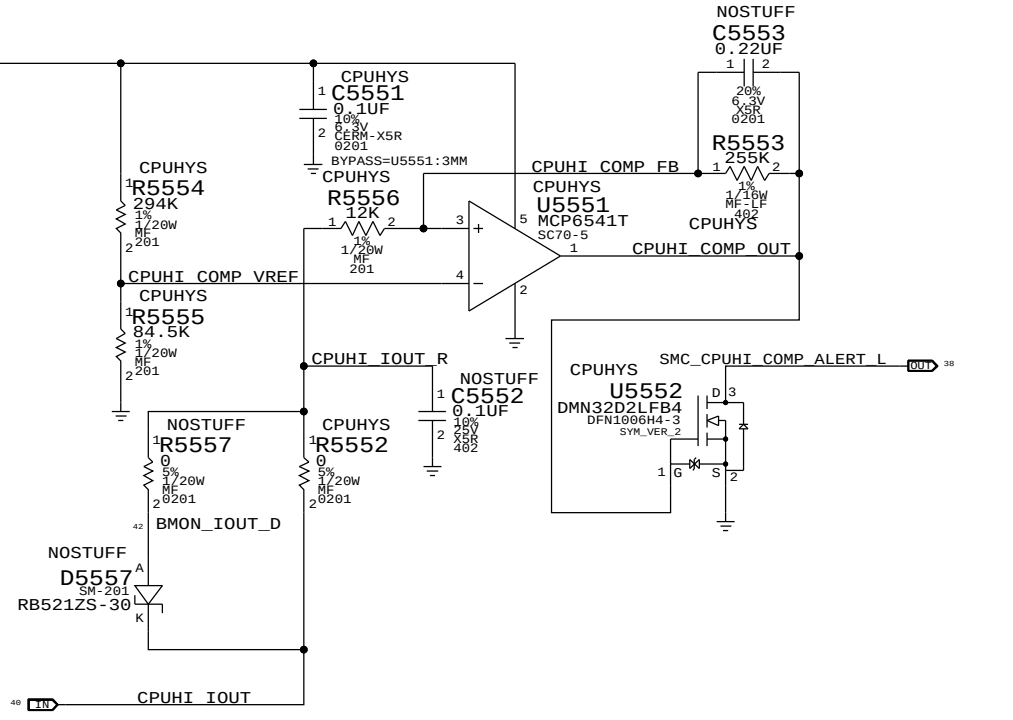
Gain: 100x, EDP: 5 A (16.5 W)
Rsense: 0.005 (R5580)
Vsense: 25 mV, Range: 6.6 A
SMC ADC: 13



CPU High Side Current (IC0R) Threshold Alert

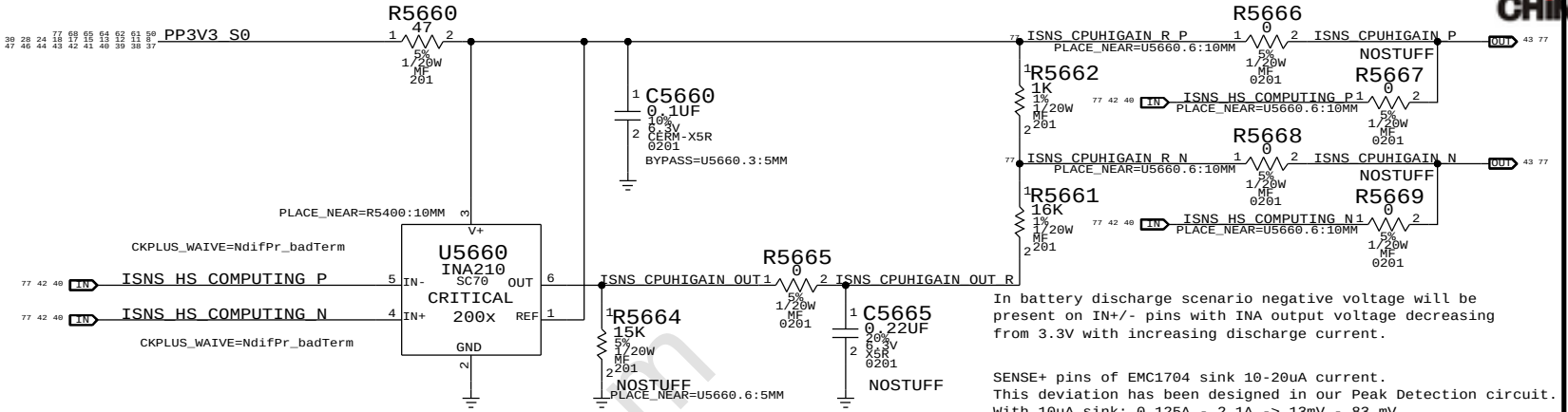
Gain: 100x
Rsense: 0.003 (R5400)

Trip Target on CPU High current: 2.5 A
Hysteresis Circuit:
Vref = 0.737 V
Vth = 0.616 V -> 2.054 A on CPU High current
Vtl = 0.771 V -> 2.571 A on CPU High current
Hysteresis Margin = 0.518 A

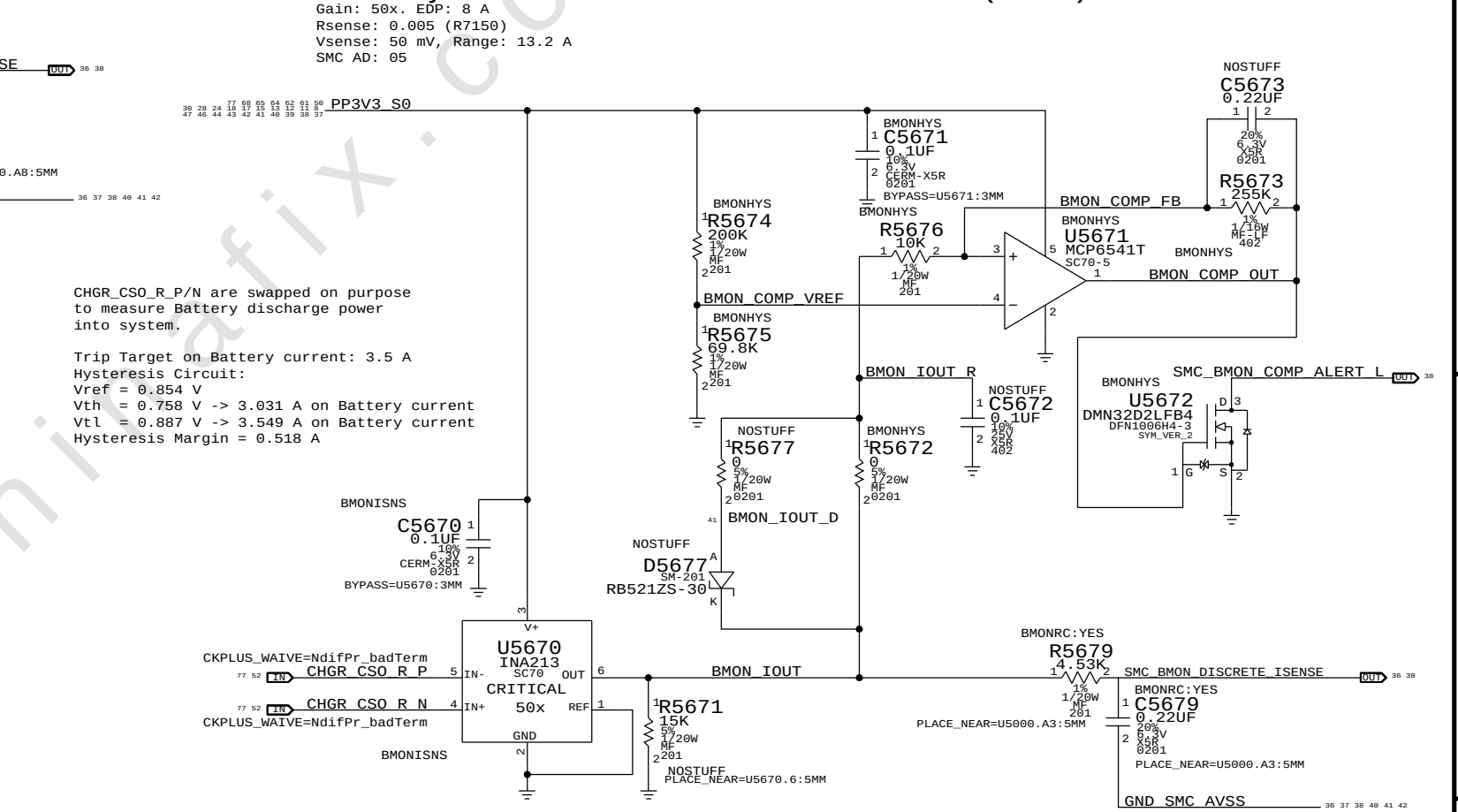




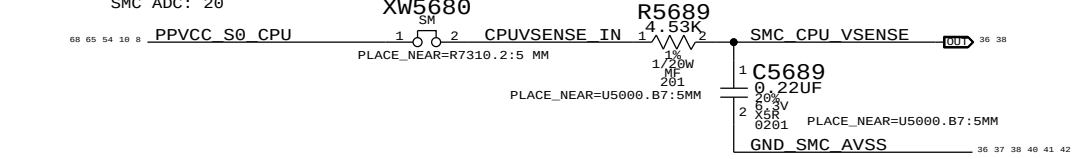
CPU High Side (IC0R) Peak Detection Support



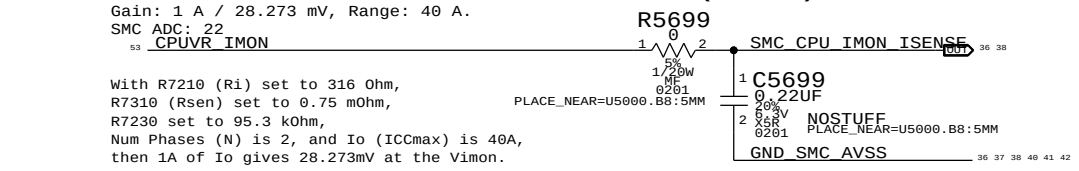
Battery BMON Discrete Current Sense (IP0R) & Threshold Alert



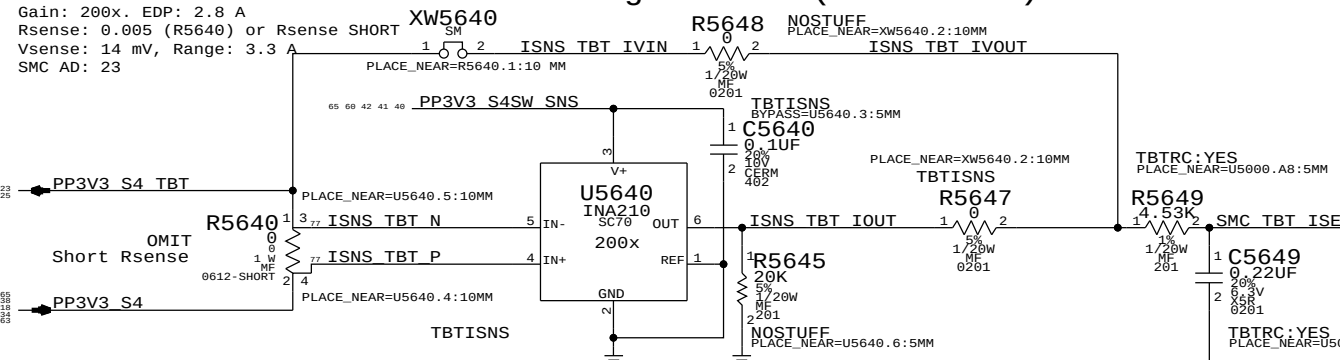
CPU Core Voltage Sense (VC0C)



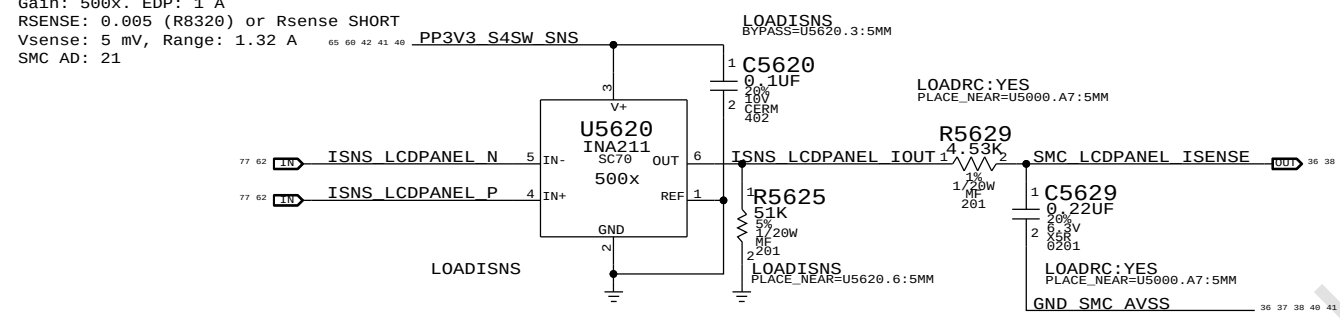
CPU Core IMON Current Sense (IC2C)



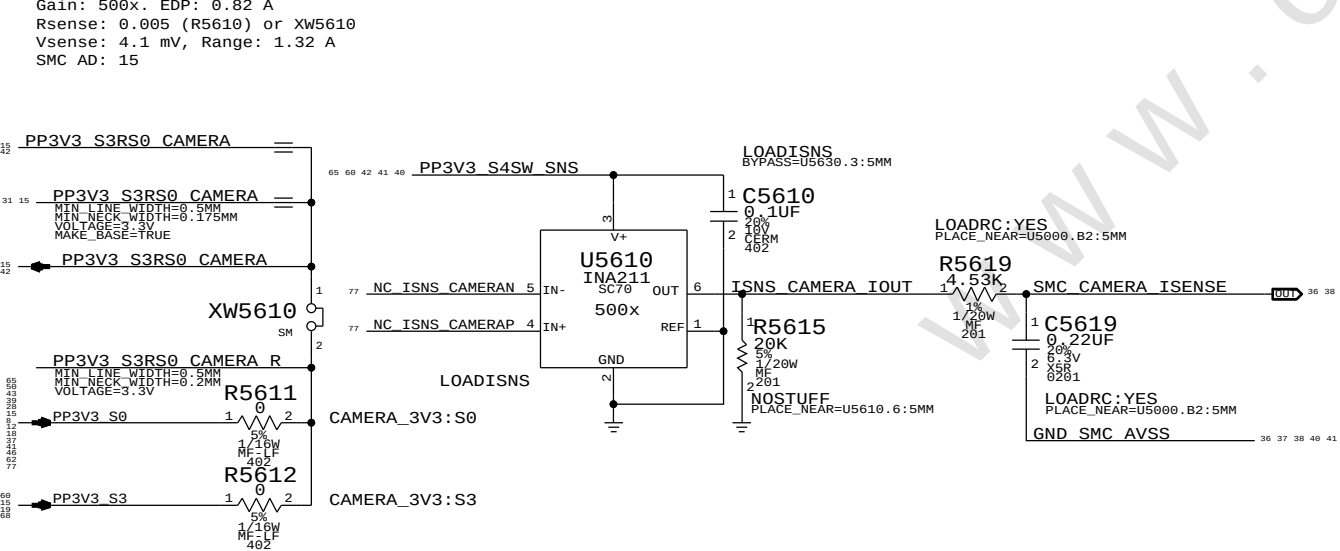
Thunderbolt TBT Current/Voltage Sense (IHSC/VHSC)



LCD Panel Current Sense (ILDC)



Camera (S2 Controller) Current Sense (ICMC)



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5619,C5629,C5649		
117S0008	1	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5679		

SYNC MASTER=144

SYNC DATE=08/12/2013

Power Sensors: Extended

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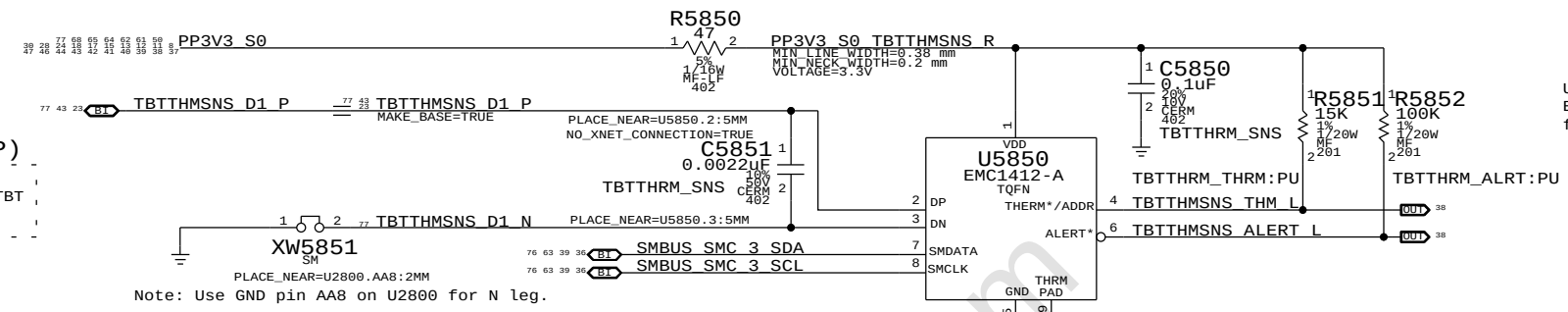
Thermal Sensor A:
Thunderbolt Die, MLB Proximity

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (THSP)

' Placement Note:

```
' The P leg connects to THERMDA pin of the TBT
' chip, the N leg connect to pin AA8.
```



U5850 I2C Address:
By setting R5851 to 15k, I2C address
for U5850 is 0xD8/0xD9.

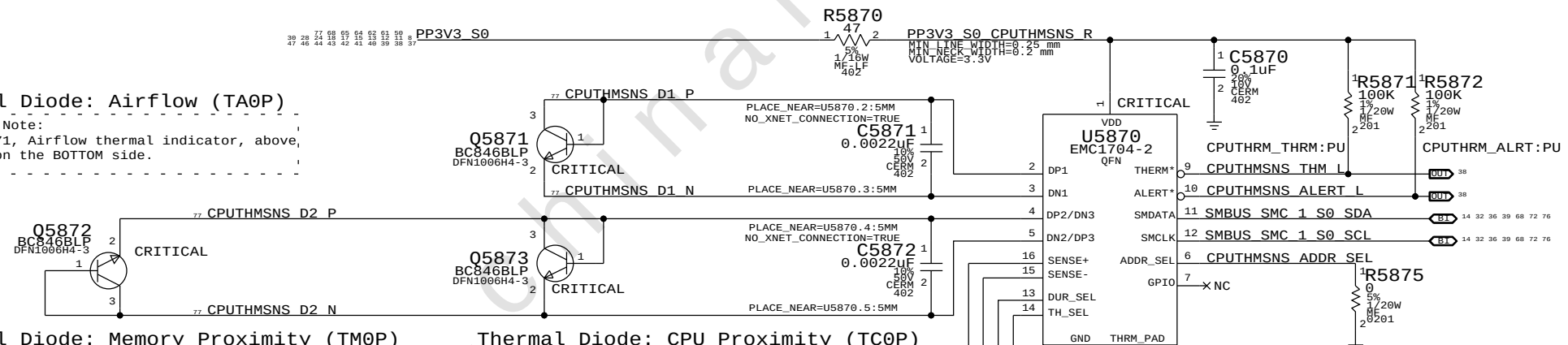
Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity

I2C Write: 0x98, I2C Read: 0x99

Thermal Diode: Airflow (TA0P)

' Placement Note:

' Place Q5871, Airflow thermal indicator, above
' the SSD, on the BOTTOM side.



Thermal Diode: Memory Proximity (TM0P)

Placement Note:

- ' Place Q5872 between two rows of Memory device
- ' between channel A and B, on the BOTTOM side.

Thermal Diode: CPU Proximity (TCOP)

Placement Note:

' Place Q5873 under the CPU,
' on the BOTTOM side.

Thermal Sensor: Fin Stack Proximity (Th1H)

```

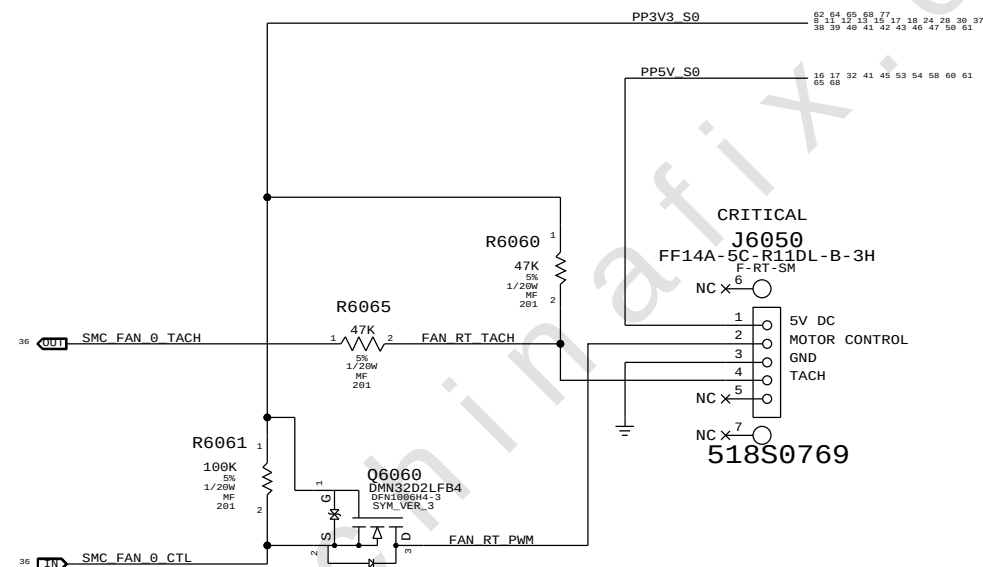
' Placement Note:
' Place U5870 at corner near Fan,
' on the TOP side.

```

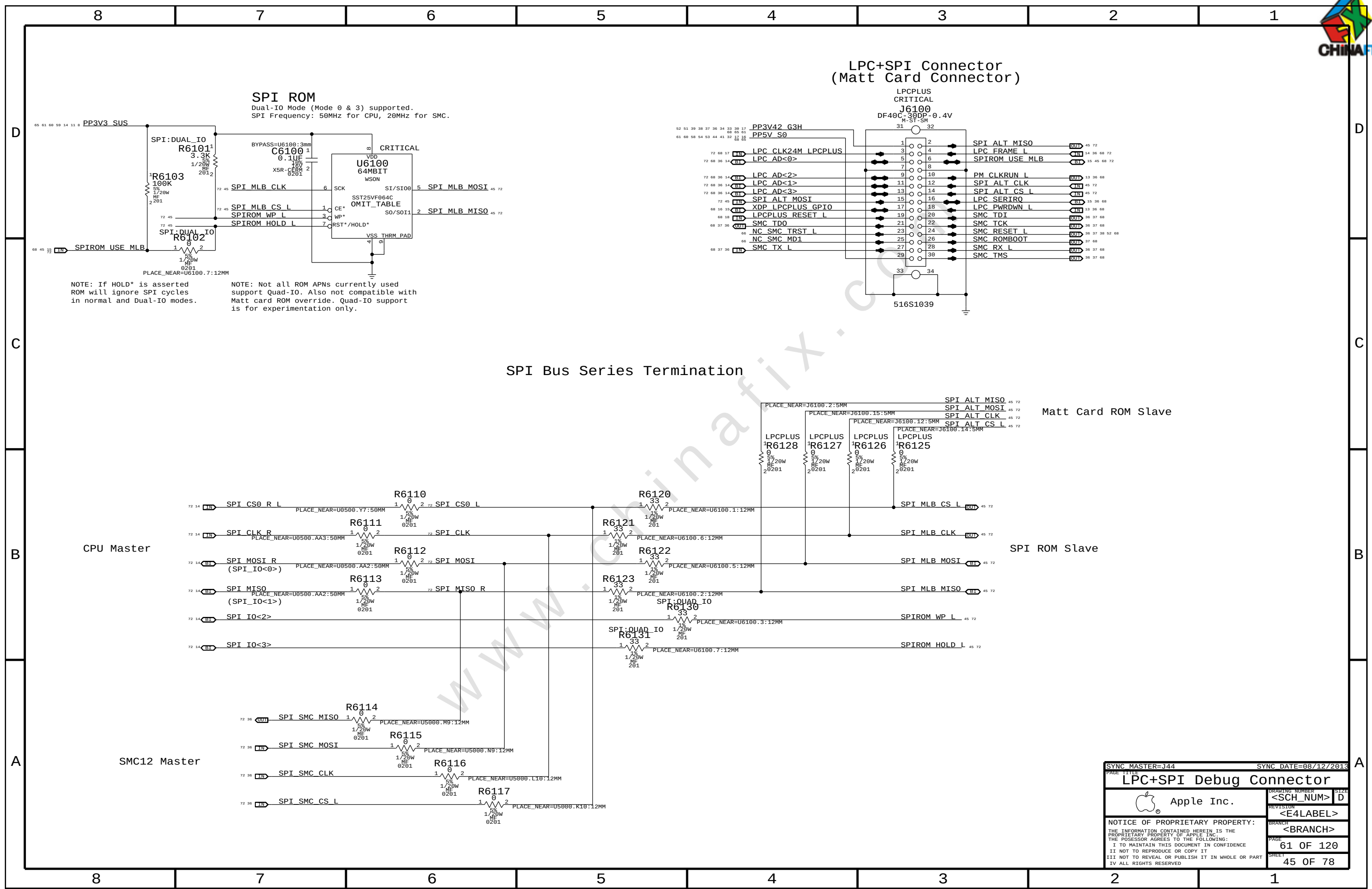
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KEEP THE 5 PIN CONNECTOR FROM D1



SYMC MASTER=J44		SYMC DATE=08/12/2013	
PAGE TITLE			
Fan			
	Apple Inc.		DRAWING NUMBER <SCH_NUM>
			SIZE D
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SPI ROM

Dual-IO Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.

LPC+SPI Connector (Matt Card Connector)

LPCPLUS
CRITICAL

J6100
DF40C-30DP-0.4V
M-ST-SM

516S1039

SPI Bus Series Termination

Matt Card ROM Slave

SPI ROM Slave

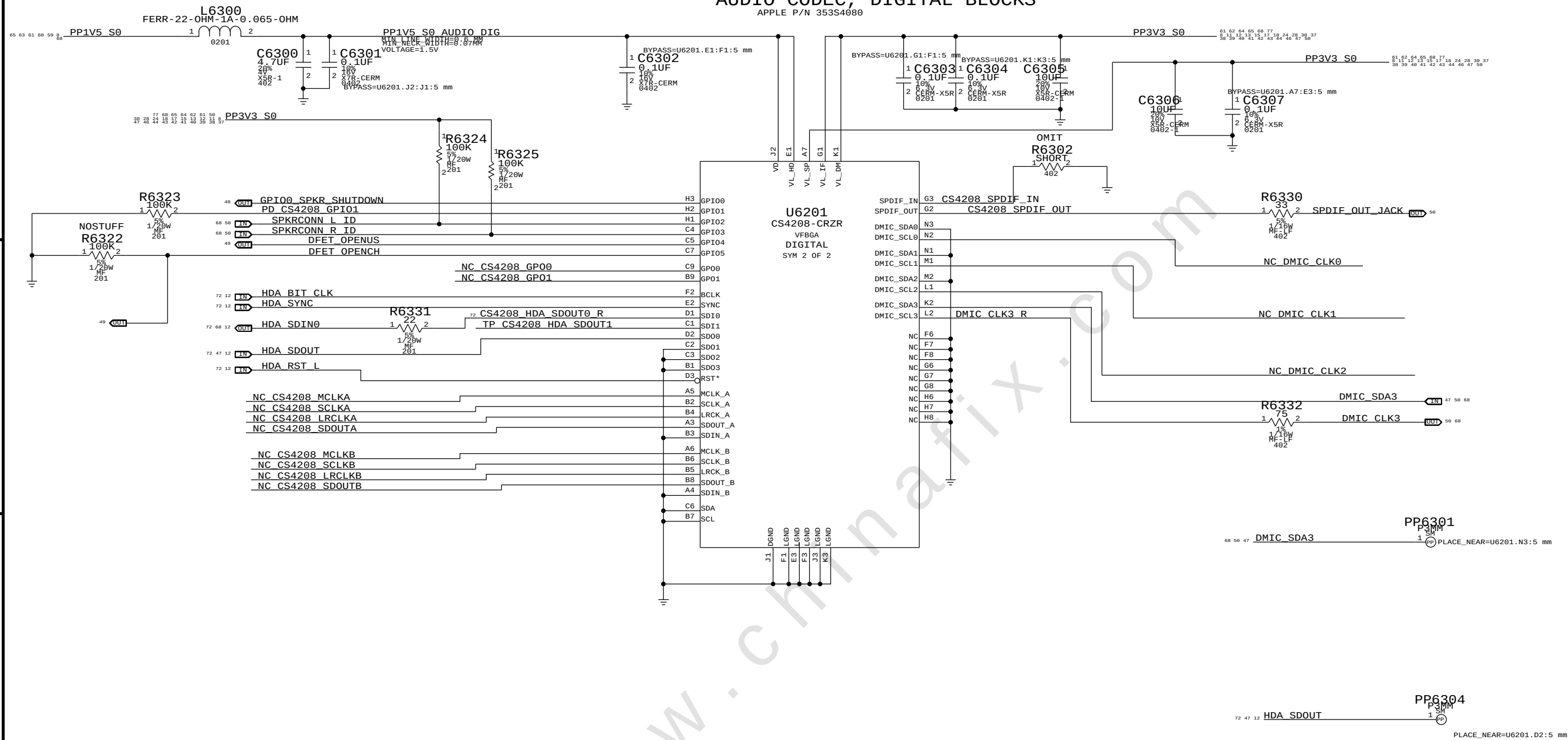
CPU Master

SMC12 Master

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LPC+SPI Debug Connector			
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AUDIO CODEC, DIGITAL BLOCKS
APPLE P/N 353S4080



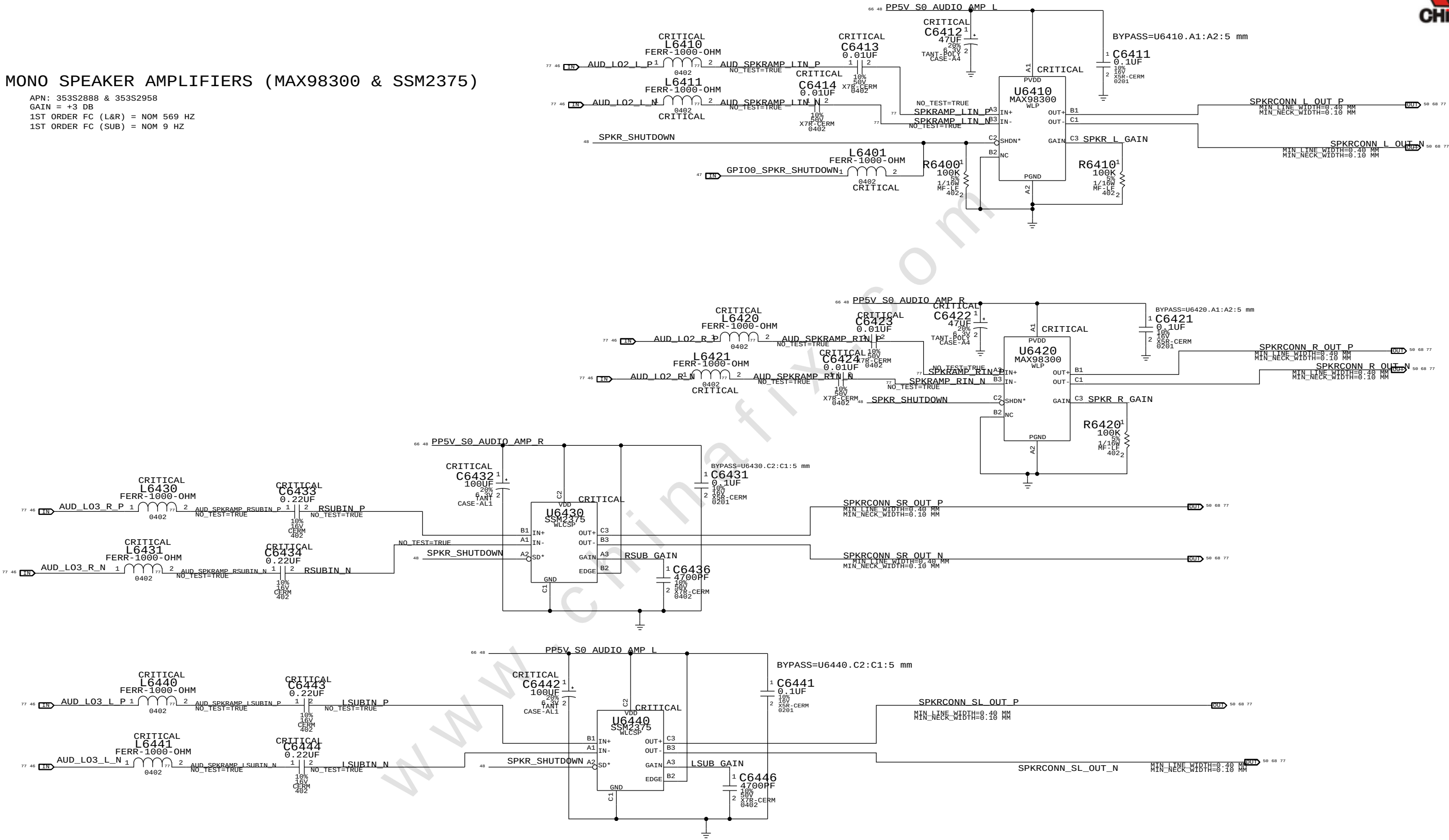
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AUDIO:CODEC, DIGITAL			
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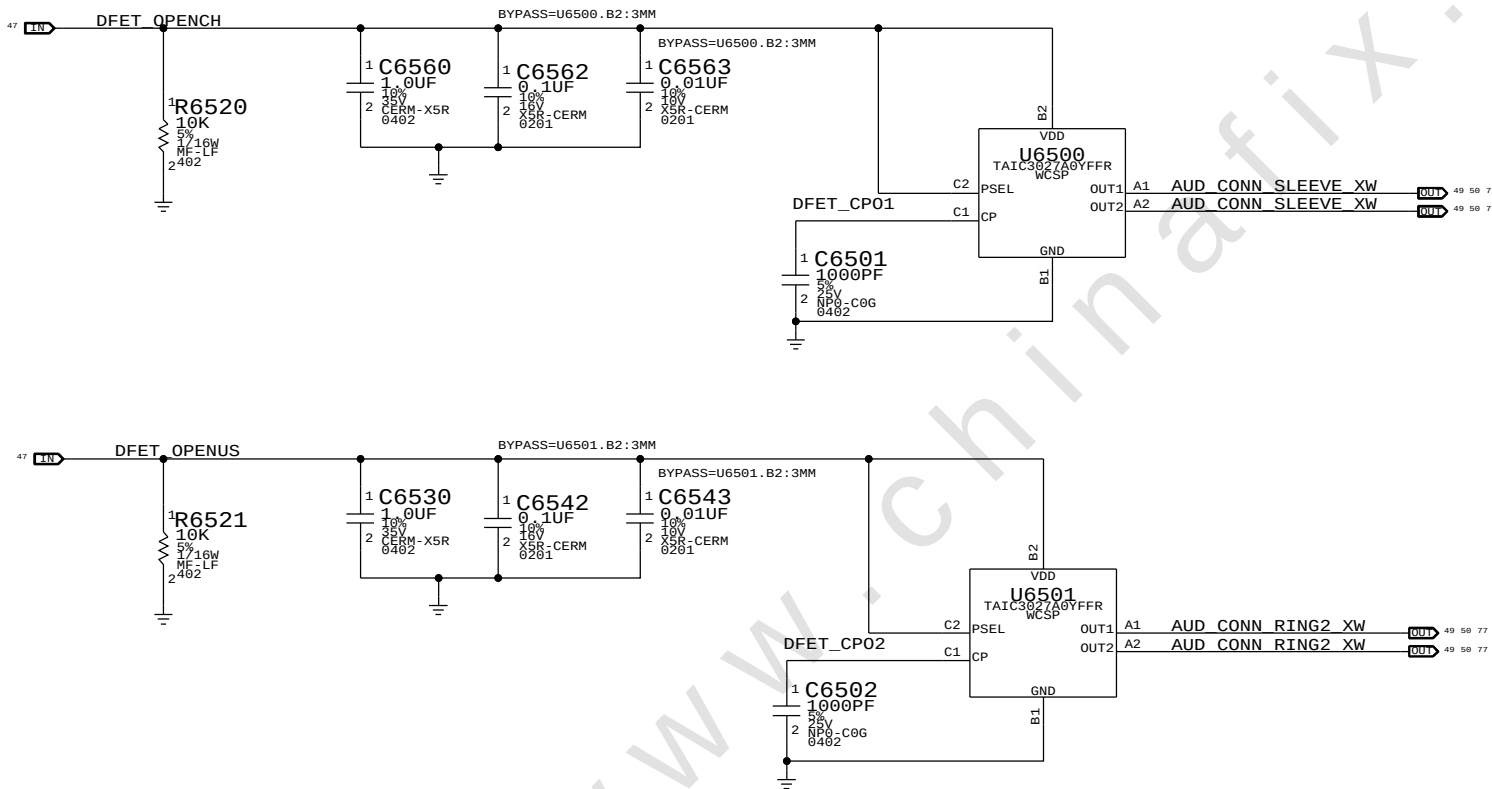
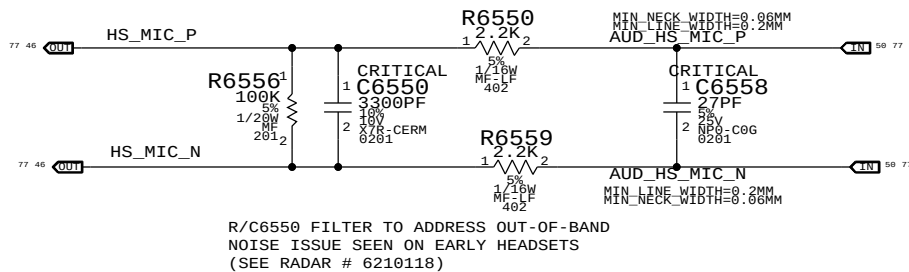
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)

APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



8 7 6 5 4 3 2 1

SYNC MASTER=J44		SYNC DATE=08/12/2013	
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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

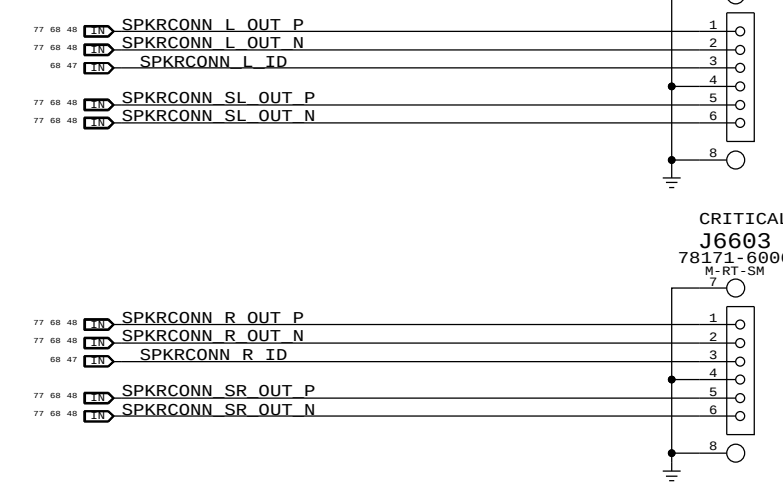
OTHER CODEC GPIO LINES

LEFT SPEAKER ID	GPIO2 INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3 INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4 OUTPUT	HIGH = DFETS OPEN

SPEAKER CONNECTOR

HP=80HZ
APN: 518S0672

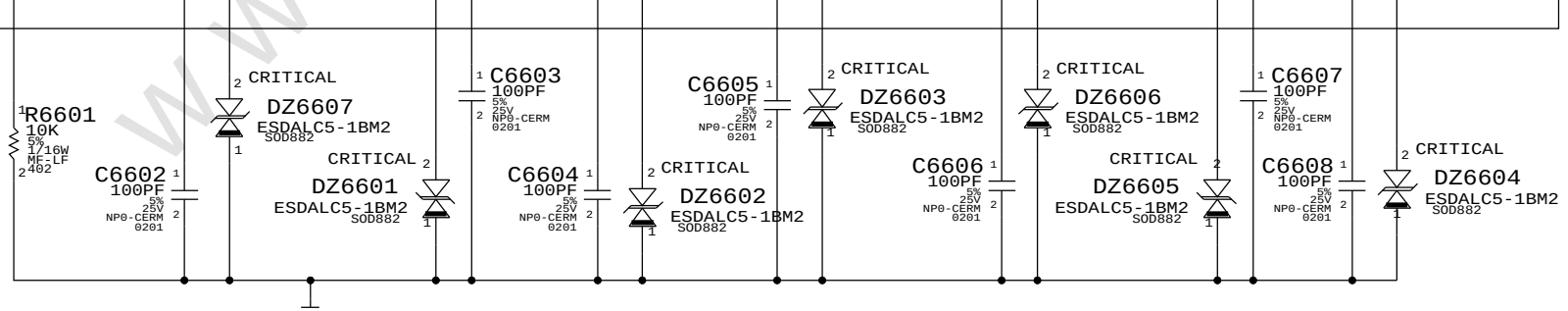
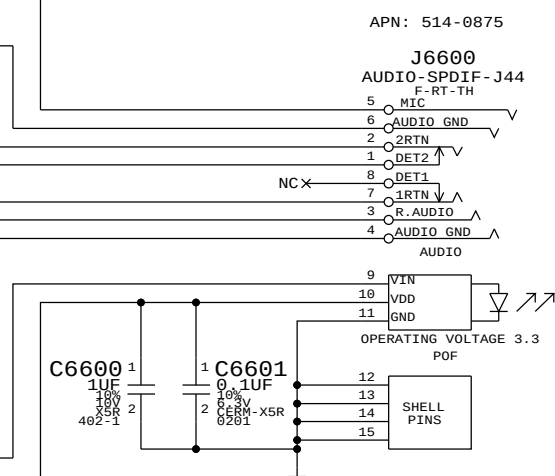
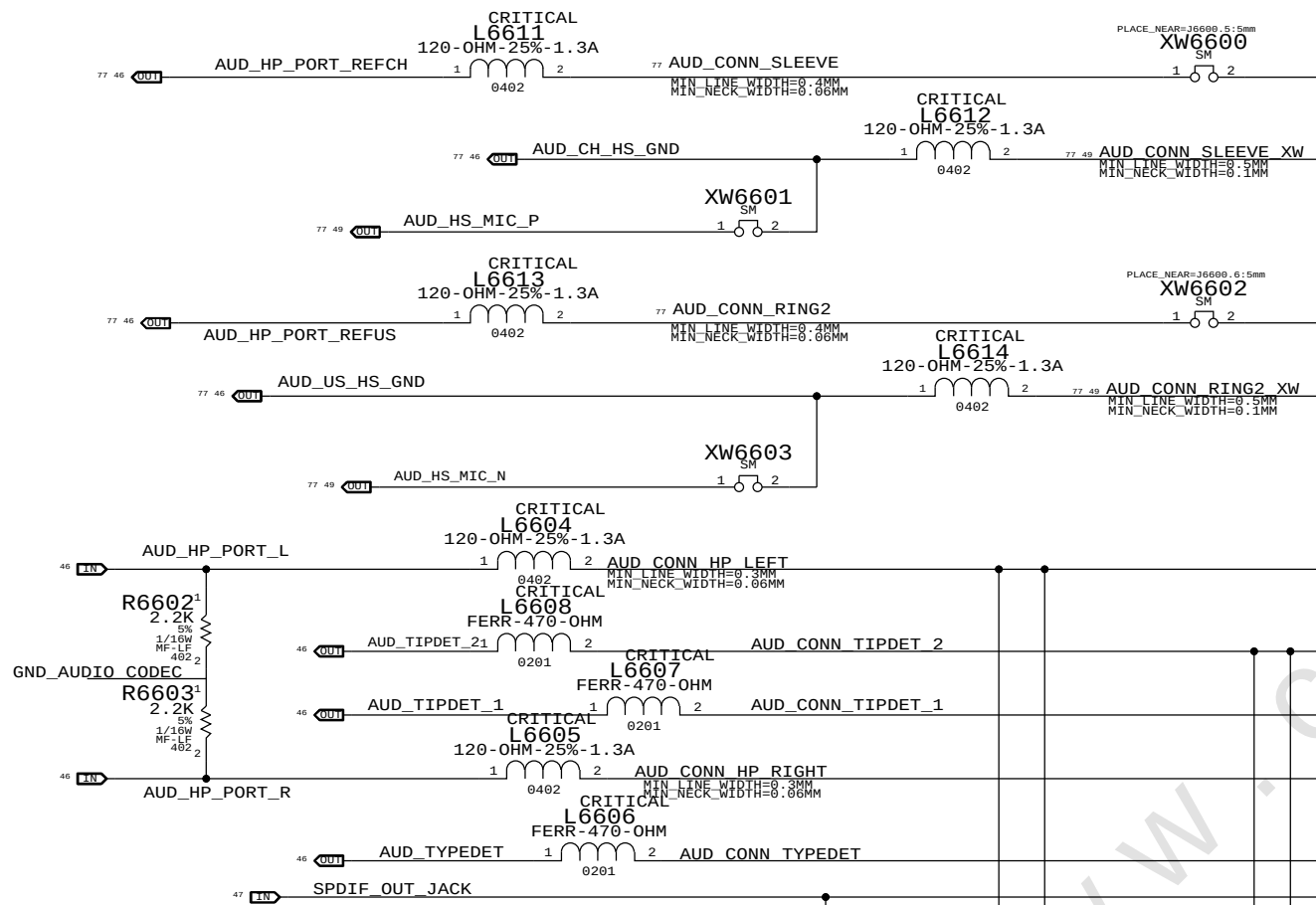
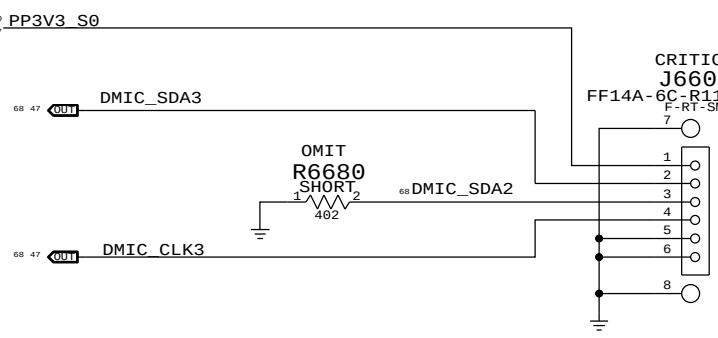
CRITICAL
J6602
78171-6006
M-RT-SM



2-MIC CONNECTOR

APN: 518S0818

CRITICAL
J6601
FF14A-6C-R11DL-B-3H
F-RT-SM



SYNC MASTER=J44

SYNC DATE=08/12/2013

AUDIO: JACK TRANSLATORS

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MagSafe DC Power Jack

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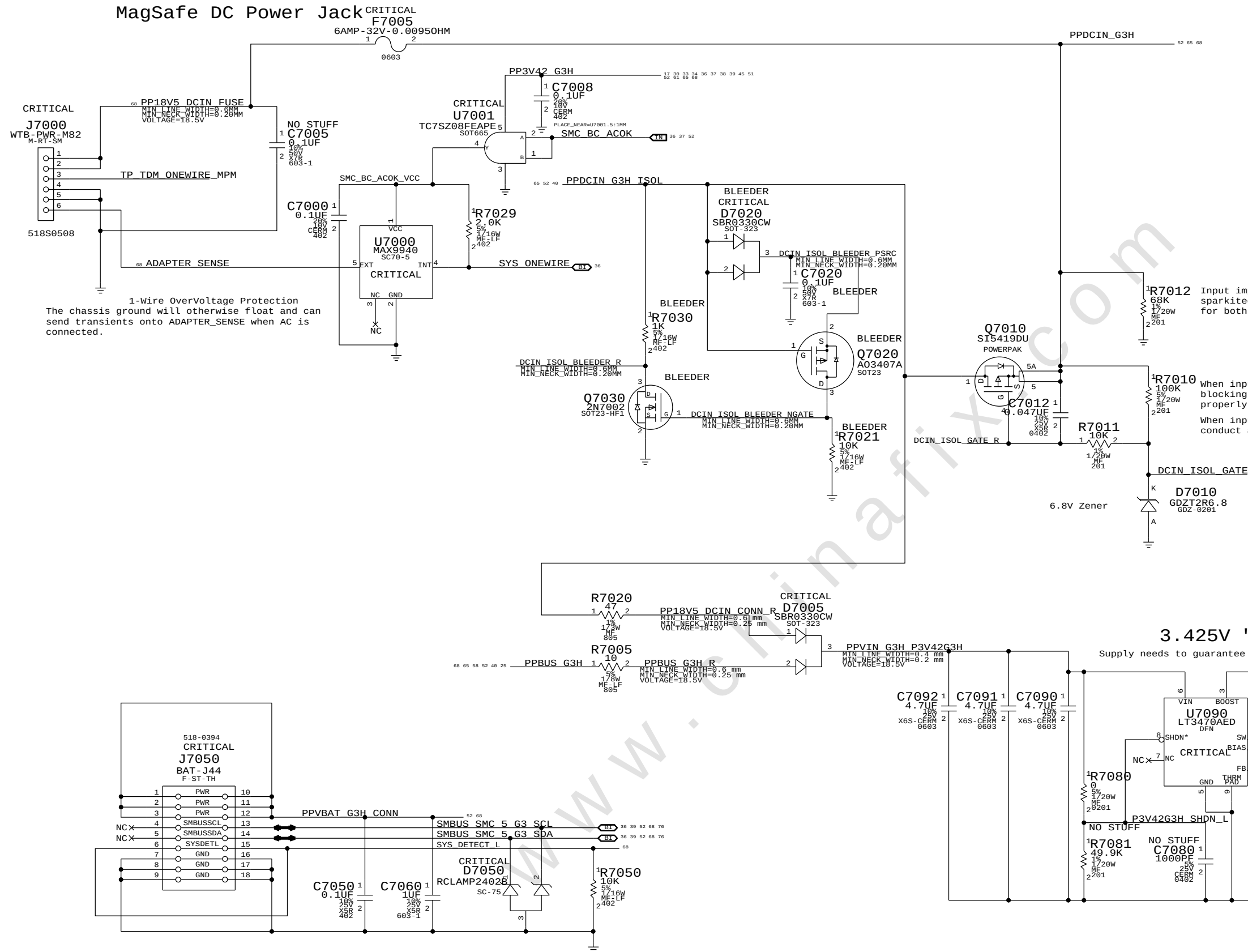
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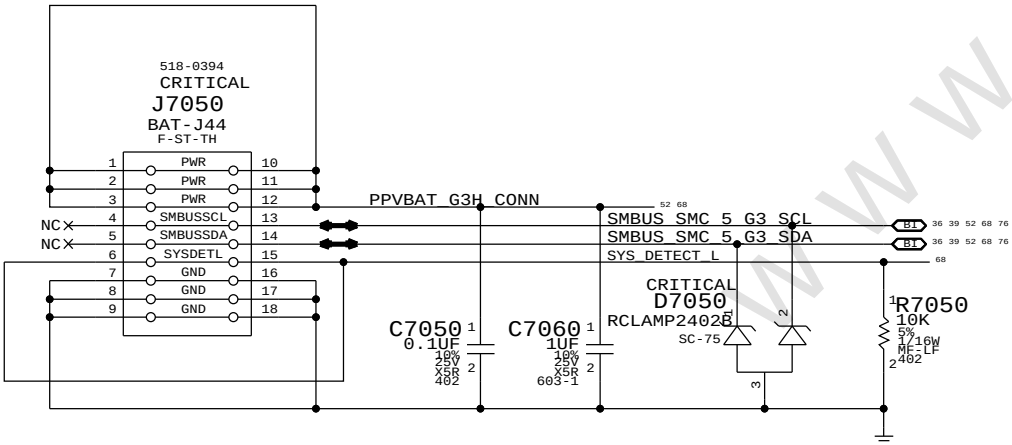
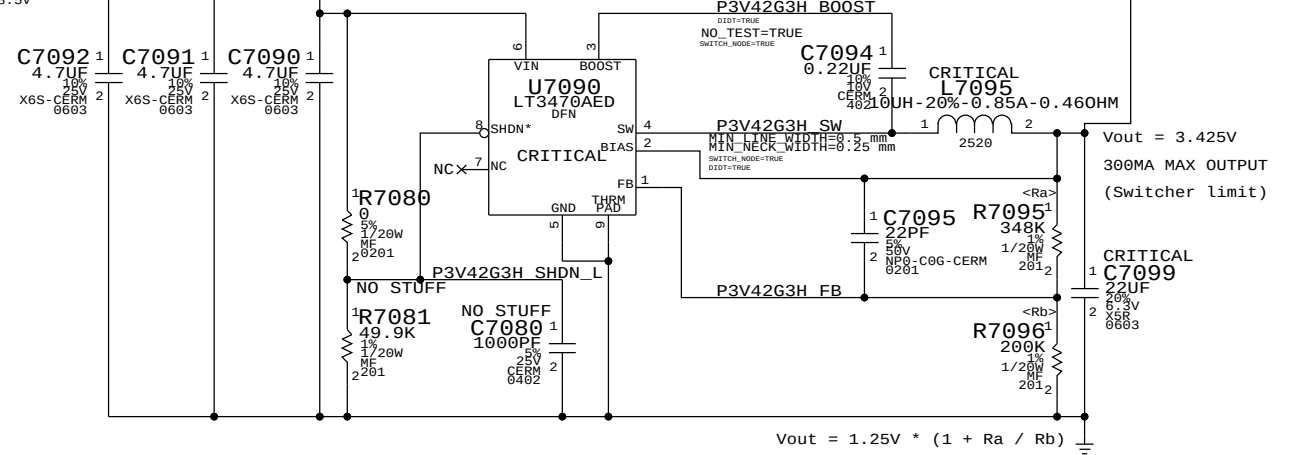
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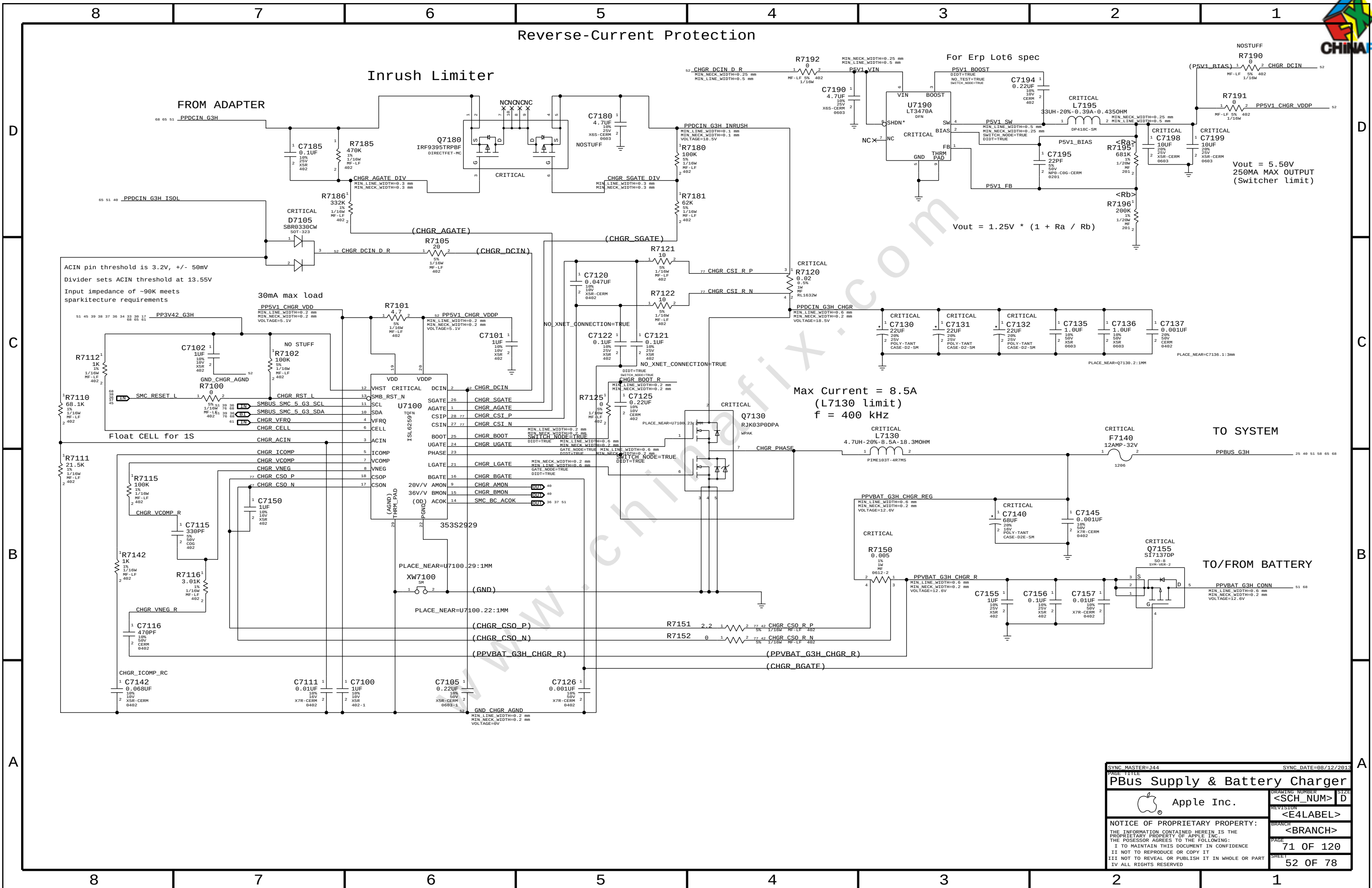


3.425V "G3Hot" Supply

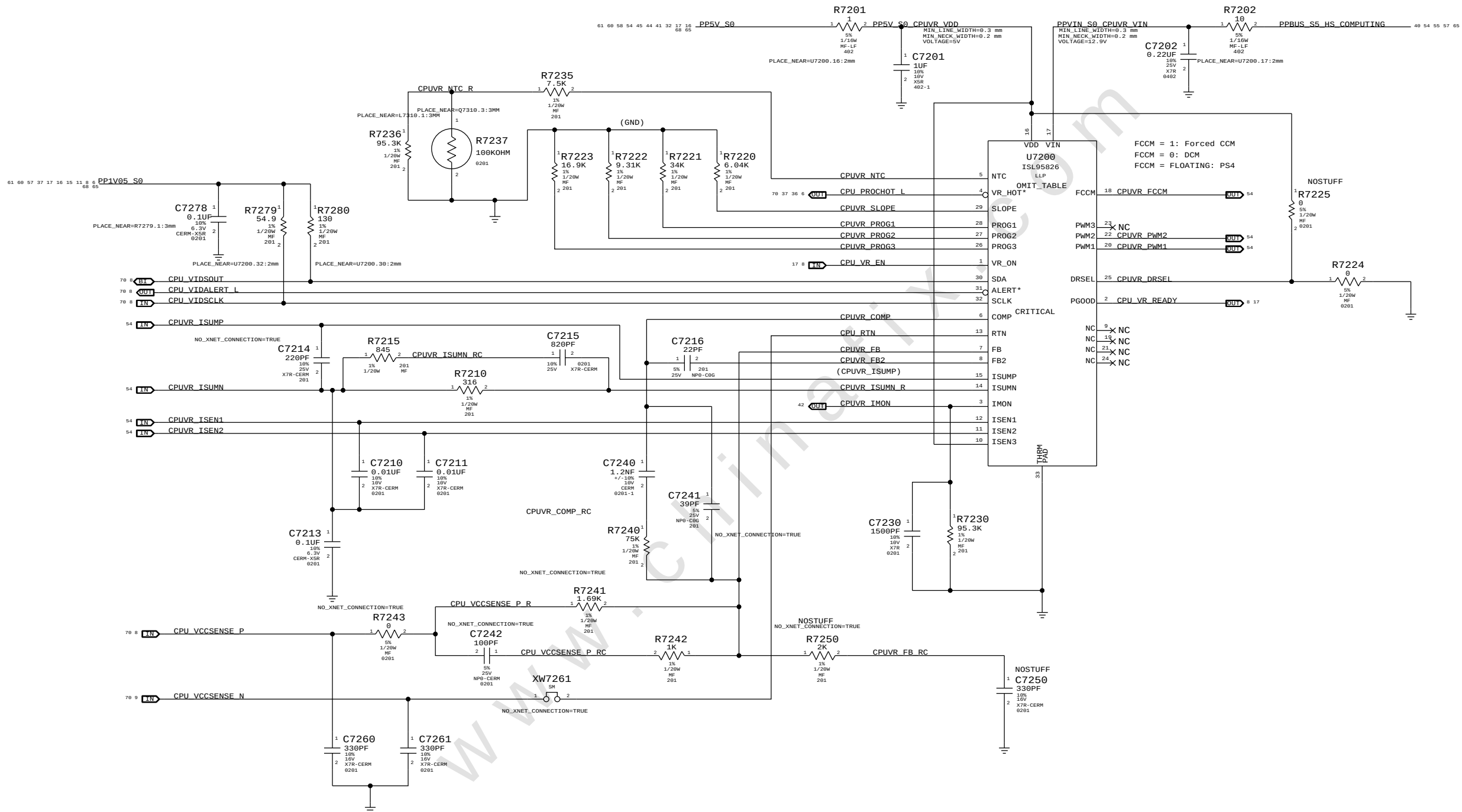
Supply needs to guarantee 3.31V delivered to SMC VRef generator



DC-In & Battery Connectors	
Apple Inc.	<SCH_NUM> D
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	70 OF 120
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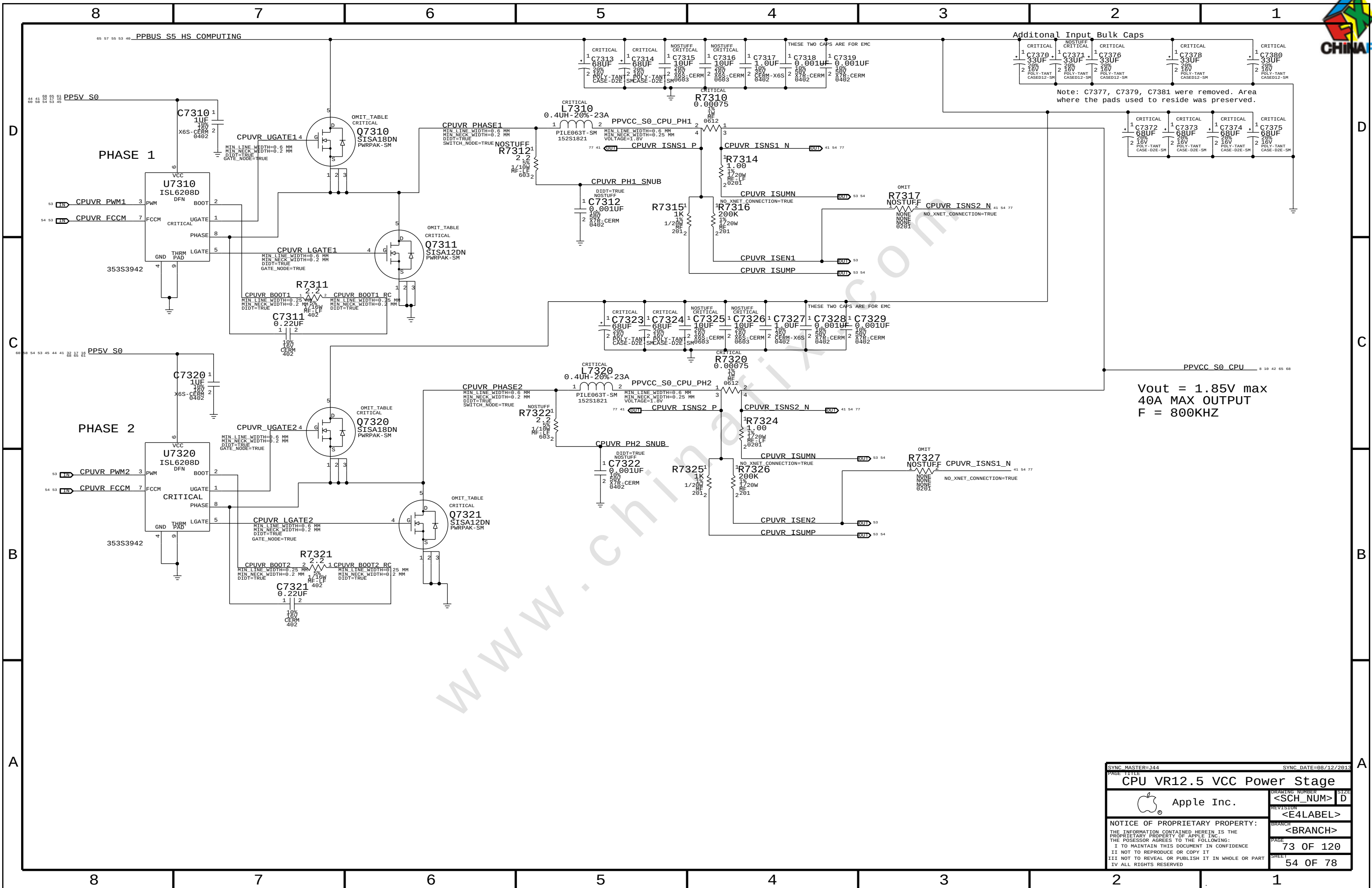


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S4170	1	IC, ISL95826R6200, PWM, PG000, SCREN, 32P, QFN	U7200	CRITICAL	



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CPU VR12.6 VCC Regulator IC			
 Apple Inc.		DRAWING NUMBER	SIZE
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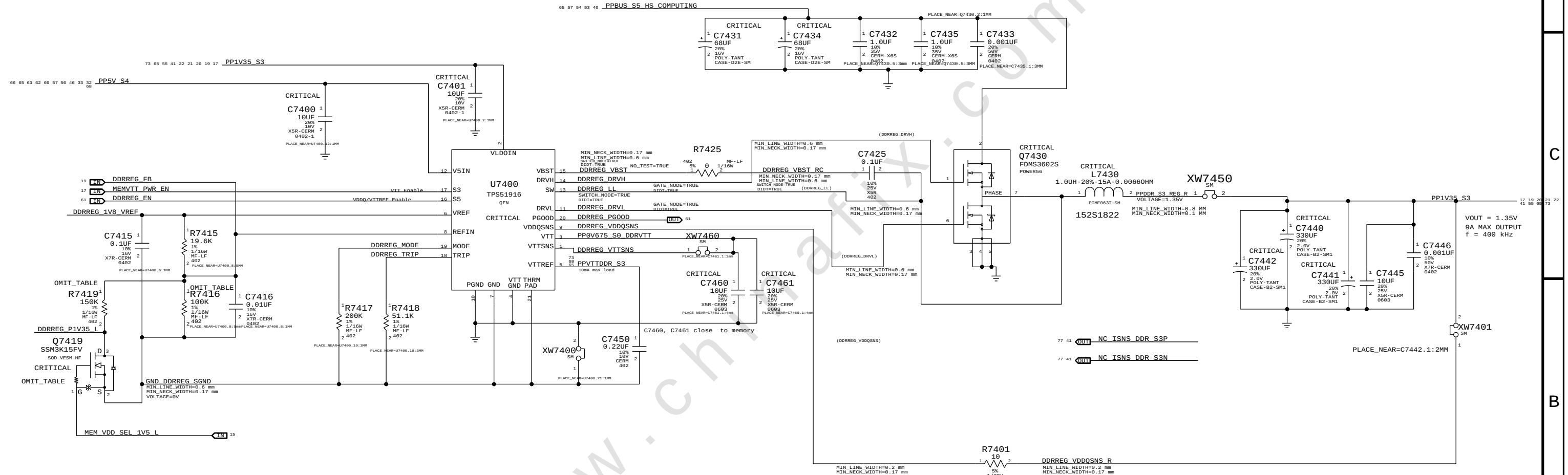
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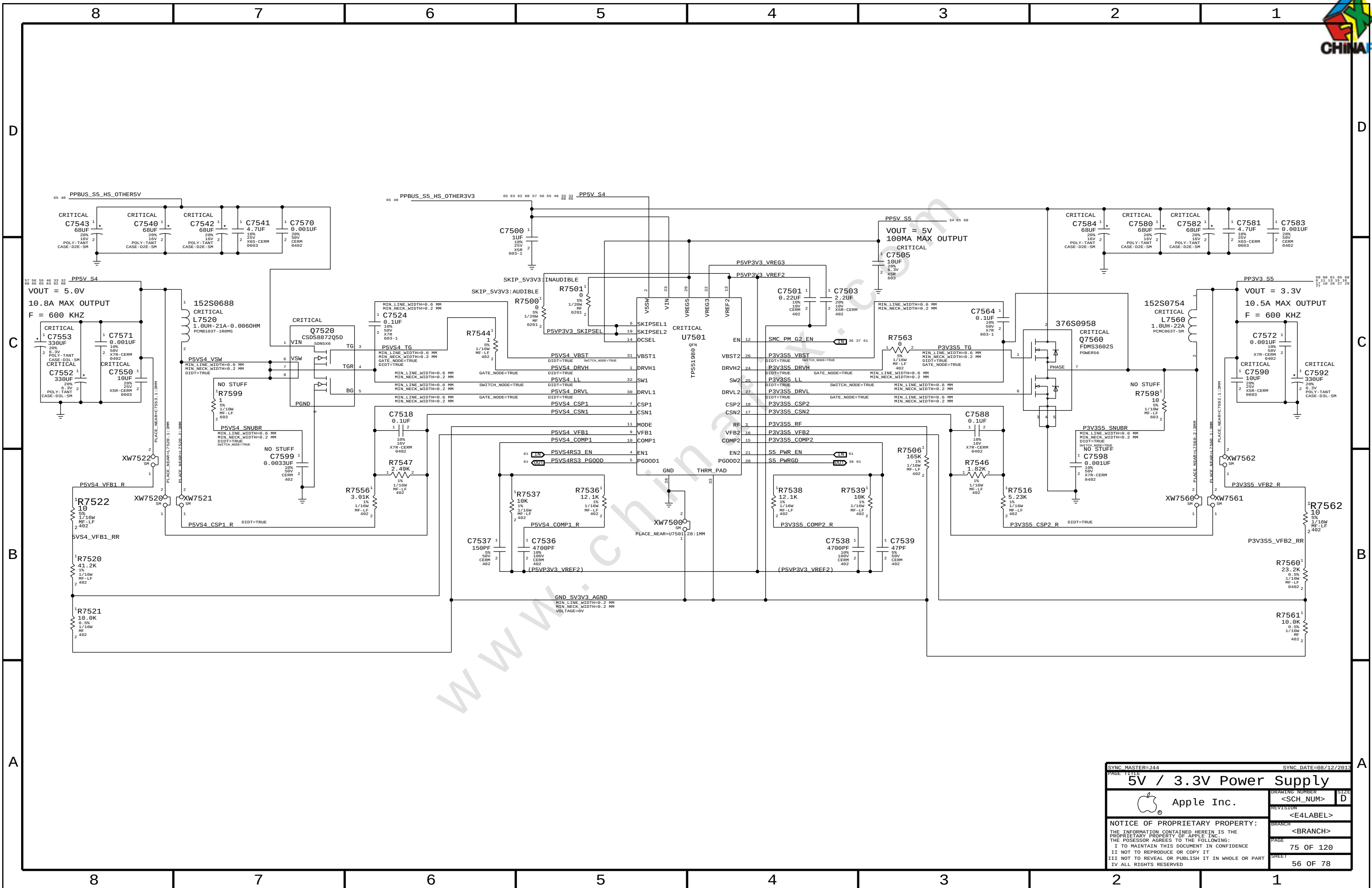
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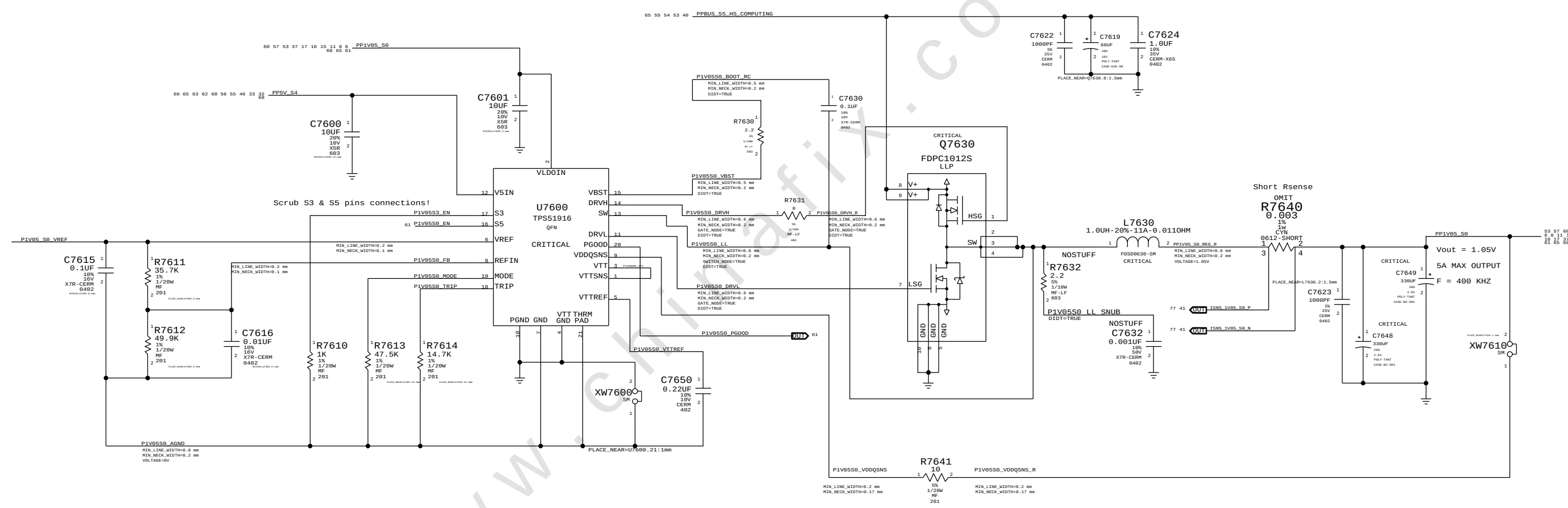


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0411	1	RES, MTL FILM, 1/16W, 100K, 1, 0402, SMD, LF	R7416	CRITICAL	PPDDR:1V5
114S0391	1	RES, MTL FILM, 1/16W, 60.4K, 1, 0402, SMD, LF	R7416	CRITICAL	PPDDR:1V35
376S0612	1	MOSFET, N-CH, 30V, 100MA, 7.00MH, SOT-723, HF	Q7419	CRITICAL	PPDDR:1V5
114S0428	1	RES, MTL FILM, 1/16W, 150K, 0402, SMD, LF	R7419	CRITICAL	PPDDR:1V5





1.05V S0 Regulator



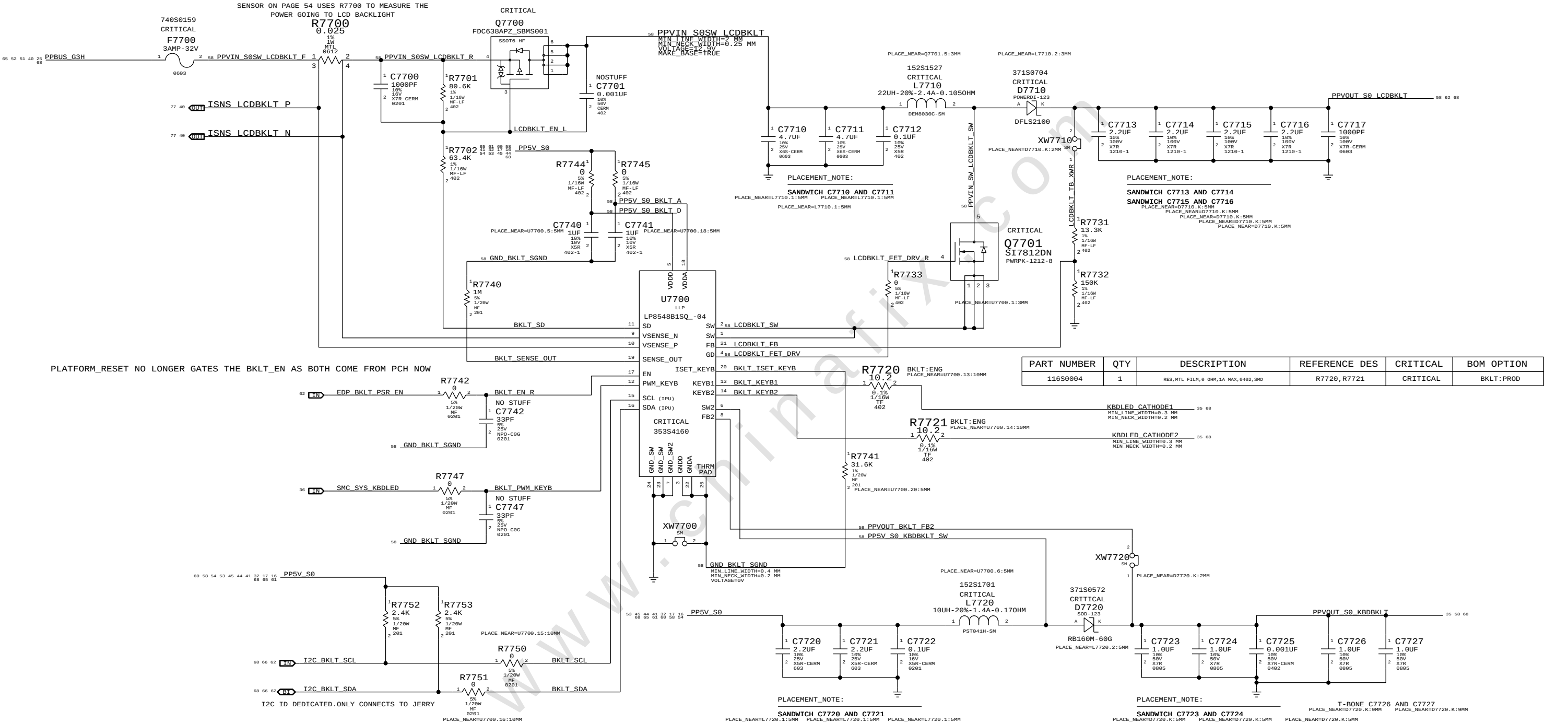
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1.05V S0 Power Supply			
 Apple Inc.	DRAWING NUMBER		SIZE
	<SCH_NUM>		D
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Page Notes

Power aliases required by this page:
- =PPVIN_S0SW_LCDBKLT_FET (9-12.6V LCD BACKLIGHT INPUT)
- =PPSV_S0_BKLT (5V BACKLIGHT DRIVER INPUT)
- =PPSV_S0SW_KBDLED (5V KEYBOARD BACKLIGHT INPUT)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FLTH,0 OHM,1A MAX,0402,SMD	R7720,R7721	CRITICAL	BKLT:PROD

PPBUS LINE WIDTHS	LCD BKLT LINE WIDTHS	KBD BKLT LINE WIDTHS
PPSV_S0_BKLT_A MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=5V	PPVIN_S0SW_LCDBKLT_F MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=12.9V	LCDBKLT_FET_DRV_R MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=5V GATE_NODE=TRUE D1D1=TRUE
PPSV_S0_BKLT_D MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=5V	PPVIN_S0SW_LCDBKLT_R MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=12.9V	PPVIN_SW_LCDBKLT_SW MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=5V SWITCH_NODE=TRUE D1D1=TRUE
	PPVIN_S0SW_LCDBKLT_FET MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=12.9V	PPVOUT_S0_LCDBKLT MIN_LINE_WIDTH=0.5 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=5V
	PPVIN_S0SW_LCDBKLT MIN_LINE_WIDTH=2 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=12.9V	PPVOUT_BKLT_FB MIN_LINE_WIDTH=0.4 MM MIN_NECK_WIDTH=0.25 MM VOLTAGE=5V

SYNC MASTER=144

SYNC DATE=08/12/2013

LCD AND KBD BKLT DRIVER

Apple Inc.

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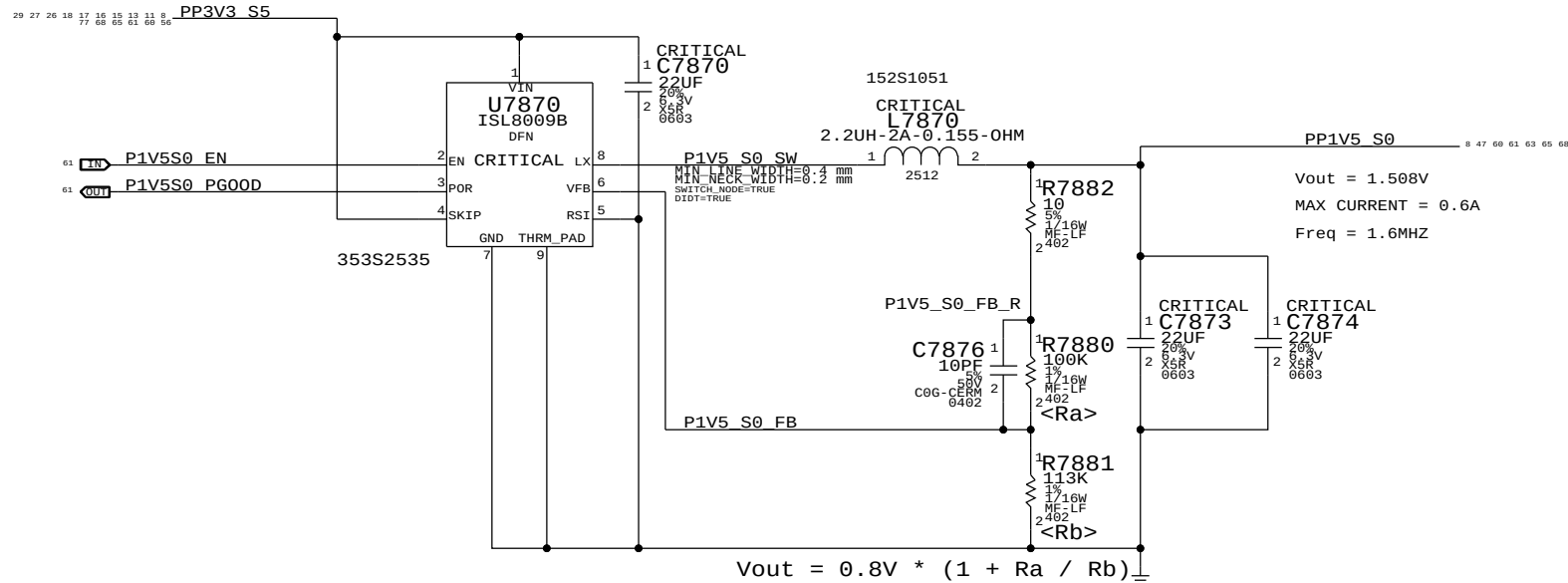
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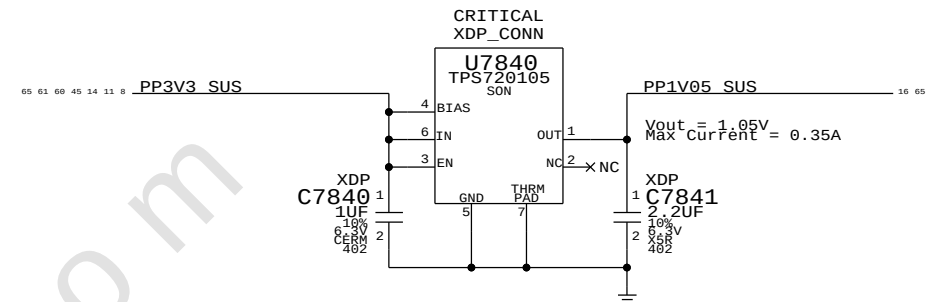


1.5V S0 Switcher



1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



SYNC MASTER=J44		SYNC DATE=08/12/2013	
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Misc Power Supplies			
 Apple Inc.		DRAWING NUMBER	SIZE
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Loading specs per J41/43_PowerBudget_Riviera_rev0.99e

1.5V S0 Audio Switch (BYPASSED)

3.3V SUS Switch

1.05V PCH HSI0 Switch

3.3V S4 Switch

3.3V SSD Switch

3.3V S3 Switch

3.3V S0 Switch

3.3V Sensor Switch

5V S0 Switch

REMOVED THE ANALOG POWER GATE AS SLG5AP1471 SHOULD BE AVAILABLE BY THEN

Part	TPS22924C
Type	Load Switch
R(on) @ 1.8V	19.6 mOhm Typ 21.8 mOhm Max
Current	2A Max

Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

Part	TPS22920
Type	Load Switch
R(on) @ 3.6V	5.5 MOHM TYP 8.8 MOHM MAX
Current	4A MAX

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 25C	7.8 mOhm Typ 8.5 mOhm Max
Current	5.3A Max

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

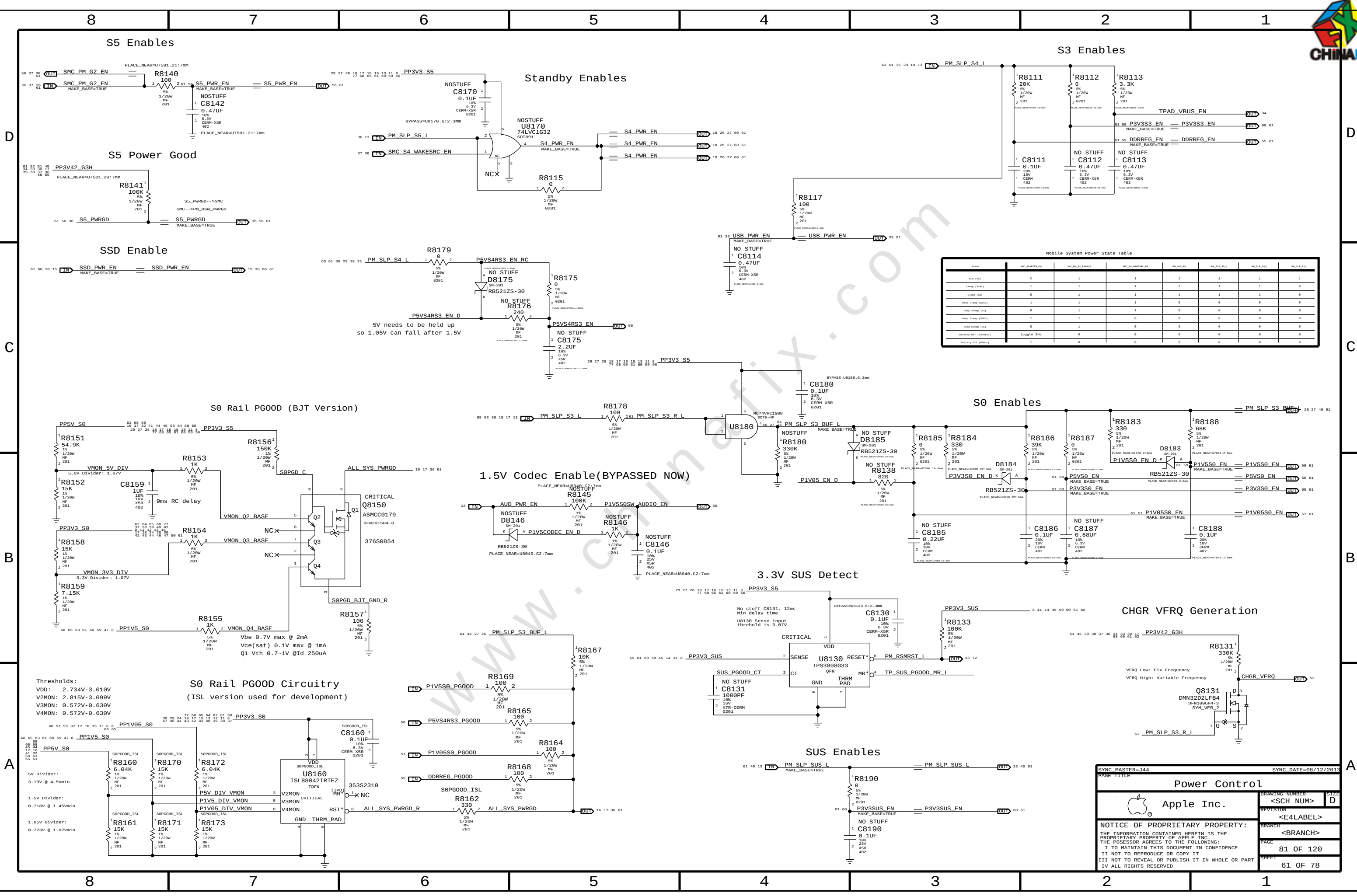
Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

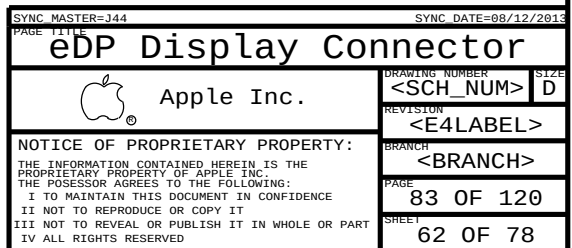
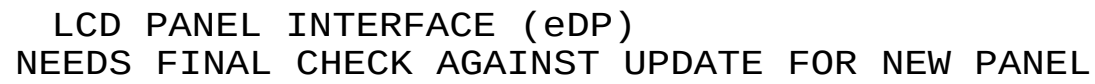
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Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

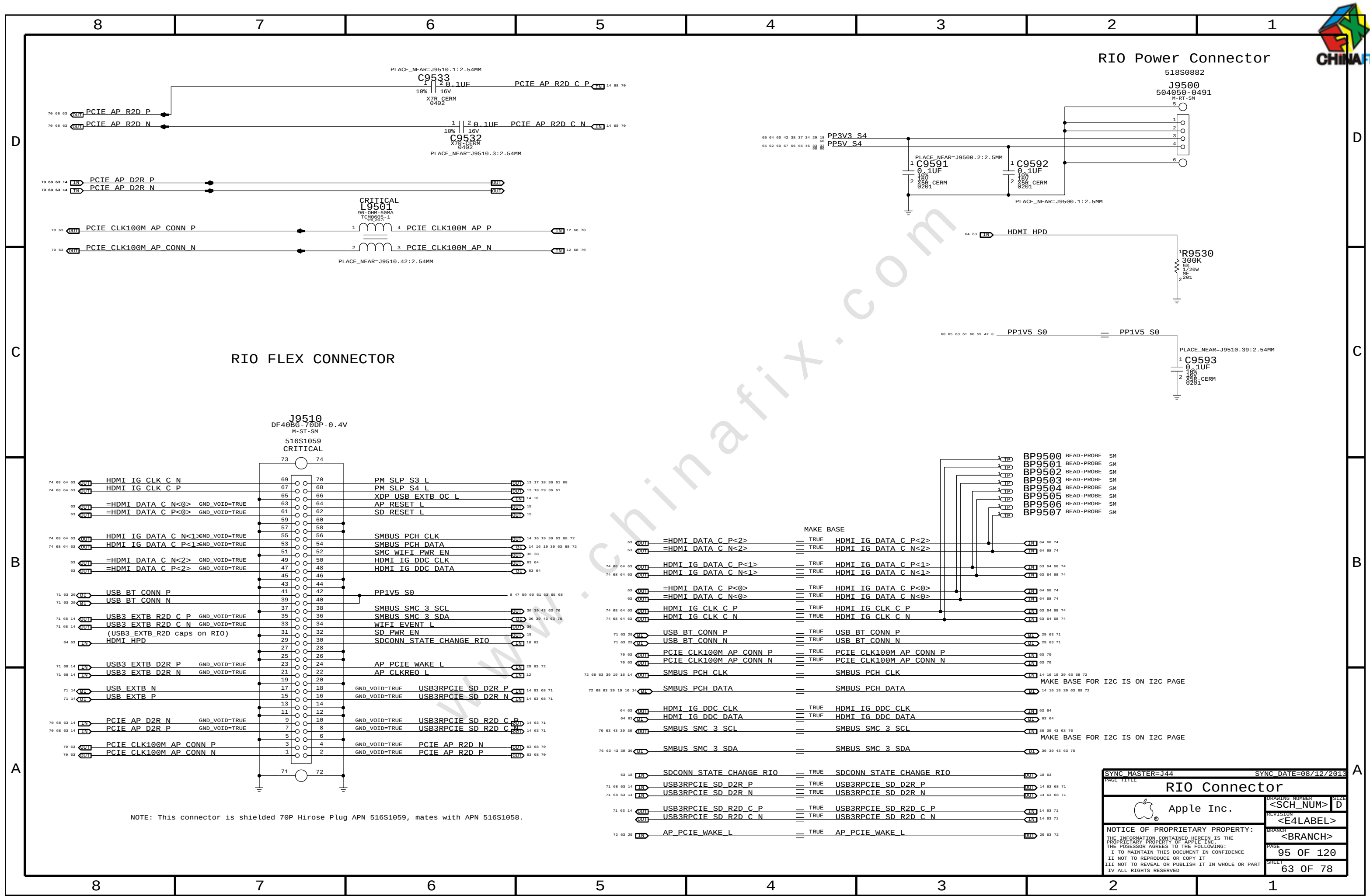
Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

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Power FETS	
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NOTE: This connector is shielded 70P Hirose Plug APN 516S1059, mates with APN 516S1058.

RIO Power Connector

518S0882

J9500

504050-0491

M-RT-SM

5

1

2

3

4

6

PLACE_NEAR=J9500.2:2.5MM

C9591

0.1UF

18V

0201

2

18V

0201

1

18V

0201

2

18V

0201

1

18V

0201

2

18V

0201

1

18V

0201

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SYNC MASTER=J44

SYNC DATE=08/12/2013

RIO Connector



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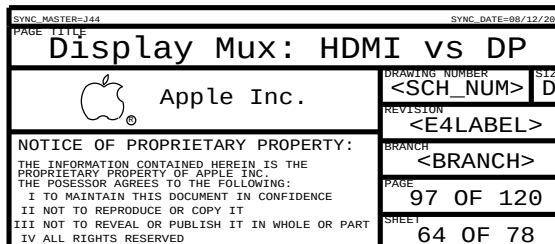
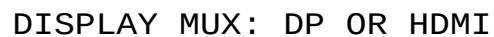
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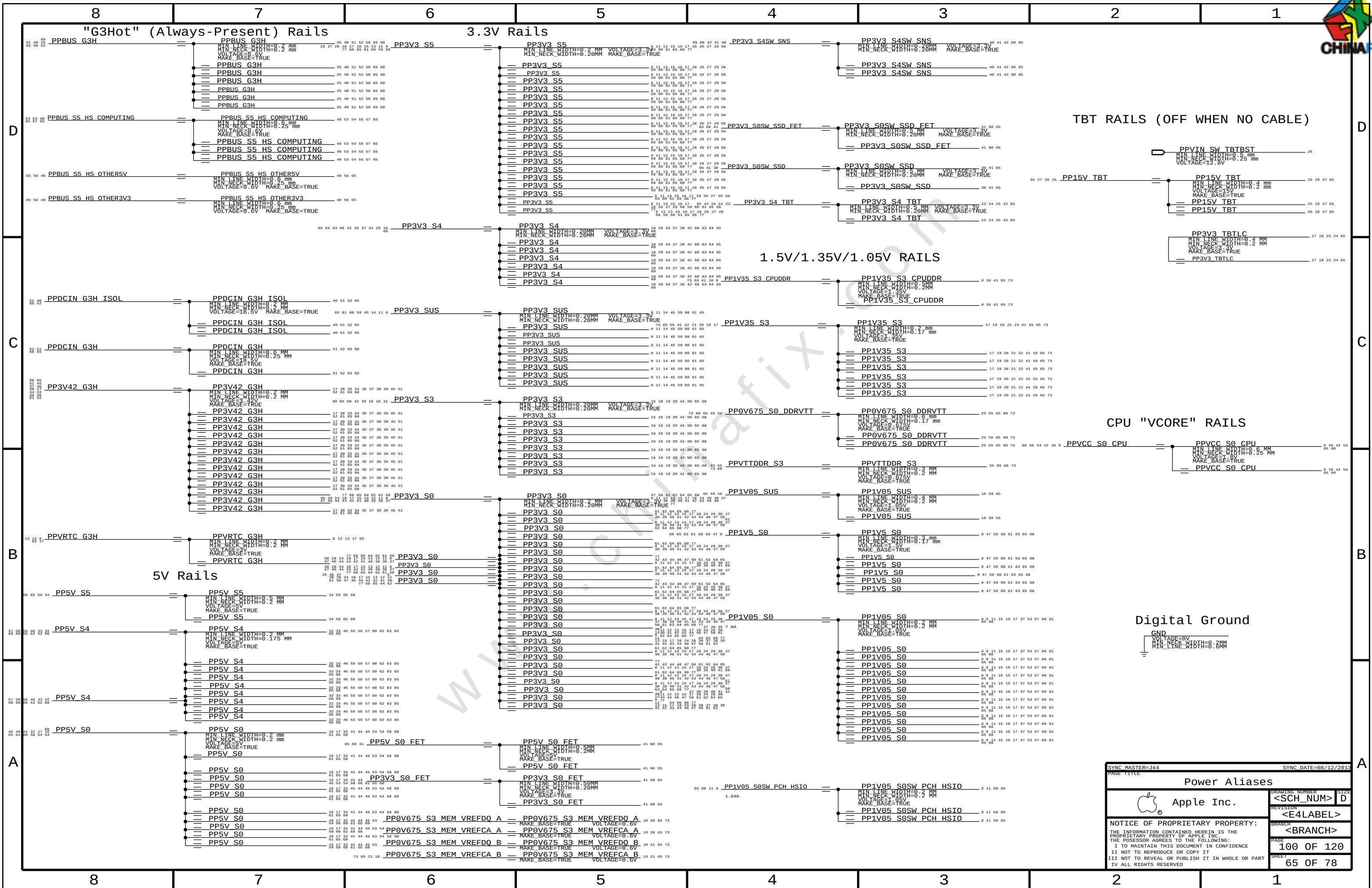
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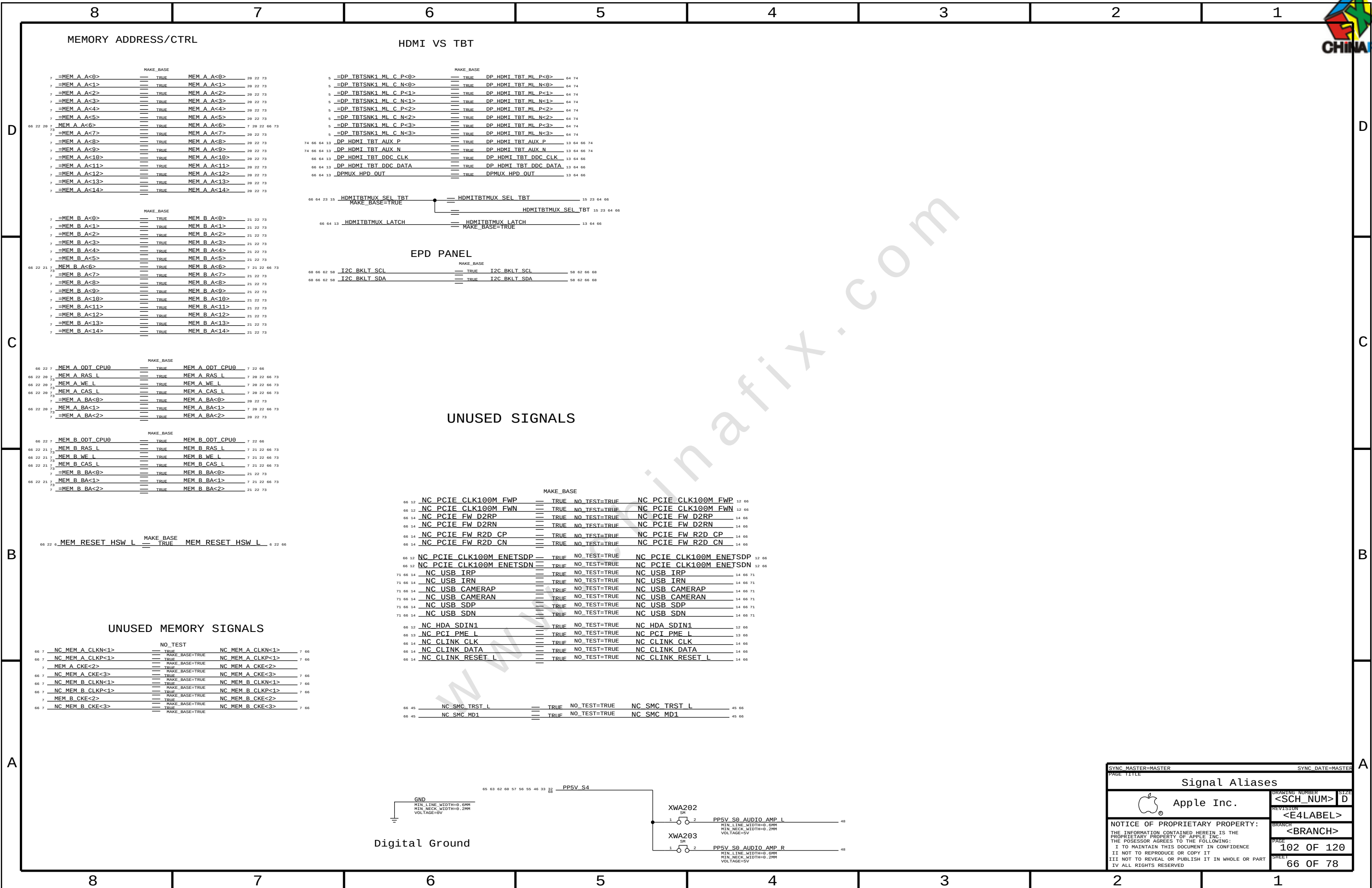
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Memory Bit/Byte Swizzle

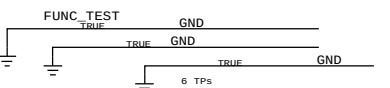
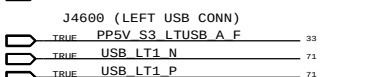
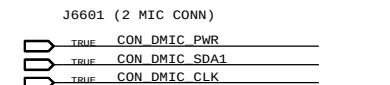
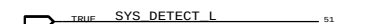
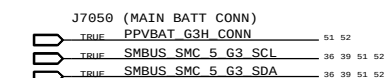
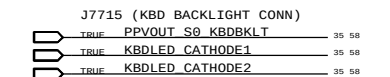
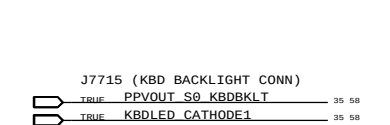
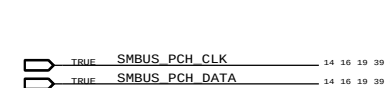
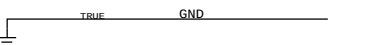
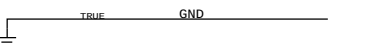
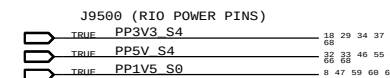
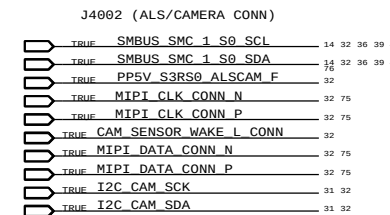
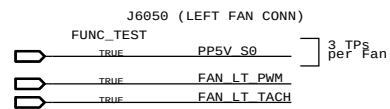
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73 7	TRUE	MEM B DQS N<5>	=MEM B DQS N<7>
73 67 21 7	TRUE	MEM B DQS P<6>	MEM B DQS P<6>
73 67 21 7	TRUE	MEM B DQS N<6>	MEM B DQS N<6>
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73 7	TRUE	MEM B DQS N<7>	=MEM B DQS N<6>

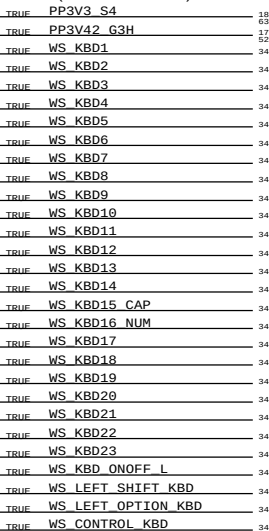
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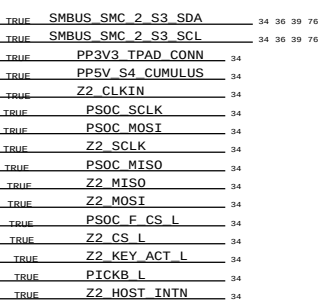
Functional Test Points



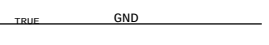
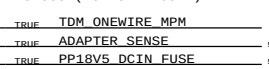
J4813 (KEY BOARD CONN)



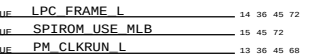
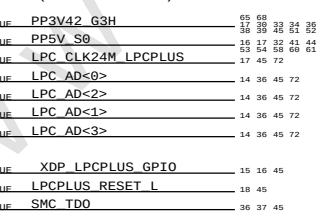
J4800 (TPAD CONN)



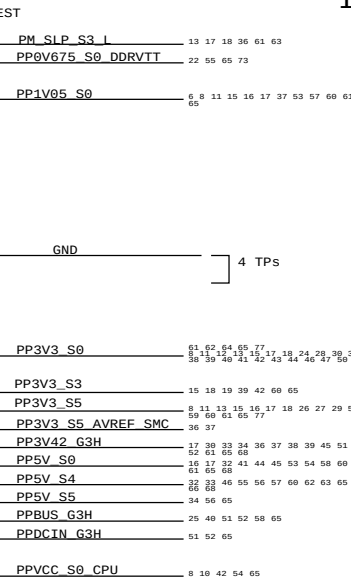
J7000 (DC POWER CONN)



J6100 (LPC + SPI CONN)



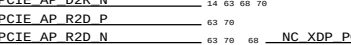
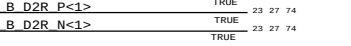
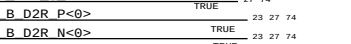
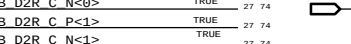
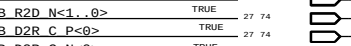
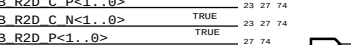
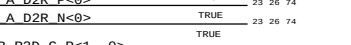
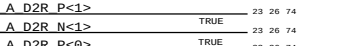
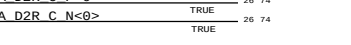
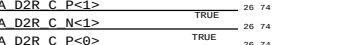
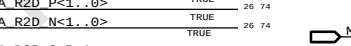
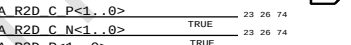
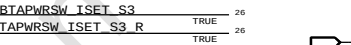
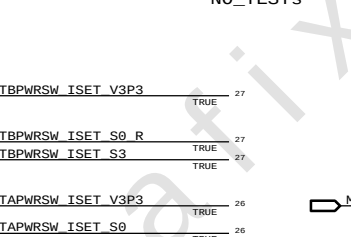
POWER RAILS



PPVTTDDR S3



NO_TESTS

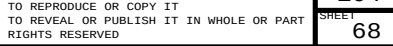
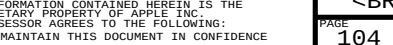
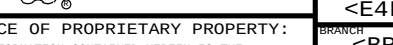
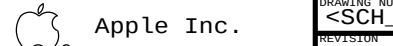
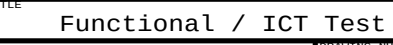
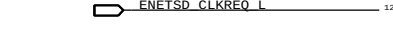
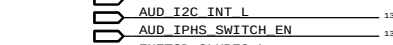
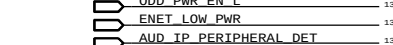
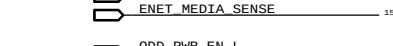
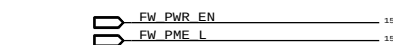
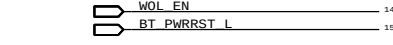
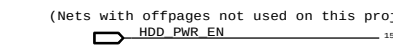
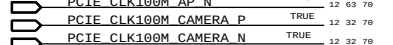
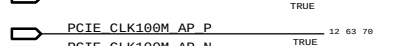
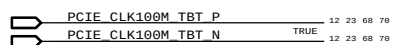
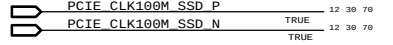
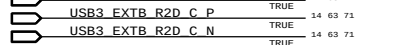
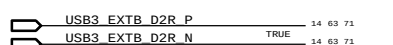
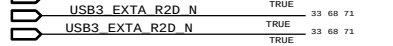
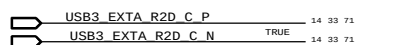
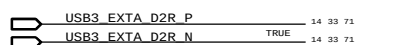
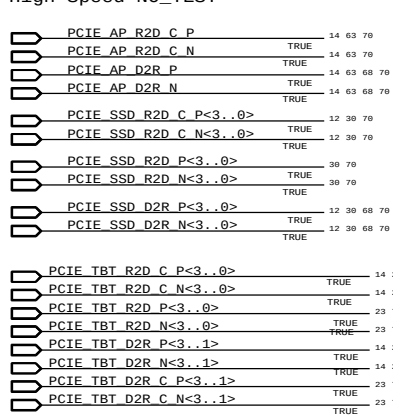


ICT Test Points



NC NO_TEST

High Speed NO_TEST



Functional / ICT Test

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SYNC DATE=08/12/2013

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J44 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS				BOARD AREAS		BOARD UNITS (ALL 0.1MM)	ALLEGRO OVERSIGHT
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO_TYPE, BGA, P65BGA, BGA_MEM		MM	16.5

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP, BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2, ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP, BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2, ISL11	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP, BOTTOM	Y	0.125 MM	0.125 MM		0.155 MM	0.155 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2, ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP, BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2, ISL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1T01_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1X_DIELECTRIC	TOP, BOTTOM	0.058 MM	?
1X_DIELECTRIC	ISL3, ISL4, ISL9, ISL10	0.053 MM	?
1X_DIELECTRIC	ISL1, ISL5, ISL6, ISL7, ISL8, ISL11	0.101 MM	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

SYNC MASTER=J44		SYNC DATE=08/12/2013	
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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CPU_VCCSENSE	*	25 MIL	?	CPU_08MIL	*	0.203 MM	?
				CPU_12MIL	*	0.305 MM	?
				CPU_18MIL	*	0.457 MM	?
				CPU_25MIL	*	0.635 MM	?

PCI Express Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE_2SAME	*	=3X_DIELECTRIC	?	PCIE_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?	PCIE_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIE_20THER	*	=4X_DIELECTRIC	?	PCIE_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?	PCIE_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIECLK_20THER	*	=7X_DIELECTRIC	?	PCIECLK_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	*	*	PCIE_20THER
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_20THER
PCIE_TX	*_RX	*	PCIE_TXRX
PCIE_RX	*_TX	*	PCIE_TXRX

CPU Signal Properties

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
XDP_TCK0	CPU_45S	CPU_18MTL	XDP_CPU_TCK 6 16
XDP_TCK0	CPU_45S	CPU_18MTL	PCH_JTAGX 12 16
XDP_TCK1	CPU_45S	CPU_18MTL	XDP_PCH_TCK 12 16
XDP_TDO	CPU_45S		XDP_CPU_TDO 6 16
XDP_TDO	CPU_45S		XDP_PCH_TDO 12 16
XDP_TDI	CPU_45S		XDP_CPU_TDI 6 16
XDP_TDI	CPU_45S		XDP_PCH_TDI 12 16
XDP_TMS	CPU_45S		XDP_CPU_TMS 6 16
XDP_TMS	CPU_45S		XDP_PCH_TMS 12 16
XDP_TRST_L	CPU_45S		XDP_TRST_L 16
XDP_TRST_L	CPU_45S		XDP_CPUPCH_TRST_L 6 12 16
XDP_PRDY_L	CPU_45S		XDP_CPU_PRDY_L 6 16
XDP_FREQ_L	CPU_45S		XDP_CPU_FREQ_L 6 16
CPU_VCCST_PWRGD	CPU_45S	CPU_08MTL	CPU_VCCST_PWRGD 8 16 17
CPU_VCCST_PWRGD	CPU_45S	CPU_08MTL	XDP_CPU_VCCST_PWRGD 16
CPU_BPM	CPU_45S	CPU_08MTL	XDP_BPM_L<1..0> 6 16
CPU_BPM_TP	CPU_45S		XDP_BPM_L<7..2> 6 16
CPU_RCOMP_SM	CPU_27P4S	CPU_25MTL	CPU_SM_RCOMP<2..0> 5
CPU_RCOMP_EDP	CPU_27P4S	CPU_25MTL	MCP_EDP_RCOMP 5
CPU_RCOMP_OPT	CPU_27P4S	CPU_12MTL	CPU_OPT_RCOMP 6
CPU_PROCHOT	CPU_45S	CPU_08MTL	CPU_PROCHOT_L 6 36 37 53
CPU_PROCHOT	CPU_45S	CPU_08MTL	CPU_PROCHOT_R_L 6
CPU_CATERR	CPU_45S	CPU_08MTL	CPU_CATERR_L 6 36
CPU_VIDALERT	CPU_45S	CPU_18MTL	CPU_VIDALERT_L 8 53
CPU_VIDALERT	CPU_45S	CPU_18MTL	CPU_VIDALERT_R_L 8
CPU_VIDSCLK	CPU_45S	CPU_18MTL	CPU_VIDSCLK 8 53
CPU_VIDSCLK	CPU_45S	CPU_18MTL	CPU_VIDSCLK_R 8
CPU_VIDSOUT	CPU_45S	CPU_18MTL	CPU_VIDSOUT 8 53
CPU_VIDSOUT	CPU_45S	CPU_18MTL	CPU_VIDSOUT_R 8
CPU_PECI	CPU_45S	CPU_18MTL	CPU_PECI 6 37
CPU_PECI	CPU_45S	CPU_18MTL	CPU_PECI_R 36 37
CPU_PECI_SMC	CPU_45S	CPU_18MTL	SMC_PECI_L 36 37
CPU_PECI_SMC	CPU_45S	CPU_18MTL	SMC_PECI_L_R 37
CPU_CFG	CPU_45S		CPU_CFG<19..11> 6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<10..8> 6 16
CPU_CFG	CPU_45S		CPU_CFG<7..5> 6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<4> 6 16
CPU_CFG_3	CPU_45S		CPU_CFG<3> 6 16
CPU_CFG	CPU_45S		CPU_CFG<2> 6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<1..0> 6 16
CPU_MEM_RESET	CPU_45S	CPU_08MTL	MEM_RESET_L 20 21 22
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P 8 53
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N 9 53

PCI Express Properties

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<3..1> 12 30 68
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<3..1> 12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<0> 12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<0> 12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_P<3..0> 12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_N<3..0> 12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_P<3..0> 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_N<3..0> 30 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<0> 14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<0> 14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<0> 23
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<0> 23
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<3..1> 14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<3..1> 14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<3..1> 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<3..1> 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_P<3..0> 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_P<3..0> 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_N<3..0> 14 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_N<3..0> 14 23 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_P 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_N 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_P 14 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_N 14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_P 14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_N 14 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P 63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N 63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P 12 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N 12 63 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P 12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N 12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P 31 32
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N 31 32
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P 12 30 68
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N 12 30 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P 12 23 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N 12 23 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_P 14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_N 14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_P 31 32
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_N 31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_P 31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_N 31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_P 14 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_N 14 32

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CPU & PCIe Constraints			
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USB 2 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAIS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?	USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAIS	*	=6X_DIELECTRIC	?	USB_RBIAIS	TOP,BOTTOM	=10X_DIELECTRIC	?

USB 3 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?	USB3_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?	USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_20THER	*	=4X_DIELECTRIC	?	USB3_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	*	*	USB3_20THER
USB3_*	=SAME	*	USB3_2SAME
USB3_TX	*_RX	*	USB3_TXRX
USB3_RX	*_TX	*	USB3_TXRX

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

SATA Interface Constraints (Not Used)

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?	SATA_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?	SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_20THER	*	=4X_DIELECTRIC	?	SATA_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	*	*	SATA_20THER
SATA_*	=SAME	*	SATA_2SAME
SATA_TX	*_RX	*	SATA_TXRX
SATA_RX	*_TX	*	SATA_TXRX

USB Constraints

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
USB_BT	USB_85D	USB	USB BT P
USB_BT	USB_85D	USB	USB BT N
USB_BT	USB_85D	USB	USB BT CONN P
USB_BT	USB_85D	USB	USB BT CONN N
USB_EXT_A	USB_85D	USB	USB EXT_A P
USB_EXT_A	USB_85D	USB	USB EXT_A N
DEFAULT	DEFAULT	DEFAULT	SMC DEBUGPRT_RX_L
DEFAULT	DEFAULT	DEFAULT	SMC DEBUGPRT_TX_L
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_P
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_N
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_F_P
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_F_N
USB_EXT_A	USB_85D	USB	USB_LT1_P
USB_EXT_A	USB_85D	USB	USB_LT1_N
USB_EXT_B	USB_85D	USB	USB_EXTB_P
USB_EXT_B	USB_85D	USB	USB_EXTB_N
USB_TPAD	USB_85D	USB	USB_TPAD_P
USB_TPAD	USB_85D	USB	USB_TPAD_N
USB_TPAD	USB_85D	USB	USB_TPAD_R_P
USB_TPAD	USB_85D	USB	USB_TPAD_R_N
USB3_EXT_A_D2R	USB_85D	USB3_RX	USB3_EXT_A D2R_P
USB3_EXT_A_D2R	USB_85D	USB3_RX	USB3_EXT_A D2R_N
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_P
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_N
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_C_P
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_C_N
USB3_EXT_B_D2R	USB_85D	USB3_RX	USB3_EXTB D2R_P
USB3_EXT_B_D2R	USB_85D	USB3_RX	USB3_EXTB D2R_N
USB3_EXT_B_R2D	USB_85D	USB3_TX	USB3_EXTB R2D_C_P
USB3_EXT_B_R2D	USB_85D	USB3_TX	USB3_EXTB R2D_C_N
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD D2R_P
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD D2R_N
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD R2D_C_P
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD R2D_C_N
USB_NC	USB_85D	USB	NC_USB_IRP
USB_NC	USB_85D	USB	NC_USB_IRN
USB_NC	USB_85D	USB	TP_USB_5P
USB_NC	USB_85D	USB	TP_USB_5N
USB_NC	USB_85D	USB	NC_USB_SDP
USB_NC	USB_85D	USB	NC_USB_SDN
USB_NC	USB_85D	USB	NC_USB_CAMERAP
USB_NC	USB_85D	USB	NC_USB_CAMERAN
PCH_USB_RBIAIS	PCH_USB_RBIAIS	USB_RBIAIS	PCH_USB_RBIAIS
SATA_85D	SATA_85D	SATA_RX	DUMMY_SATA_D2R_P
SATA_85D	SATA_85D	SATA_RX	DUMMY_SATA_D2R_N
SATA_85D	SATA_85D	SATA_TX	DUMMY_SATA_R2D_P
SATA_85D	SATA_85D	SATA_TX	DUMMY_SATA_R2D_N
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R

Notes:
This is here to keep the SATA rules.

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USB Constraints			
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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2X_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2X_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPT_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
PCH_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_12MIL	*	0.305 MM	?
PCH_15MIL	*	0.381 MM	?
PCH_18MIL	*	0.457 MM	?
PCH_20MIL	*	0.508 MM	?

PCH Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 36 45 68
LPC_AD	LPC_45S	LPC	LPC_FRAME_L	14 36 45 68
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 36 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS	17 45 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS_R	12 17
SMBUS_PCH	SMB_45S	SMR	SMBUS_PCH_CLK	14 16 19 39 63 68
SMBUS_PCH	SMB_45S	SMR	SMBUS_PCH_DATA	14 16 19 39 63 68
SML_PCH_0	SMB_45S	SMR	SML_PCH_0_CLK	14 39
SML_PCH_0	SMB_45S	SMR	SML_PCH_0_DATA	14 39
	SMB_45S	SMR	SMBUS_SMC_1_S0_SCL	14 32 36 39 43 68 76
	SMB_45S	SMR	SMBUS_SMC_1_S0_SDA	14 32 36 39 43 68 76
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 47
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 47
HDA_SYNC	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST	HDA_45S	HDA	HDA_RST_R_L	12
HDA_RST	HDA_45S	HDA	HDA_RST_L	12 47
HDA_SDIN	HDA_45S	HDA	HDA_SDIN0	12 47 68
HDA_SDIN	HDA_45S	HDA	CS4208_HDA_SDOUT0_R	47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12 47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT_R	12 17
SPT_MLB	SPT_45S	SPT	SPI_ALT_CLK	45
SPT_MLB	SPT_45S	SPT	SPI_CLK	45
SPT_MLB	SPT_45S	SPT	SPI_CLK_R	14 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_CLK	45
SPT_MLB	SPT_45S	SPT	SPI_SMC_CLK	36 45
SPT_MLB	SPT_45S	SPT	SPI_ALT_CS_L	45
SPT_MLB	SPT_45S	SPT	SPI_CS0_L	45
SPT_MLB	SPT_45S	SPT	SPI_CS0_R_L	14 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_CS_L	45
SPT_MLB	SPT_45S	SPT	SPI_SMC_CS_L	36 45
SPT_MLB	SPT_45S	SPT	SPI_ALT_MISO	45
SPT_MLB	SPT_45S	SPT	SPI_MISO	14 45
SPT_MLB	SPT_45S	SPT	SPI_MISO_R	45
SPT_MLB	SPT_45S	SPT	SPI_MLB_MISO	45
SPT_MLB	SPT_45S	SPT	SPI_SMC_MISO	36 45
SPT_MLB	SPT_45S	SPT	SPI_ALT_MOSI	45
SPT_MLB	SPT_45S	SPT	SPI_MOSI	45
SPT_MLB	SPT_45S	SPT	SPI_MOSI_R	14 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_MOSI	45
SPT_MLB	SPT_45S	SPT	SPI_SMC_MOSI	36 45
SPT_MLB_I02	SPT_45S	SPT	SPI_I0<2>	14 45
SPT_MLB_I02	SPT_45S	SPT	SPIROM_WP_L	45
SPT_MLB_I03	SPT_45S	SPT	SPI_I0<3>	14 45
SPT_MLB_I03	SPT_45S	SPT	SPIROM_HOLD_L	45
SPT_MLB_I03	SPT_45S	SPT	SPIROM_USE_MLB	15 45 68
PCH_RTCX	PCH_45S	PCH_15MTI	PCH_CLK32K_RTCX1	12 17
PCH_SRTCST	PCH_45S	PCH_15MTI	PCH_SRTCST_L	12
PCH_RTCRST	PCH_45S	PCH_15MTI	RTC_RESET_L	12
PCH_THRMTRIP	PCH_45S	PCH_18MTI	PM_THRMTRIP_L	15 37
PCH_THRMTRIP	PCH_45S	PCH_18MTI	PM_THRMTRIP_R_L	37
	PCH_45S	PCH_15MTI	PCH_INTRUDER_L	12
	PCH_45S	PCH_15MTI	PCH_INTVRMEN	12
	PCH_45S	PCH_15MTI	PCH_DSWVRMEN	13
	PCH_45S	PCH_15MTI	PM_RSMRST_L	13 61
	PCH_45S	PCH_15MTI	PM_SYSRST_L	13 17 36 68
	PCH_45S	PCH_15MTI	XDP_DBRESET_L	16 17
	PCH_45S	PCH_15MTI	PM_PCH_SYS_PWROK	13 16 17 36
	PCH_45S	PCH_15MTI	XDP_SYS_PWROK	16
	PCH_45S	PCH_15MTI	SYS_PWROK_R	17
	PCH_45S	PCH_15MTI	PM_PCH_PWROK	13 17
	PCH_45S	PCH_15MTI	PM_S0_PG00D	17
	PCH_45S	PCH_15MTI	SMC_DELAYED_PWRGD	17 24 25 36 37
	PCH_45S	PCH_15MTI	PM_DSW_PWRGD	13 36
	PCH_45S	PCH_15MTI	PM_PWRBTN_L	13 16 36
	PCH_45S	PCH_15MTI	XDP_CPU_PWRBTN_L	16
	PCH_45S	PCH_15MTI	PCIE_WAKE_L	13 29 31
	PCH_45S	PCH_15MTI	AP_PCIE_WAKE_L	29 63
	PCH_45S	PCH_15MTI	CAM_PCIE_WAKE_L	31
	PCH_45S	PCH_15MTI	TBT_CIO_PLUG_EVENT_L	15 18 23
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALIN	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALOUT	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALOUT_R	17
PCH_RCOMP_PCIE	PCH_27P4S	PCH_12MTI	PCH_PCIE_RCOMP	14
PCH_RCOMP_OPT	PCH_27P4S	PCH_12MTI	PCH_OPI_COMP	15
PCH_RCOMP_SATA	PCH_27P4S	PCH_12MTI	PCH_SATA_RCOMP	12

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PAGE TITLE			
PCH Constraints			
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTL	*	=2x_DIELECTRIC	?
MEM_CTL2CTL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_20THERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?
MEM_CMD2CMD_BM	*	=2x_DIELECTRIC	?
MEM_CMD2CTL_BM	*	=2x_DIELECTRIC	?
MEM_CTL2CTL_BM	*	=2x_DIELECTRIC	?
MEM_12MIL	*	0.305 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTL2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_20THERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CMD_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CMD2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CTL2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	*	*	MEM_20THER
MEM_*_DQS_*	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_CTL	*	*	MEM_20THER
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTL	*	MEM_CMD2CTL
MEM_CTL	MEM_CTL	*	MEM_CTL2CTL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_20THERMEM
MEM_CMD	MEM_CMD	BGA_MEM	MEM_CMD2CMD_BM
MEM_CMD	MEM_CTL	BGA_MEM	MEM_CMD2CTL_BM
MEM_CTL	MEM_CTL	BGA_MEM	MEM_CTL2CTL_BM

Haswell ULT Memory Down DDR3L 1x8 Length Matching

DDR3 Signal Group	Unit	Min Length	Max Length
CTLmax - CTLmin	mils	0	100
CTL to CLK	mils	CLK - 500	CLK + 500
CMDi to CMDj	mils	CMDj - 100	CMDj + 100
CMD to CLK	mils	CLK - 500	CLK + 500
(DQmax - DQmin) per byte	mils	0	250
(DQS - DQmax) per byte	mils	-100	150
DQS to DQS#	mils	-5	5
DQS to CLK (Rule 1)	mils	CLK - 6500	CLK + 500
Max(CLK-DQS) - Min(CLK-DQS)	mils	0	5500
CLK to CLK#	mils	-5	5

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<0>	7 20 22
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM A CKE<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM A CS L<0>	7 20 22
MEM_A_ODT0	MEM_40S	MEM_CTL	MEM A ODT<0>	20 22
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	7 20 22 66
MEM_A_DQBYTE0	MEM_45S	MEM_A_DQBYTE_0	MEM A DQ<7..0>	7 67 68
MEM_A_DQBYTE1	MEM_45S	MEM_A_DQBYTE_1	MEM A DQ<15..8>	7 67 68
MEM_A_DQBYTE2	MEM_45S	MEM_A_DQBYTE_2	MEM A DQ<23..16>	7 67 68
MEM_A_DQBYTE3	MEM_45S	MEM_A_DQBYTE_3	MEM A DQ<31..24>	7 67 68
MEM_A_DQBYTE4	MEM_45S	MEM_A_DQBYTE_4	MEM A DQ<39..32>	7 20 67 68
MEM_A_DQBYTE5	MEM_45S	MEM_A_DQBYTE_5	MEM A DQ<47..40>	7 67 68
MEM_A_DQBYTE6	MEM_45S	MEM_A_DQBYTE_6	MEM A DQ<55..48>	7 67 68
MEM_A_DQBYTE7	MEM_45S	MEM_A_DQBYTE_7	MEM A DQ<63..56>	7 67 68
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM A DQS P<0>	7 67
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM A DQS N<0>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM A DQS P<1>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM A DQS N<1>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM A DQS P<2>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM A DQS N<2>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM A DQS P<3>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM A DQS N<3>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM A DQS P<4>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM A DQS N<4>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM A DQS P<5>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM A DQS N<5>	7 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM A DQS P<6>	7 20 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM A DQS N<6>	7 20 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM A DQS P<7>	7 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM A DQS N<7>	7 67
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<0>	7 21 22
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM B CKE<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM B CS L<0>	7 21 22
MEM_B_ODT0	MEM_40S	MEM_CTL	MEM B ODT<0>	21 22
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	7 21 22 66
MEM_B_DQBYTE0	MEM_45S	MEM_B_DQBYTE_0	MEM B DQ<7..0>	7 67 68
MEM_B_DQBYTE1	MEM_45S	MEM_B_DQBYTE_1	MEM B DQ<15..8>	7 67 68
MEM_B_DQBYTE2	MEM_45S	MEM_B_DQBYTE_2	MEM B DQ<23..16>	7 67 68
MEM_B_DQBYTE3	MEM_45S	MEM_B_DQBYTE_3	MEM B DQ<31..24>	7 67 68
MEM_B_DQBYTE4	MEM_45S	MEM_B_DQBYTE_4	MEM B DQ<39..32>	7 21 67 68
MEM_B_DQBYTE5	MEM_45S	MEM_B_DQBYTE_5	MEM B DQ<47..40>	7 67 68
MEM_B_DQBYTE6	MEM_45S	MEM_B_DQBYTE_6	MEM B DQ<55..48>	7 67 68
MEM_B_DQBYTE7	MEM_45S	MEM_B_DQBYTE_7	MEM B DQ<63..56>	7 67 68
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM B DQS P<0>	7 67
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM B DQS N<0>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM B DQS P<1>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM B DQS N<1>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM B DQS P<2>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM B DQS N<2>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM B DQS P<3>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM B DQS N<3>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM B DQS P<4>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM B DQS N<4>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM B DQS P<5>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM B DQS N<5>	7 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM B DQS P<6>	7 21 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM B DQS N<6>	7 21 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM B DQS P<7>	7 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM B DQS N<7>	7 67
		MEM_PWR	PP1V35 S3	17 19 20 21 22 41 55 65
		MEM_PWR	PP1V35 S3 CPUDDR	8 10 41 65
		MEM_PWR	PP0V675 S0 DDRVTT	22 55 65 68
		MEM_PWR	PPVTTDDR S3	55 65 68
		MEM_12MIL	CPU DIMMA VREFD0	7 19
		MEM_12MIL	CPU DIMMA VREFDQ A ISOL	19
		MEM_12MIL	CPU DIMMB VREFD0	7 19
		MEM_12MIL	CPU DIMMB VREFDQ B ISOL	19
		MEM_12MIL	CPU DIMM VREFCA	7 19
		MEM_12MIL	CPU DIMM VREFCA A ISOL	19
		MEM_12MIL	CPU DIMM VREFCA B ISOL	19
		MEM_12MIL	PP0V675 S3 MEM VREFDQ A	19 20 65
		MEM_12MIL	PP0V675 S3 MEM VREFDQ B	19 21 65
		MEM_12MIL	PP0V675 S3 MEM VREFCA A	19 20 65
		MEM_12MIL	PP0V675 S3 MEM VREFCA B	19 21 65

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Thunderbolt, DP, HDMI Constraints

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt & DisplayPort Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBDP_2SAME	*	=3X_DIELECTRIC	?
TBDP_TXRX	*	=6X_DIELECTRIC	?
TBDP_20THER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBDP_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
TBDP_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
TBDP_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBDP_*	*	*	TBDP_20THER
TBDP_*	=SAME	*	TBDP_2SAME
TBDP_TX	*_RX	*	TBDP_TXRX
TBDP_RX	*_TX	*	TBDP_TXRX

DisplayPort & HDMI Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?
DP_20THER	*	=4x_DIELECTRIC	?
HDMICLK_20THER	*	=7x_DIELECTRIC	?
HDMICLK_2DPHDMI	*	=4x_DIELECTRIC	?
HDMIDATA_2SAME	*	=3x_DIELECTRIC	?
HDMIDATA_20THER	*	=4x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_20THER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICLK_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICLK_2DPHDMI	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMIDATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMIDATA_20THER	TOP,BOTTOM	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDMI_DATA	*	*	HDMIDATA_20THER
HDMI_DATA	=SAME	*	HDMIDATA_2SAME
HDMI_DATA	TBDP_TX	*	HDMIDATA_2SAME
HDMI_DATA	TBDP_RX	*	TBDP_2SAME
HDMI_CLK	*	*	HDMICLK_20THER
HDMI_CLK	HDMI_DATA	*	HDMICLK_2DPHDMI
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DPHDMI
HDMI_CLK	TBDP_TX	*	HDMICLK_2DPHDMI

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_20THER
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	HDMI_DATA	*	DP_2SAME
DISPLAYPORT	TBDP_TX	*	DP_2SAME
DISPLAYPORT	TBDP_RX	*	TBDP_TXRX

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
SOURCE: Calpella SFF D6 Rev 1.5 (407364) and Family GPU D6-04202-001-v04.
MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING
DP_85D	DISPLAYPORT	DP TBTSRC ML C P<3..0>
DP_85D	DISPLAYPORT	DP TBTSRC ML C N<3..0>
DP_85D	DISPLAYPORT	DP TBTSRC AUXCH C P
DP_85D	DISPLAYPORT	DP TBTSRC AUXCH C N
SPI_TBT_CLK	TBT_SPI_45S	TBT_SPI
SPI_TBT_MOST	TBT_SPI_45S	TBT_SPI
SPI_TBT_MISO	TBT_SPI_45S	TBT_SPI
SPI_TBT_CS_L	TBT_SPI_45S	TBT_SPI
DP_HDMI_TBT_ML	DP_85D	DISPLAYPORT
DP_HDMI_TBT_MI	DP_85D	DISPLAYPORT
DP_HDMI_TBT_AUX	DP_85D	DP_HDMI_TBT_AUX_P
DP_HDMI_TBT_AUX	DP_85D	DP_HDMI_TBT_AUX_N
HDMI_CLOCK	HDMI_85D	HDMI_CLK
HDMI_CLOCK	HDMI_85D	HDMI_CLK
HDMI_DATA	HDMI_85D	HDMI_DATA
HDMI_DATA	HDMI_85D	HDMI_DATA

Only used on hosts supporting Thunderbolt video-in

Thunderbolt, DP, HDMI Net Properties

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING
TBT_A_R2D	TBDP_85D	TBDP_TX
TBT_A_R2D	TBDP_85D	TBDP_TX
TBT_A_R2D	TBDP_85D	TBDP_TX
TBT_A_R2D	TBDP_85D	TBDP_TX
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_A_LSX_ML	DP_85D	DISPLAYPORT
DP_TBTPA_ML	DP_85D	DISPLAYPORT
DP_TBTPA_ML	DP_85D	DISPLAYPORT
DP_TBTPA_ML	DP_85D	DISPLAYPORT
DP_TBTPA_ML	DP_85D	DISPLAYPORT
DP_TBTPA_ML	DP_85D	DISPLAYPORT
TBT_A_D2R_0	TBDP_85D	TBDP_RX
TBT_A_D2R_0	TBDP_85D	TBDP_RX
TBT_A_D2R_0	TBDP_85D	TBDP_RX
TBT_A_D2R_0	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
TBT_A_D2R_1	TBDP_85D	TBDP_RX
DP_TBTPA_AUXCH	DP_85D	DP_TBTPA_AUXCH_C_P
DP_TBTPA_AUXCH	DP_85D	DP_TBTPA_AUXCH_C_N
DP_TBTPA_AUXCH	DP_85D	DP_TBTPA_AUXCH_P
DP_TBTPA_AUXCH	DP_85D	DP_TBTPA_AUXCH_N
TBT_B_R2D	TBDP_85D	TBDP_TX
TBT_B_R2D	TBDP_85D	TBDP_TX
TBT_B_R2D	TBDP_85D	TBDP_TX
TBT_B_R2D	TBDP_85D	TBDP_TX
DP_B_LSX_ML	DP_85D	DISPLAYPORT
DP_B_LSX_ML	DP_85D	DISPLAYPORT
DP_B_LSX_ML	DP_85D	DISPLAYPORT
DP_B_LSX_ML	DP_85D	DISPLAYPORT
DP_B_LSX_ML	DP_85D	DISPLAYPORT
DP_B_LSX_ML	DP_85D	DISPLAYPORT
DP_TBTPB_ML	DP_85D	DISPLAYPORT
DP_TBTPB_ML	DP_85D	DISPLAYPORT
DP_TBTPB_ML	DP_85D	DISPLAYPORT
DP_TBTPB_ML	DP_85D	DISPLAYPORT
TBT_B_D2R_0	TBDP_85D	TBDP_RX
TBT_B_D2R_0	TBDP_85D	TBDP_RX
TBT_B_D2R_0	TBDP_85D	TBDP_RX
TBT_B_D2R_0	TBDP_85D	TBDP_RX
TBT_B_D2R_1	TBDP_85D	TBDP_RX
TBT_B_D2R_1	TBDP_85D	TBDP_RX
TBT_B_D2R_1	TBDP_85D	TBDP_RX
TBT_B_D2R_1	TBDP_85D	TBDP_RX
TBT_B_D2R_1	TBDP_85D	TBDP_RX
TBT_B_D2R_1	TBDP_85D	TBDP_RX
DP_TBTPB_AUXCH	DP_85D	DP_TBTPB_AUXCH_C_P
DP_TBTPB_AUXCH	DP_85D	DP_TBTPB_AUXCH_C_N
DP_TBTPB_AUXCH	DP_85D	DP_TBTPB_AUXCH_P
DP_TBTPB_AUXCH	DP_85D	DP_TBTPB_AUXCH_N
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK0_AUXCH_C_P
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK0_AUXCH_C_N
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK0_AUXCH_P
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK0_AUXCH_N
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK1_AUXCH_C_P
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK1_AUXCH_C_N
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK1_AUXCH_P
DP_TBTSNK_AUXCH	DP_85D	DP_TBTSNK1_AUXCH_N
DP_INT_ML	DP_85D	DISPLAYPORT
DP_INT_ML	DP_85D	DISPLAYPORT
DP_INT_ML	DP_85D	DISPLAYPORT
DP_INT_ML	DP_85D	DISPLAYPORT
DP_INT_AUXCH	DP_85D	DP_INT_AUXCH_C_P
DP_INT_AUXCH	DP_85D	DP_INT_AUXCH_C_N
DP_INT_AUXCH	DP_85D	DP_INT_AUXCH_P
DP_INT_AUXCH	DP_85D	DP_INT_AUXCH_N

Notes:
AUX and DDC was removed from DISPLAYPORT or
TBDP_RX/TX because it's not high speed, and
to save routing space.

Only used on dual-port hosts.

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Camera Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	31 32
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	31 32
S2_MEM_CKE	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	31 32
S2_MEM_CS	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	31 32
S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	31 32
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	32 68
		S2_MEM_PWR	PP1V35_CAM	31 32
		S2_MEM_PWR	PP0V675_CAM_VREF	31 32
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	32
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	32

MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?	MIPI_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?	MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_20THER	*	=7X_DIELECTRIC	?	MIPICLK_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?	S2_DATA2SELF	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_DQS20WNDATA	*	=2X_DIELECTRIC	?	S2_DQS20WNDATA	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?	S2_CMD2CMD	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?	S2_CMD2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?	S2_CTRL2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?	S2_20THERMEM	TOP,BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?	S2MEM_2PWR	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?	S2MEM_2GND	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?	S2MEM_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER	S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS*	*	*	S2MEM_20THER	S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA
S2_MEM_CMD	*	*	S2MEM_20THER				
S2_MEM_CTRL	*	*	S2MEM_20THER				
S2_MEM_CLK	*	*	S2MEM_20THER				
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF				
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD				
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL				
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL				
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM				

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

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SMC SMBus & Charger Net Properties

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING
SMBUS_SMC_2	SMB_45S	SMB
SMBUS_SMC_2	SMB_45S	SMB
SMBUS_SMC_1	SMB_45S	SMB
SMBUS_SMC_1	SMB_45S	SMB
SMBUS_SMC_0	SMB_45S	SMB
SMBUS_SMC_0	SMB_45S	SMB
SMBUS_SMC_5	SMB_45S	SMB
SMBUS_SMC_5	SMB_45S	SMB
SMBUS_SMC_3	SMB_45S	SMB
SMBUS_SMC_3	SMB_45S	SMB

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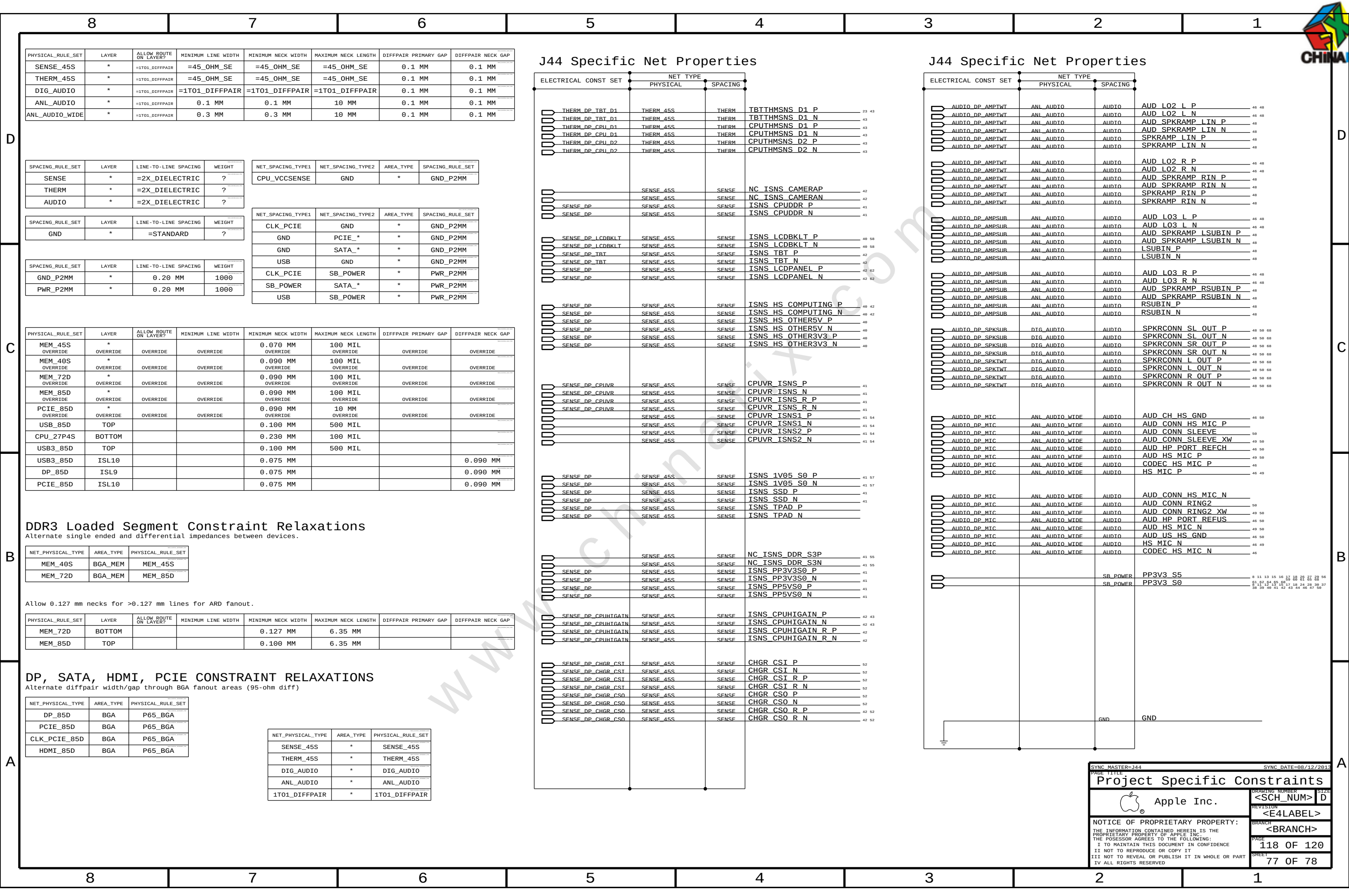
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